The Transformation of Accumulation into Productive Capacities: Capacity Utilization in the Center and Periphery*

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September 29, 2026

Abstract

Chapter 1 demonstrated that the bivariate output–capital relation fails system-level cointegration and that a stable long-run capacity path requires the rate of exploitation to enter the cointegrating vector, yielding a transformation elasticity below unity ($\hat{\theta} < 1$) indicative of structural overaccumulation. This chapter identifies *how* distribution conditions the conversion of accumulated capital into productive capacity and extends the analysis to the periphery. I estimate the transformation elasticity $\theta \equiv \partial \ln Y^p / \partial \ln K$ as a state-dependent cointegrating relation: for the United States (1931–2024), using Integrated Modified OLS with fixed-b inference and endogenously dated structural breaks (Vogelsang and Wagner, 2024; Bai and Perron, 2003); for Chile (1943–2010), using a threshold Cointegrating Multivariate Polynomial Regression with a composite balance-of-payments index. In the United States, θ remains strictly below unity throughout the sample. It rises from 0.37 in 1960 to 0.43 in 2024 as effective wage pressure falls with the diversion of investment into intellectual property. Capacity utilization declines from its 1944 wartime peak to 93.8% in 2024, never re-anchoring to its Fordist level. In Chile, a balance-of-payments threshold at $\hat{\gamma} = 0.425$ switches the mechanization response off: the wage-share elasticity of the capital composition gap falls from +3.29 to +0.14 once foreign exchange binds. Peripheral capitalists retain the incentive to substitute machinery for labor and lose the means to do so, because the machinery is imported. The center–periphery comparison establishes that the same transformation relation is mediated by distributive conflict in the center and truncated by the balance-of-payments constraint in the periphery. Neither economy operates near $\theta = 1$. Capacity utilization is not a norm toward which economies converge; it is the residual of a politically conditioned conversion of accumulation into productive capacity.

Keywords: Capacity Utilization; Transformation Elasticity; Choice of Technique; Cointegration; Okishio Viability; Balance-of-Payments Constraint; Center–Periphery.
JEL Codes: B51; C32; E01; E22; E32; P16.

*This paper benefited from AI-assisted editing for syntax refinement, structural suggestions, and stylistic consistency checks. All analytical arguments, historical interpretations, and substantive content remain the author's sole responsibility.

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1 Introduction

Capacity utilization is among the most widely reported macroeconomic ratios and among the least identified. Central banks read it as a gauge of inflationary pressure. Growth models treat it as the variable that absorbs the gap between accumulation and effective demand. A four-decade heterodox debate over whether firms converge on a normal rate has been conducted almost entirely on the assumption that the rate is observed (Kurz, 1986; Blecker and Setterfield, 2019). It is not. The numerator of the ratio—realized output—is measured. The denominator—productive capacity—is not. Every claim about utilization is therefore a claim about an object no statistical agency records, resting on whatever convention supplied the denominator.

The utilization ratio has been supplied in three ways, and all three fail identically. The first takes utilization as a *given* macro ratio—an observed series to be used as a regressor or a policy input—when its denominator was never observed. The second anchors it to an *inherited benchmark*: an engineering maximum or a survey-reported normal rate imported from outside the accounting framework that is supposed to give the ratio meaning. Managerial surveys do not support the premise this requires (Singh et al., 2022; Nikiforos, 2016). The third recovers utilization as an *empirical residual*: whatever remains after output has been fitted to capital by a filter or a production function. This third treatment is where most current empirical work sits, and the objection to it is structural. Fitting a trend through output conditional on capital sets the capacity path to move one-for-one with accumulation—$\theta = 1$ by construction—and then defines utilization as the deviation from a benchmark that assumed away the very relation needed to locate it. The three treatments are not three difficulties but one: each supplies the denominator by convention, and each conceals the transformation relation that determines it.

Chapter 1 demonstrated that the bivariate output–capital relation fails system-level cointegration and that a stable long-run capacity path requires the rate of exploitation to enter the cointegrating vector, yielding a transformation elasticity below unity ($\hat{\theta} < 1$) indicative of structural overaccumulation. That finding established *that* the output–capital relation is distributionally conditioned. It did not identify *how* distribution conditions the conversion of accumulated capital into productive capacity, nor did it extend the analysis beyond the closed-economy core. This chapter supplies both.

I estimate the transformation elasticity $\theta \equiv \partial \ln Y^p / \partial \ln K$ as a state-dependent cointegrating relation. For the United States over 1931–2024, I use the Integrated Modified OLS estimator of Vogelsang and Wagner (2024) with fixed-b inference and endogenously dated structural breaks (Bai and Perron, 2003); distribution enters through effective wage pressure. For Chile over 1943–2010, the balance-of-payments constraint enters as a threshold that switches the choice of technique. The estimates locate θ well below unity in both economies. In the United States the elasticity moves from 0.37 in 1960 to 0.43 in 2024 as effective wage pressure falls with the diversion of investment into intangibles; utilization declines from its 1944 wartime peak to 93.8% in 2024. In Chile a composite balance-of-payments threshold at $\hat{\gamma} = 0.425$ switches the mechanization response off: the wage-share elasticity of the capital composition gap falls from +3.29 to +0.14 once foreign exchange binds (Hansen, 2000). The center–periphery comparison establishes that the same transformation relation is mediated by distributive conflict in the center and truncated by the balance-of-payments constraint in the periphery.

Two literatures have circled this problem without closing it. The induced-innovation tradition following Kennedy (1964) made distribution govern the *direction* of technical change along a frontier whose *scale* stayed exogenous; its own later assessments concede that it explains the composition of innovation while remaining close to silent on its intensity (Tavani and Zamparelli, 2018). The

measurement literature has established convincingly that reported capacity is a construct rather than an observation (Shaikh, 2016; Nikiforos, 2016), but has stopped short of supplying the relation that would replace it. Okishio (1961, 2022) supplies half of that relation—decentralized investment produces disproportionality as a structural norm, not a friction—and Vidal (2022, 2019) supplies the other half, showing that the firm converting capital into usable capacity is internally divided over control and cooperation and settles into configurations that reproduce production without resolving the conflict. Neither framework was carried to the point where the transformation elasticity could be estimated. Basu (2022, 2026) closes part of the gap by tying the profitability outcome of viable technical change to the labor-market closure, which maps onto historical accumulation regimes; and for the periphery, Prebisch (1981) and Kaldor (1959) established the external ceiling without connecting it to the choice of technique that runs into it.

This chapter makes three contributions against that background. First, it reconstructs capacity utilization as a *derived* quantity, identified downstream of a transformation elasticity that is itself estimated rather than assumed, extending the distributional conditioning established in Chapter 1 from a cointegration-survival result to a structural elasticity with a historical trajectory. Second, it supplies an econometric identification of that elasticity that does not presuppose the relation it recovers, using a cointegrating specification in which distribution enters as a state-dependent interaction and structural change is dated endogenously. Third, it extends the relation to the periphery by letting the balance-of-payments constraint enter as a *threshold* that switches the choice of technique, rather than as a control variable added to it.

The argument runs as follows. The transformation relation cannot be posited without a theory of the unit that performs the transformation, which is what Section 2 supplies by replacing the unified firm with decentralized accumulation and internal division. Section 3 then asks what has to be true of the choice of technique for that transformation to have a determinate elasticity, and derives θ under homogeneous capital, under heterogeneous capital with a mechanization gap, and under an external constraint that can bind. Section 4 confronts the resulting specifications with reconstructed capital stocks for both economies and reports what the estimates say about the historical path of θ and μ. Section 5 sets the two histories against each other, and Section 6 states what the reconstruction settles and what it leaves open.

2 The Internally Divided Firm and Capacity Formation

Capacity utilization is not a technical parameter. It is an inter-scalar political-economic settlement. The macroeconomic capacity ceiling—the denominator of the utilization rate—is not a neutral engineering constraint but a historically contingent outcome conditioned by the balance of class power. Yet identifying this distributional conditioning at the macro level leaves the underlying transmission mechanism black-boxed: how, precisely, does distributive conflict physically alter the conversion of accumulated capital into productive capacity?

Answering this question requires dismantling a hidden premise shared by virtually all heterodox growth models: the unified firm. Whether treating the normal rate of capacity utilization as an endogenous behavioral target or an exogenous gravitational anchor, standard frameworks presume a coherent organization capable of selecting a utilization benchmark, enforcing it across the labor process, and sustaining it over time. Formally, the utilization rate is defined as:

$$\mu_t \equiv \frac{Y_t}{Y_t^p}, \tag{1}$$

where Y_t is realized output and Y_t^p is potential productive capacity. But this ratio conceals a deeper question: who determines Y_t^p, and under what organizational conditions?

The unified-firm premise collapses when confronted with the social organization of capitalist production. In a decentralized economy, individual capitals commit fixed investments without *ex-ante* coordination, confronting one another in competitive markets. Within the enterprise, production decisions encounter contradictory managerial and worker imperatives. Capacity utilization cannot be reduced to a target selected by a single optimizing entity. To theorize the political economy of productive capacity, I replace the unified firm with two interconnected frameworks: Okishio's theory of decentralized market anarchy, and Vidal's theory of the internally divided firm.

2.1 Smooth Reproduction and the Unified-Firm Premise

Proportional reproduction is a condition of feasibility, not a tendency of capitalist accumulation. Smooth reproduction requires strict material proportionality between departments of production—a condition Basu (2022) and Okishio (2022) formally established defined by the latter as the equilibrium accumulation trajectory. Bergmann et al. (2009) call this balanced trajectory, where realized output meets productive capacity ($\mu_t = 1$), the *"golden path."* But the golden path is a methodological closure, not a gravitational attractor. It identifies the conditions under which realization can reproduce itself smoothly; it does not describe the path capitalist accumulation actually follows. Neither the post-Keynesian assumption of long-run balanced growth (Blecker and Setterfield, 2019) nor the *"Keynesian in the short run, classical in the long run"* synthesis (Duménil and Lévy, 1999) captures this persistence of unbalanced expansion.

Nothing in the logic of decentralized competition establishes that private investment decisions will reproduce these proportions. A stylized illustration makes the point concrete. If productive capacity grows at 5% per year while aggregate demand for output grows at 2%, capacity outpaces demand by three percentage points each year. After a decade, utilization falls to roughly 75% of capacity—not because of a temporary shock, but because uncoordinated investment persistently builds capacity faster than demand can absorb it. Balanced growth ($g_Y = g_K$) is merely one configuration of the transformation elasticity θ; it is not a restoring force. Smooth reproduction ($\mu_t = 1$) does not require $\theta = 1$; it holds for any transformation elasticity. Balanced growth is the narrower special case in which $\theta = 1$ and $\mu_t = 1$ coincide in the long run.

Because private investment decisions are made without *ex-ante* coordination, real capitalist economies routinely violate the proportional conditions of the golden path. Okishio (2022) demonstrates that chronic disproportionality is not a temporary friction but the structural norm of decentralized accumulation. Treating the transient benchmark of $\mu_t = 1$ as a behavioral target therefore relies on a hidden premise: that the firm is a unified planning subject capable of selecting a utilization norm, enforcing it across the labor process, and sustaining it over time. This premise—the unified firm—is the target of the present section.

The denominator of the utilization ratio (productive capacity Y_t^p) is not an engineering constant. It is a contested settlement of divided management and shop-floor struggle. The numerator (realized output Y_t) is hostage to the anarchy of demand. Smooth reproduction supplies a reference case against which this historical non-coincidence can be studied. It does not supply the mechanism that produces it.

2.2 Decentralized Accumulation and Market Anarchy

If the "golden path" of smooth reproduction is merely a benchmark of feasibility, the actual trajectory of capitalist accumulation is governed by the decentralized organization of investment. Capitalist production is coordinated through exchange, meaning that the social validation of private decisions occurs only *ex-post*. Because money mediates exchange, sale and purchase are separated moments of the capital circuit (Basu, 2022). Surplus value must be realized in circulation, not by construction (Foley, 1985; Trigg, 2020). This constitutes the logic of capital as impersonal domination: capitalists commit to production and investment with the law of value operating *"behind their backs."*[1] They are subject to market forces and value relations they neither control nor consciously direct. Critically, this anarchy is not a coordination failure in an otherwise functional market. The market is a socially constituted institution whose normal operation produces disproportionality as a structural feature (Vidal and Peck, 2012). Okishio (2022) locates the source of macroeconomic instability precisely in this constitutive anarchy. Individual capitals commit to fixed investment on the basis of private profitability expectations, yet the market validates those commitments only after production has begun. No firm can know in advance whether its expansion fits the investment and consumption decisions taken elsewhere. The coordination problem is not one of imperfect information surrounding an otherwise coherent plan; no economy-wide plan exists.

Smooth reproduction requires proportional expansion across sectors producing means of production and sectors producing consumption goods. Because private investment does not enforce this proportionality, firms extrapolate past prices, profit rates, and sales. A signal that induces expansion in one branch routinely produces excess capacity before complementary demand materializes elsewhere. Contraction works through the same cumulative mechanism: falling sales cut investment, reducing aggregate demand and validating further cuts. Removing Say's Law reintroduces realization as a substantive condition of reproduction: accumulation may generate capacity that demand fails to absorb, or demand may temporarily exceed productive capacity (Foley, 1985). Prices register the resulting imbalance, but they do not prevent the sunk commitments produced by it.

Fixed capital gives this temporal lag its structural force. Structures and machinery embody long-horizon commitments that cannot be reversed when relative prices or demand conditions shift. Once installed, excess capacity leaves the firm to operate below its anticipated scale, defer maintenance, hold equipment idle, or devalue the asset. These responses distribute the costs of market anarchy among firms, workers, and creditors, but they do not restore the missing *ex-ante* coordination.

Idle capacity is therefore a recurrent, structural form of decentralized accumulation rather than a temporary friction. Private investment physically builds capacity before the market can realize its social use, recurrently a gap between installed productive capacities (Y_t^p) and effective demand (Y_t). The capacity utilization rate (μ_t) registers this gap at the macro level. Unbalanced growth is not a disturbance around proportionality; it is the structural consequence of distributively conditioned investment under decentralized decision-making. The analytical problem is therefore not how proportionality re-emerges, but how persistent imbalance remains historically reproducible. The condition $\mu_t = 1$ is at most a transient point through a given trajectory proportionality, while persistent divergence records the ongoing reality of market anarchy.

Okishio's macro framework stops one step short: it does not explain how firms translate this

[1] Karl Marx, *Capital*, Volume I, Chapter 1, Section 4 (*"The Fetishism of Commodities and the Secret thereof"*), where the law of value operates as an overriding law of nature when private producers exchange products (Marx, 1999).

external anarchy into internal operating decisions. The firm does not manage market anarchy; it internalizes it. A firm does not receive a price signal and respond as a frictionless, unified unit. It processes the signal through an organization divided over control, cooperation, effort, and the allocation of resources. Vidal's internally divided firm provides that missing mediation.

2.3 The Internally Divided Firm

Vidal builds on Marx's Volume I theory of the labor process, drawing on contemporary organizational theory and sociological institutionalism to develop an organizational political economy of the capitalist firm (Vidal and Peck, 2012; Vidal, 2022). Marx established that capitalist management faces a constitutive contradiction: it must simultaneously organize the cooperative labor required for production and enforce the coercive control required for valorization. Vidal (2022) demonstrates how this contradiction fractures the modern capitalist firm, treating it as a divided social organization whose internal structure is shaped by cognitive, normative, regulative, and organizational institutions. Management must extract surplus value through control over labor, but production also requires workers' cooperation, tacit knowledge, and problem-solving capacity. Standardization and surveillance secure a measurable pace of work while weakening the discretion needed to sustain complex production. The firm cannot maximize both imperatives at once because the labor process that produces value is also the site where workers contest its extraction. This contradiction is not a universal constant of capitalism; it is institutionally mediated, taking variegated forms across distinct accumulation regimes throughout historical capitalism (Vidal, 2019).

Workers respond to this contradiction by actively calibrating effort, cooperation, machine pacing, and the deployment of tacit knowledge. Capitalist management extracts potential capacity through concrete shop-floor mechanisms: increasing machine speed, deferring maintenance to hit short-term throughput targets, and enforcing rigid surveillance. Workers contest this extraction through labor turnover, collective exit, or the deliberate withdrawal of problem-solving cooperation and effort. Throughput, quality, and maintenance therefore depend on social relations that no engineering specification fixes in advance. A production target that raises machine speed by deferring maintenance may increase current output while degrading sustainable capacity. A directive that suppresses worker discretion may meet a short-term labor-control objective while worsening workflow bottlenecks. Labor effort is consequently not a fixed technical parameter applied to a pre-existing capital stock; it is an active, contested determinant of the capacity utilization itself.

This dynamic enforces a strict wedge between *engineering capacity* and *practical capacity*. Engineering capacity records the theoretical maximum output of equipment under ideal technical conditions. Practical capacity records the sustainable output rate dictated by actual maintenance schedules, product-mix complexity, and the negotiated intensity of labor. When managers report capacity to national statistical agencies, they rely on conventions shaped by their organizational position and the immediate pressures of the shop floor. Aggregate capacity statistics therefore do not reveal a neutral physical ceiling; they aggregate situated, institutional judgments (Shaikh, 2016; Nikiforos, 2016). Complementarily, empirical surveys of managerial practice confirm that firms do not systematically organize production decisions around a coherent capacity utilization target (Singh et al., 2022). For the purposes of macroeconomic modeling, the productive capacity ceiling (Y_t^p) is not an exogenous engineering constraint but an endogenous settlement of workplace conflict. This settlement is the micro-foundation through which class struggle and distribution transmit into capacity utilization, setting up the distributive–technological mapping developed in the following subsection.

Because the valorization–labor process contradiction cannot be solved, firms settle into organizational configurations of *"mediocre sufficiency"* (Vidal, 2022). These are hybrid arrangements that reproduce production without resolving the underlying conflict between control and cooperation. Partial worker autonomy coexists with rigid discipline; continuous improvement programs coexist with brutal throughput pressure; machinery might remain idle alongside binding productive bottlenecks. Following Vidal (2014), these arrangements are best understood as *institutional incoherence*: the firm's internal settlements are functional enough to sustain accumulation yet contradictory enough to generate persistent friction. This stands in direct contrast to the Varieties of Capitalism assumption of institutional complementarity, where national institutions are presumed to form coherent, mutually reinforcing systems (Hall and Soskice, 2003). Here, incoherence is not a deviation from an optimizing norm; it is the permanent, structural form through which capitalist management physically converts accumulation demand into capacity usage.

2.4 Distribution, Technique, and the Transformation of Accumulation

The choice of technique connects the divided firm to distribution, governing how accumulated capital is physically transformed into productive capacity. Capitalists adopt a new technique only when it reduces the cost of production at prevailing prices and wages; such a technique is capitalistically viable (Okishio, 1961). Okishio (1961) demonstrates that a viable capital-using, labor-saving (CU-LS) technique raises the equilibrium profit rate when the real wage remains constant—a direct challenge to Marx's law of the tendential fall in the rate of profit. Marx's account, by contrast, allows the same form of technical change to depress profitability when rising capital intensity increases the capital advanced relative to surplus value. Basu (2022) reconciles this seemingly mutually exclusive theorizations, by formalizing the *Marx–Okishio threshold*: the rate of real-wage growth defines the result driven by the balance of class forces.

This threshold situates the present framework within the induced-innovation tradition while departing from it. The Innovation Possibility Frontier (IPF) literature, initiated by Kennedy (1964), treats distribution as selecting the *direction* of technical change along a given frontier: relative factor costs determine whether innovation is labor-augmenting or capital-augmenting, while the frontier itself—the scale of feasible productivity growth—remains exogenous (Julius, 2005; Tavani, 2012). Later assessments sharpen this limitation: IPF-based models explain the composition of technical change but remain largely silent on innovation intensity (Tavani and Zamparelli, 2018). Zamparelli (2024a) partially endogenizes scale by mapping a family of IPF levels to R&D effort, yet the analytical structure continues to rely on frontier geometry and steady-growth closure (Zamparelli, 2024b). An alternative strand reframes inducement through mechanization rather than frontier selection: Kaldor (1957) links productivity growth to capital deepening, while Cajas-Guijarro (2024) reformulates the IPF with an explicit choice over mechanization growth under distributive and labor-market conditions. The present framework builds upon a mechanization productive frontier, treating the distributive–technological relation as a regime-specific transformation between accumulation and productive capacity. Rather than selecting along a given frontier, distribution conditions mechanization intensity, which in turn determines the transformation elasticity linking accumulation to the formation of productive capacities.

The Marx–Okishio threshold gives wage pressure a precise role in mechanization, but the profitability outcome depends on the labor market closure. Basu (2026) demonstrates that the equilibrium profit rate after viable CU-LS technical change is governed by the institutional structure of capital–labor bargaining. Under a constant-wage-share closure—what Basu (2026) formalizes

as the "strong labor" regime, applicable to Atlantic Fordism—any viable CU-LS technique unambiguously *lowers* the equilibrium profit rate. Marx's LTFRP reigns supreme precisely where labor is powerful enough to compel real wages to grow in tandem with productivity. Under this closure, rapid mechanization is induced by distributive pressure, yet the same pressure contests capital's claim on the resulting productivity gains (Vidal, 2015, 2019). A stylized illustration makes the mechanism concrete. If a 10% real wage increase raises mechanization intensity by 8% under strong-labor conditions, the transformation elasticity θ rises from 0.70 to 0.85: accumulation translates more efficiently into productive capacity because firms substitute machinery for labor. Wage-led mechanization was not a harmonious productivity bargain; but a a contested settlement in which the distributive pressure that induced technical change simultaneously eroded the profit rate it was meant to restore.

The weakening of organized labor after the Fordist crisis altered that mechanism. Under a *"weak labor"* closure—where the unemployment rate remains constant such that workers lack the institutional power to enforce a target wage share—the distributive pressure to replace labor is suppressed (Basu, 2026). Yet mechanization does not necessarily cease. Lower relative machinery prices and technical change in capital-goods production can continue to favor capital-using techniques through a second, independent channel. Distribution and machinery prices therefore constitute a dual-channel mechanism in the choice of technique. Their relative force changes across accumulation regimes, which prevents the wage share from being treated as an external control added to an otherwise autonomous technical relation. Extending the stylized example: under Post-Fordist conditions, a 15% decline in relative machinery prices raises mechanization intensity by 6%, partially compensating for the weakened distributive channel and sustaining θ near 0.75 despite wage suppression. The same 10% wage increase that produced an 8% mechanization response under Fordism produces only a 2% response under Post-Fordism, leaving θ near 0.72. The two channels do not simply add; they interact through the institutional configuration that governs their relative force.

These channels still operate through the contradictions identified in the preceding subsections. Firms choose techniques before demand validates the associated capacity, and they operate those techniques through labor processes that divide control from productive cooperation. Investment consequently does not translate one-for-one into a stable productive ceiling. The transformation depends on the composition of capital, the distributive conditions governing mechanization, and the organization through which installed equipment becomes usable capacity. In a labor surplus economy—the third closure identified by Basu (2026), applicable to peripheral capitalism—the real wage remains constant and the equilibrium profit rate rises unambiguously after viable CU-LS technical change. Yet the peripheral transformation is structurally truncated by external financial subordination and the balance of payments constraint, a mechanism developed in the following subsection.

Okishio and Vidal jointly rule out a direct passage from accumulation to productive capacity. Decentralized investment generates structural disproportionality across firms and sectors; the internally divided firm internalizes this disproportionality as contested operating conditions rather than a coherent utilization target. The analytical object that captures this mediated transformation is the *transformation elasticity* ($\theta \equiv dy^p/dk$), which governs how accumulation converts into productive capacity under a context of distributive conflict and market anarchy. The empirical analysis limits the original inter-departamental formulation by Marx in Capital Volume II into a single-sector aggregate, masking internal sectoral divergences while preserving the core logic.

2.5 Historical Regimes of Institutional Mediation and the Transformation Elasticity

The transformation elasticity is an emergent property of class struggle that mediates the process of capital accumulation and its transformation into productive capacities under the prevailing institutional environment. Decentralized accumulation does not automatically guarantee proportional reproduction ($\theta = 1$). Institutional mediation—wage-setting norms, financial architectures, and competitive structures—conditions how distributive pressures translate into mechanization choices (Aglietta, 2000).

Basu (2026) demonstrates that the equilibrium profit rate after viable capital-using, labor-saving (CU-LS) technical change is governed by the labor market closure, which maps directly onto distinct historical accumulation regimes. Under a *constant-wage-share* closure—the institutional signature of Atlantic Fordism (1947–1973)—workers possess the bargaining power to grasp productivity gains, forcing real wages to grow in tandem with labor productivity. In this strong-labor regime, viable mechanization unambiguously lowers the equilibrium profit rate, so that Marx's law of the tendential fall in the rate of profit holds (Basu, 2026). This stagnation tendency was institutionally offset: the class compromise, mass consumption, oligopolistic market structures, and military Keynesianism of the Fordist era absorbed the surplus and sustained effective demand (Vidal, 2019).

Under a *constant-unemployment* closure—which resembles the persistent unemployment feature of the post-Fordist accumulation regime—an era during which the class compromise that took place during the Fordist period is dismantled and labor's bargaining power is severely weakened. The wage share falls, and mechanization is driven not by distributive pressure but by the relative cheapening of ICT and the shift toward financially reversible, short-horizon assets, patents and intellectual property of intangible assets (Lapavitsas, 2011; Pagano, 2014). Because wages no longer track productivity, the profit rate may stabilize or rise, but the suppression of wage-led demand generates a chronic under-consumptionist stagnation tendency. Following Vidal (2019), I hypothesize that under structural conditions, the post-fordist accumulation regime produces a dysfunctional utilization configuration: accumulation decouples from productive capacity formation, generating chronic idle capacity that is temporarily masked by debt-led consumption and financial fragility rather than sustainable realization (Stockhammer, 2012; Vidal, 2019).

In the capitalist periphery, different closures represent distinct historical configurations throughout Chilean history. First, under an assumption of a *constant-real-wage*, which corresponds to Okishio's original formulation maps onto a surplus-labor economy (Lewis, 1954; Basu, 2026). During the import-substituting industrialization (ISI) era, the *rural-urban* and *demographic* transition enabled by the transformation of sanitary conditions and a massive drop of child mortality in the early XX century, regulated late industrialization, providing an unlimited supply of labor from the countryside that disciplined urban wages and allowed the profit rate to reproduce itself alongside mechanization. However, peripheral mechanization is structurally import-biased: widening the mechanization gap requires imported capital goods and technologically non-substitutable inputs (Prebisch, 1981). The balance of payments constraint therefore acts as a hard external ceiling on the transformation of accumulation into capacity. Under a developmentalist regime with surplus labor (1933–1957), this operated as an *import capacity ceiling*. Second, a *constant-wage-share* or more accurately *rising-wage-share* closure, represents the intensification of stagnation tendencies leading to structural crisis of capitalism once the unlimited supply of labor depleted and the incentives for capitalist firms to mechanize risen due distributive pressures while increasing import requirements and stressing further the external constraint. Third, the discipline imposed by the neoliberal

counter-revolution (Casals, 2021) (1974–present) a *constant-unemployment* closure coupled with trade and capital account liberalization relieved the external constraint, but locked the periphery into extractive specialization and premature deindustrialization, such that the WTO and bilateral free trade agreements legally encoded a *policy space ceiling* (Ahumada, 2019; Palma, 2019).

These historical configurations establish that the productive ceiling is not a neutral, trans-historical engineering constraint but a contingent outcome of decentralized accumulation, class power, and a country's relative positionality within the capitalist world system. The center–periphery comparison reveals a fundamental asymmetry: in the center, structural crises arise from the *exhaustion* of institutional mediation (e.g., the 1970s profit squeeze); in the periphery, arise from the *absence* of sovereign mediation and the truncation imposed by the hierarchically oranized financial order imposing an external constraint that is not necessarily binding.

3 The Analytical Framework: Formalizing the Transformation Elasticity

Distributive pressure conditions the choice of technique, and the choice of technique determines how much capacity a unit of accumulation produces. Section 2 established that claim historically and institutionally. It carries no elasticity until the cost-minimization problem behind it is written down, because only then does θ acquire a determinate value rather than a direction of movement. Cajas-Guijarro (2024) first posed that problem as a choice over mechanization growth, and I extend it in three steps.

The steps are nested rather than alternative. A firm that treats capital as homogeneous faces a single margin, and the elasticity that emerges from its cost minimization is a scalar. Allow the capital stock to separate into nonresidential structures and machinery, and a second margin opens: the mechanization gap κ_t, whose marginal capacity payoff depends on distribution and on relative asset prices at once. Impose a foreign-exchange bottleneck on the machinery half of that gap, and the interior solution can fail entirely, truncating θ at a corner. Setting the heterogeneity parameter to zero recovers the first model from the second; relaxing the foreign-exchange constraint recovers the center from the periphery. One structure, evaluated under three constraint sets.

A limitation is worth stating before the derivations rather than after them. The mechanism is multi-sectoral in principle—disproportionality is a relation between departments of production—while the empirical implementation bounds it to a single-sector aggregate. That bound masks internal sectoral divergence. It preserves the transformation logic, which is what the estimation requires, but it means θ is recovered as an economy-wide average of a relation that operates unevenly across branches.

3.1 The Baseline Model of Cost-Minimizing Mechanization (Homogeneous Capital Stock)

Actual aggregate output is represented by a Leontief production function with fixed technical coefficients:

$$Y = \min\left(AL,\ \mu BK\right) \tag{2}$$

where $A = Y/L$ is actual labor productivity, $\mu = Y/Y^p$ is capacity utilization, and $B = Y^p/K$ is maximum capital productivity. This production structure governs the rate of profit through the

fundamental heterodox accounting identity:

$$r = \mu B(1 - \omega) \tag{3}$$

where ω is the wage share. Standard macroeconomic models treat B as an exogenous technological given and μ as a behavioral constant smoothly targeted by a unified firm. I reject both assumptions. The components of (3) are endogenous outcomes of class struggle and decentralized market anarchy, as established in Section 2. This subsection formalizes how distributive pressure forces cost-minimizing firms to alter their choice of technique, endogenously determining B.

3.1.1 Tactical Choice of Technique and Induced Innovation

Capitalist firms choice of technique is given by aiming a mechanization growth rate q under competitive cutting cost-pressures, looking at the net productivity surplus of technical change as their objective function. Let a denote the growth rate of labor productivity. The effective distributive cost of deploying mechanization growth is $q(1 - \omega)$. Hence, the choice of technique of capitalist mechanization can be represented by:

$$\min_{q} \; c(q) = q(1 - \omega) - a \tag{4}$$

This objective captures the induced-innovation logic: firms accelerate mechanization to expand labor productivity growth, but the distributive cost of capital intensity holds them back. The cost-minimizing objective function of capitalist becomes constrained by a mechanization function that can be represented as:

$$a = \Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2 \tag{5}$$

Assumption 1. *Concavity of the mechanization frontier.* The frontier parameters satisfy $\lambda_0 > 0$, $\lambda_1 > 0$, and $\lambda_2 < 0$. The base productivity growth λ_0 sets the floor without advanced machinery. The linear coefficient λ_1 captures the direct productivity payoff of mechanization. The strict concavity $\lambda_2 < 0$ ensures diminishing marginal returns to mechanization.

Substituting (5) into (4) and solving the first-order condition yields the optimal mechanization growth:

$$c(q) = \lambda_0 + [\lambda_1 - (1 - \omega)]\, q + \lambda_2 q^2 \tag{6}$$

$$\frac{\partial c}{\partial q} = \lambda_1 - (1 - \omega) + 2\lambda_2 q^* = 0 \tag{7}$$

$$q^*(\omega) = \frac{(1 - \omega) - \lambda_1}{2\lambda_2} \tag{8}$$

The strict concavity ($c''(q) = 2\lambda_2 < 0$) guarantees a unique maximum.

Proposition 1. *Induced innovation.* Under Assumption 1, the optimal mechanization growth is strictly increasing in the wage share:

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2} > 0 \tag{9}$$

A rise in the wage share forces cost-minimizing firms to substitute machinery for labor. Rising wages

drive mechanization directly through the first-order condition, pushing optimal capital intensity growth upward. *Proof.* Immediate from differentiating (8). ■

Remark 1. The objective (4) does not assert that the firm is a coherent optimizer. Following Vidal (2022), the firm is internally divided between the *control imperative* (valorization: cost reduction through mechanization) and the *cooperation imperative* (production: the need for worker effort, tacit knowledge, and problem-solving capacity). The formal model captures the cost-minimizing pole—the Okishio viability logic—while the constraints on the optimization problem capture the limits imposed by the cooperation imperative and by external conditions. The firm maximizes net productivity surplus—labor productivity growth minus the distributive cost of mechanization—subject to diminishing returns. The firm's actual mechanization choice is a contested settlement between these imperatives, not a coherent optimum. What the model demonstrates is the cost-reducing direction in which distributive pressure pushes the settlement, not the settlement itself.

3.1.2 Endogenous Transformation Elasticity

Substituting the optimal technique back into the technological frontier recovers the normal growth rate of labor productivity under smooth reproduction:

$$a^N(\omega) = \lambda_0 + \lambda_1 q^*(\omega) + \lambda_2 \left[q^*(\omega)\right]^2 \tag{10}$$

This optimal technique dictates the structural trajectory of the Leontief parameter B. By definition, maximum capital productivity is $B = Y^p/K$. In dynamic growth rates, the growth of capital productivity is:

$$b \equiv g_{Y^p} - g_K = a - q \tag{11}$$

Evaluated at the smooth reproduction benchmark: $b^N(\omega) = a^N(\omega) - q^*(\omega)$.

Definition 1. *Transformation elasticity.* The transformation elasticity θ is the structural parameter that translates capital accumulation into productive capacity growth:

$$g_{Y^p} = \theta\, g_K \tag{12}$$

Substituting (12) into (11) reveals the link between the level coefficient B and the elasticity θ:

$$\theta = 1 + \frac{b}{g_K} \tag{13}$$

To evaluate this elasticity, I anchor the choice of technique to a baseline rate of capital accumulation. Following the normal reproduction path, the normal accumulation rate is driven by the normal profit rate and the normal rate of re-capitalization net of depreciation. I follow Foley (1986) definition of the re-capitalization rate, $\chi = \frac{I}{\Pi}$, and interpret its normal conditions as continuous realization and re-investment of the surplus.[2] Hence,

$$g_K^N = \chi^N r^N - \delta \tag{14}$$

In dynamic growth accounting, changes in the growth rate of capital represent an acceleration term $\ddot{k} \equiv dg_K/dt$. In historical time, $\ddot{k}$ is kept strictly bounded by corporate liquidity constraints, debt-

[2] A simple accounting decomposition can demonstrate that the multiplier of profitability that drives investment, is precisely χ. In a monetary economy with credit, the re-capitalization rate does not require to be bounded by the surplus Π order of magnitude.

servicing ceilings, and Central Bank credit cycles, preventing explosive capital growth divergence. Expressed dynamically relative to this normal path, the aggregate transformation elasticity becomes:

$$\theta(\omega) = 1 + \frac{b^N(\omega)}{g_K^N} = 1 + \frac{a^N(\omega) - q^*(\omega)}{g_K^N} \tag{15}$$

Proposition 2. *Endogeneity of the transformation elasticity.* Under Assumption 1, the transformation elasticity θ is not an exogenous technical constant. It emerges endogenously from the cost-minimizing choice of technique. When distributive pressure forces firms to alter mechanization via (9), it simultaneously shifts the structural efficiency of accumulation via (15). The elasticity θ is therefore a function of the wage share ω, the frontier parameters $\lambda_0, \lambda_1, \lambda_2$, and the normal accumulation rate g_K^N.

Econometric note on capital measurement error. Because the aggregate capital stock K_t is constructed via Perpetual Inventory (GPIM) methods involving service-life and discard schedules, any classical right-hand-side measurement error in K_t introduces standard econometric attenuation bias that pulls the estimated elasticity $\hat{\theta}$ downward toward zero. Classical measurement noise cannot account for the empirical finding that $\theta < 1$; rather, it reinforces that the true structural transformation elasticity is bounded strictly below unity even under conservative measurement assumptions.

3.2 Heterogeneous Capital, the Mechanization Gap, and State-Dependent Viability

Mainstream macroeconomics treats the capital stock as a homogeneous aggregate, abstracting from the internal architecture of the firm. Following the theory of organizational political economy and the internally divided firm (Vidal, 2022), I unbundle aggregate productive capital into and extensive margin given by nonresidential structures (K^{NRC}) and intensive one, given by the accumulation of machinery and equipment (K^{ME}). These components do not unfold throughout the labor process symmetrically. Structures form the built environments that setup the "envelope"of accumulation — i.e. the factory floor and logistics shell of the productive process. Capital accumulation in Machinery and Equipment is the active component that restructures the production process and the extraction of relative surplus value (Marx, 1999, Vol. I, Ch. 15).

Assumption 2. *Heterogeneous capital decomposition.* Aggregate productive capital decomposes as $K_t^{cap} = K_t^{NRC} + K_t^{ME}$. Nonresidential structures evolve independently of short-run distributive conflict (ω) and its rate of accumulation of accumulation is governed by its long-run relation with the rate of accumulation in machinery and equipment under normal conditions of accumulation in the long-run

$$g^N_{K^{NRC},t} = f(\chi^N, r_t^N, g^N_{K^{ME},t}, \delta) \tag{16}$$

This extensive envelope $\Theta(K^{NRC})$ becomes a baseline for aggregate elasticity of transformation θ such that the rate of accumulation of capital stock in non-residential construction is a byproduct of the intensive margin of accumulation decided by a tactical technique choice made by capitalists in response to distributive pressures:

3.2.1 The Extensive Envelope and the Intensive Margin

Given the fixed infrastructural envelope, the **intensive margin** of accumulation is the tactical choice of the mechanization gap, defined in natural logs as:

$$\kappa_t = k_t^{ME} - k_t^{NRC} \tag{17}$$

This gap measures mechanization intensity relative to the extensive plant scale. The capacity identity decomposes potential output into the extensive envelope and the intensive capacity yield Ψ_t:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \Psi_t(\kappa_t, \omega_{t-1}, p_t^\kappa) \tag{18}$$

The intensive yield Ψ_t is not a fixed engineering constant. It is the outcome of a decentralized cost-minimization problem governed by the regime of accumulation, as formalized below.

3.2.2 Cost-Surplus Minimization and the Okishio Viability Yield

In this case, the capitalist firm chooses a mechanization gap κ_t under cutting-cost pressures, subject to the mechanization frontier:

$$\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2 \tag{19}$$

The *realized* capacity yield is governed by a viability condition: a cost-minimizing firm deploys a new technique only if it reduces unit costs at current distributive and price parameters (Okishio, 1961; Basu, 2022). I formalize this as a cost-minimization as problem:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) - \Phi(\kappa_t) \tag{20}$$

The marginal viability cost $MC(\omega_{t-1}, p_t^\kappa)$ is determined by two channels:

1. **The distributive dividend (labor saving).** Increasing κ_t substitutes machinery for labor. The value of this displacement is proportional to the lagged wage share (ω_{t-1}), reflecting the wage settlement period prior to current capacity realization. A higher wage share *reduces* the net cost of mechanization.

2. **The relative price penalty (capital cost).** Increasing κ_t requires accumulating machinery faster than structures. The cost of this composition shift is proportional to the log relative price of machinery ($p_t^\kappa = \ln P_t^{ME} - \ln P_t^{NRC}$). A higher relative price *increases* the marginal cost.

I specify the marginal viability cost as:

$$MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa \tag{21}$$

where $\gamma_1 > 0$ captures the wage-savings dividend and $\gamma_2 > 0$ captures the relative price penalty.

Substituting (19) and (21) into (20) and solving the first-order condition yields the optimal

mechanization gap:

$$c(\kappa_t) = \lambda_0 + [\lambda_1 - (\gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^{\kappa})]\,\kappa_t + \lambda_2\kappa_t^2 \tag{22}$$

$$\frac{\partial c}{\partial \kappa_t} = \lambda_1 - \gamma_0 + \gamma_1\omega_{t-1} - \gamma_2 p_t^{\kappa} + 2\lambda_2\kappa_t^* = 0 \tag{23}$$

$$\kappa_t^*(\omega_{t-1}, p_t^{\kappa}) = \frac{(\gamma_0 - \lambda_1) - \gamma_1\omega_{t-1} + \gamma_2 p_t^{\kappa}}{2\lambda_2} \tag{24}$$

Proposition 3. *Dual-channel induced innovation.* Under Assumption 1 (concavity) and Assumption 2 (heterogeneous capital), the optimal mechanization gap satisfies:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\gamma_1}{2\lambda_2} > 0 \tag{25}$$

$$\frac{\partial \kappa_t^*}{\partial p_t^{\kappa}} = \frac{\gamma_2}{2\lambda_2} < 0 \tag{26}$$

A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. A rising relative price of machinery suppresses the optimal gap, as the structural cost of the composition shift penalizes mechanization. *Proof.* Immediate from differentiating (24). ■

3.2.3 The Marginal Capacity Payoff and the Econometric Bridge

The macroeconomic translation from the mechanization gap to potential capacity is given by the *marginal capacity payoff* (MP_κ), the derivative of the intensive capacity yield Ψ_t with respect to κ_t. By the envelope theorem:

$$MP_\kappa = \frac{\partial \Psi_t}{\partial \kappa_t} = \frac{\partial \Phi}{\partial \kappa_t} - MC(\omega_{t-1}, p_t^{\kappa}) \tag{27}$$

$$= (\lambda_1 + 2\lambda_2\kappa_t) - (\gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^{\kappa}) \tag{28}$$

$$= \underbrace{(\lambda_1 - \gamma_0 + 2\lambda_2\kappa_t)}_{\text{autonomous yield}} + \gamma_1\omega_{t-1} - \gamma_2 p_t^{\kappa} \tag{29}$$

Integrating the marginal payoff with respect to κ_t (treating ω_{t-1} and p_t^{κ} as predetermined shifters) and defining θ_κ, $\theta_{\omega\kappa} \equiv \gamma_1$, and $\theta_{\kappa p\kappa} \equiv -\gamma_2$ yields the intensive capacity yield function:

$$\Psi_t = \theta_\kappa\kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^{\kappa}\kappa_t) \tag{30}$$

Substituting (30) back into (18) recovers the structural capacity equation:

$$y_t^p = \theta_{NRC}k_t^{NRC} + \theta_\kappa\kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^{\kappa}\kappa_t) \tag{31}$$

The interaction terms are not ad-hoc econometric additions. They are the macroeconomic footprint of the divided firm's Okishio viability constraint. Distributive pressure and relative price enter because they alter the marginal capacity payoff of the mechanization gap.

3.2.4 The Transformation Elasticity under Heterogeneous Capital Stock

Specification B recovers separate capacity elasticities for nonresidential structures (θ_{NRC}) and machinery (θ_κ), but evaluating aggregate capacity dynamics requires defining an aggregate transformation elasticity ($\theta_t^{\text{Aggregate}}$). In a single-asset model (Specification A), $\theta_A = \partial y^p / \partial k^{\text{cap}}$ is a scalar. Under heterogeneous capital, the total differential of productive capacity (31) separates extensive expansion from intensive composition shifters:

$$dy_t^p = \theta_{\text{NRC}}\, dk_t^{\text{NRC}} + MP_\kappa\, d\kappa_t \tag{32}$$

$$= \theta_{\text{NRC}}\, dk_t^{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\, \omega_{t-1} + \theta_{\kappa p\kappa}\, p_t^\kappa)\, d\kappa_t. \tag{33}$$

Standard continuous-time growth accounting evaluates aggregate capital variation using a Divisia index of heterogeneous components (Divisia, 1925). Since $s_{\text{ME},t} + s_{\text{NRC},t} = 1$ and $d\kappa_t = dk_t^{\text{ME}} - dk_t^{\text{NRC}}$, aggregate capital growth decomposes as:

$$dk_t^{\text{cap}} \approx s_{\text{ME},t}\, dk_t^{\text{ME}} + s_{\text{NRC},t}\, dk_t^{\text{NRC}} \tag{34}$$

$$= dk_t^{\text{NRC}} + s_{\text{ME},t}\, d\kappa_t. \tag{35}$$

Dividing (33) by (35) yields the conventional total-derivative aggregate elasticity along the empirical growth path:

$$\theta_{\text{total},t}^{\text{Aggregate}} = \frac{\theta_{\text{NRC}} + (\theta_\kappa + \theta_{\omega\kappa}\, \omega_{t-1} + \theta_{\kappa p\kappa}\, p_t^\kappa)\, \frac{d\kappa_t}{dk_t^{\text{NRC}}}}{1 + s_{\text{ME},t}\, \frac{d\kappa_t}{dk_t^{\text{NRC}}}}. \tag{36}$$

Equation (36) suffers from a severe theoretical flaw illuminated by the Cambridge Capital Controversy (Robinson, 1953; Sraffa, 1960). Heterogeneous physical assets cannot be aggregated into a single scalar k^{cap} without value weights that depend on distribution and prices. Differentiating such a value aggregate and placing dk_t^{cap} in the denominator re-introduces the circularity that the model with heterogeneous capital was designed to eliminate. Empirically, the denominator $1 + s_{\text{ME},t}(d\kappa_t / dk_t^{\text{NRC}})$ vanishes whenever physical accumulation stalls or shifts composition, producing spurious zero-denominator singularities ($\theta \to \pm\infty$).

Alternatively, I construct the aggregate transformation elasticity as an asset weighted from the heterogeneous capital stock system. Rather than differentiating a value aggregate, the structural asset-weighted transformation elasticity weights identified heterogeneous elasticities by gross capital asset shares:

$$\theta_t^{\text{Aggregate}} = \theta_{\text{NRC}}^{\text{str}} + s_{\text{ME},t} \cdot MP_{\kappa,t}^{\text{str}}, \tag{37}$$

where $\theta_{\text{NRC}}^{\text{str}}$ is the base elasticity of structures, $s_{\text{ME},t} = K_{\text{ME},t} / K_{\text{cap},t}$ is the physical equipment share, and $MP_{\kappa,t}^{\text{str}} = \theta_\kappa^{\text{str}} + \theta_{\omega\kappa}\, (\omega_{t-1} - \bar{\omega}) + \theta_{\kappa p}\, (p_t^\kappa - \bar{p}^\kappa)$ is the marginal capacity yield of mechanization. Equation (37) is immune to the Cambridge critique because the parameters are identified in a disaggregated system without prior scalar aggregation, and the weighting share $s_{\text{ME},t}$ is a physical asset ratio. This provides the equation for the trajectory to be implcitly identified from the co-integration relation estimated in empirically in Section 4.3.

3.3 The Peripheral Extension: Balance-of-Payments Constraint and the Corner Solution

In the capitalist periphery, the intensive margin of accumulation is structurally tethered to the world market and constrained by domestic social friction. Following Prebisch (1981) and Kaldor (1959), peripheral capitalism is characterized by high consumption propensities out of profits, structural inelasticities in agriculture, and a chronic vulnerability to balance-of-payments crises (Ahumada, 2019). This introduces a dual realization bottleneck: the macroeconomic capacity to import and the inelastic domestic supply of wage-goods. Section 2.5 established this constraint qualitatively as the peripheral accumulation wedge. This subsection formalizes it as a strict macroeconomic boundary constraint on the cost-minimizing firm's choice of technique, integrating the structuralist insights of Prebisch (1981) into the heterogeneous capital framework of Section 3.2.

3.3.1 The Import-Biased Intensive Margin and the Balance of Payments Constraint

In Section 3.2, the cost-minimizing firm chooses the mechanization gap κ_t to maximize the net viability surplus:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) - \Phi(\kappa_t) \tag{38}$$

where $\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2$ is the regime-conditioned mechanization frontier, and $MC(\omega_{t-1}, p_t^{\kappa}) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^{\kappa}$ is the marginal viability cost of the gap. In the core, this optimization proceeds unconstrained, yielding the optimal mechanization gap κ_t^*. In the periphery, however, active mechanization is strictly import-biased: widening the mechanization gap requires imported capital goods, specialized machinery, and technologically non-substitutable spare parts (Vernengo, 2006).

Assumption 3. *Import-biased mechanization.* The physical import requirement per unit of output (m_t^{req}) is a linear function of the chosen technique:

$$m_t^{req} = \nu \cdot \kappa_t \tag{39}$$

where $\nu > 0$ captures the structural import-intensity of the mechanization gap. Simultaneously, the physical capacity to satisfy this requirement is constrained by the *capacity to import* per unit of output ($m_t^{avail} = M_t^{avail}/Y_t$). Following Prebisch (1981), peripheral export growth is governed by the center's income elasticity of demand for primary goods ($\epsilon_{m,c} < 1$), structurally lagging behind domestic import demand. This external asymmetry, compounded by high consumption propensities out of capitalist profits (Kaldor, 1959), severely squeezes the foreign exchange available for productive accumulation. A necessary condition for the re-investment of the surplus. This imposes a strict ceiling on the choice of technique:

$$\nu \cdot \kappa_t \leq m_t^{avail} \implies \kappa_t \leq \bar{\kappa}_t \equiv \frac{m_t^{avail}}{\nu} \tag{40}$$

3.3.2 The Macroeconomic Allocation of Foreign Exchange

Assumption 4. *Balance-of-payments constraint.* Total import demand cannot exceed export capacity. The total supply of foreign exchange per unit of output is driven by primary exports ($x_t = p_t^x X_t / Y_t$). Due to the center's low income elasticity for primary goods, x_t is structurally rigid. The total demand for foreign exchange per unit of output consists of three competing components:

1. **Wage-goods imports** (m_t^W): driven by the wage share ω_{t-1} and the structural inelasticity of domestic agriculture (Kaldor, 1959). $m_t^W = m_w\omega_{t-1}$.

2. **Conspicuous consumption imports** (m_t^C): driven by the profit share π_t and the high propensity of the peripheral upper strata to imitate the consumption patterns of the center (Prebisch, 1981; Kaldor, 1959). $m_t^C = m_c c_k \pi_t$.

3. **Productive capital imports** (m_t^K): driven by the mechanization gap. $m_t^K = \nu\kappa_t$.

Assumption 5. *Surplus leakage.* The propensity to consume out of profits satisfies $c_k > c_k^{\min}$, where $c_k^{\min}$ is the threshold below which domestic savings suffice to finance capital formation without external borrowing. Under the developmentalist regime, the leakage channel is conspicuous consumption imports by domestic elites (Prebisch, 1981; Kaldor, 1959). Under the neoliberal regime, the leakage channel is profit repatriation by multinational corporations and capital flight (Ahumada, 2019). In both cases, $c_k > c_k^{\min}$ ensures that the effective profit share available for productive reinvestment is compressed below the nominal level: $\pi_t^{eff} = \pi_t \cdot \phi_t < \pi_t$.

The balance of payments constraint requires:

$$m_w\omega_{t-1} + m_c c_k \pi_t + \nu\kappa_t \leq x_t \tag{41}$$

Using the distributive identity $\pi_t = 1 - \omega_{t-1}$, I substitute and solve for the maximum feasible mechanization gap:

$$m_w\omega_{t-1} + m_c c_k(1 - \omega_{t-1}) + \nu\kappa_t \leq x_t \tag{42}$$

$$\nu\kappa_t \leq x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1} \tag{43}$$

$$\bar{\kappa}_t(\omega_{t-1}) = \frac{1}{\nu}\left[x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1}\right] \tag{44}$$

This equation formalizes the structuralist critique: the base capacity to import capital goods is determined by export earnings minus the baseline luxury consumption of the capitalist class ($x_t - m_c c_k$). The sign of $(m_c c_k - m_w)$ determines whether a rising wage share expands or contracts the feasible mechanization gap.

3.3.3 The Constrained Cost-Surplus Maximization Problem

The peripheral firm's tactical problem is therefore a cost-minimization problem constrained by the mechanization function and the external constraint, such that the minimization problem can be stated as a Kuhn–Tucker optimization problem:

$$\min_{\kappa_t} \; c(\kappa_t) = \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) - \Phi(\kappa_t) \tag{45}$$

$$\text{subject to:} \quad \kappa_t \leq \bar{\kappa}_t \tag{46}$$

This formulation distinct a regime shifting behavior of the choice of technique in the capitalist periphery given by a double constraint in the capitalists cost-minimizing behavior: i) Endogenous to domestic distributive conflict, ii) and to the dependency of foreign currency to fulfill import requirements at a given choice of technique. Therefore, if the external constraint becomes binding due lack of foreign currency, a regime shifting behavior of technological change takes place.

3.3.4 Endogenous Regime Shifts: The Corner Solution

The firm's unconstrained optimal technique $\kappa_t^*(\omega_{t-1})$ is strictly increasing in the wage share ($\partial\kappa_t^*/\partial\omega_{t-1} > 0$) due to the Okishio labor-saving bias established in (25).

Proposition 4. *Peripheral regime shifts.* Two regimes are endogenous to the interaction between domestic distributive conflict and the global market:

1. **Regime A (Slack BoP):** When $\kappa_t^* \leq \bar{\kappa}_t$, the firm solves the standard first-order condition, yielding κ_t^*. The labor-saving bias successfully extracts the Okishio distributive dividend.

2. **Regime B (Peripheral realization trap):** When $\kappa_t^*(\omega_{t-1}) > \bar{\kappa}_t(\omega_{t-1})$, the constraint binds and the firm is forced into a corner solution:

$$\kappa_t = \bar{\kappa}_t \tag{47}$$

The corner solution arises through two intersecting structural vulnerabilities. First, if $m_c c_k$ is large (the conspicuous consumption drain), the baseline FX availability is severely squeezed, lowering the intercept of $\bar{\kappa}_t$. Second, if domestic agriculture is highly inelastic ($m_w > m_c c_k$), a rising wage share simultaneously increases the *demand* for capital imports (via κ_t^*) while *contracting* the supply of available FX (via $\bar{\kappa}_t$). *Proof.* Immediate from the Kuhn–Tucker conditions applied to (45)–(46). ■

3.3.5 Effective Profit Compression and the Blocked Surplus

Definition 3. *FX-Conversion Ratio.* The FX-conversion ratio ϕ_t is the ratio of the maximum feasible technique to the optimal unconstrained technique:

$$\phi_t = \min\left(1, \frac{\bar{\kappa}_t}{\kappa_t^*}\right) \tag{48}$$

When the external constraint binds ($\kappa_t^* > \bar{\kappa}_t$), $\phi_t < 1$. This generates a direct compression of the effective profit share available for intensive accumulation.

Definition 4. *Blocked surplus.* Even if domestic wage suppression yields a high nominal profit share (π_t), the physical inability to convert that surplus into imported machinery squeezes the effective profit rate:

$$\pi_t^{eff} = \pi_t \cdot \phi_t = (1 - \omega_{t-1}) \cdot \phi_t \tag{49}$$

The difference constitutes the *blocked surplus*:

$$\text{Blocked Surplus} = \pi_t(1 - \phi_t) \tag{50}$$

Because the accumulation of machinery is driven by the *effective* profit share, the constraint decouples nominal profitability from capacity formation. If capitalists squeeze wages to raise π_t, they inadvertently increase conspicuous consumption (m_t^C), which lowers $\bar{\kappa}_t$ and compresses ϕ_t.

3.3.6 The Regime-Shifting Aggregate Transformation Elasticity

Definition 5. *BoP binding ratio.* The BoP binding ratio τ_t is the ratio of productive capital imports to import capacity:

$$\tau_t \equiv \frac{\nu \cdot I_t^{ME}}{\text{ToT}_t \times X_t} \tag{51}$$

where I_t^{ME} is gross investment in machinery and equipment, and $\text{ToT}_t \times X_t$ proxies the import capacity (FX supply). Under Assumption 3, productive capital imports $m_t^K = \nu\kappa_t$, since mechanization is import-biased in the periphery. The regime switch is governed by τ_t:

- $\tau_t < \bar{\tau}$: BoP constraint is **slack** (Regime A: unconstrained optimum).
- $\tau_t \geq \bar{\tau}$: BoP constraint is **binding** (Regime B: corner solution).

The threshold $\bar{\tau}$ is determined empirically by the threshold regression in Section 4.

The peripheral extension generates a regime-shifting aggregate transformation elasticity. Under **Regime A** (slack BoP), the constraint does not bind, and the elasticity coincides with the unconstrained expression from Section 3.2 for the aggregate elasticity:

$$\theta_t^A == \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\,\frac{d\kappa_t^*}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\kappa_t^*}{dk_t^{NRC}}} \tag{52}$$

Under **Regime B** (BoP binding), the surplus of capitalist enterprises becomes blocked, and the firm is forced into a corner solution $\kappa_t = \bar{\kappa}_t$. The marginal capacity payoff at the corner is strictly positive because the firm would like to mechanize further but cannot:

$$MP_\kappa\big|_{\kappa=\bar{\kappa}} = \Phi'(\bar{\kappa}_t) - MC(\omega_{t-1}, p_t^\kappa) > 0 \tag{53}$$

The Kuhn–Tucker multiplier on (46) is strictly positive, confirming that the constraint binds. The aggregate transformation elasticity under the binding regime becomes:

$$\theta_t^B = \frac{\theta_{NRC} + MP_\kappa\big|_{\kappa=\bar{\kappa}} \cdot \frac{d\bar{\kappa}_t}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\bar{\kappa}_t}{dk_t^{NRC}}} \tag{54}$$

Since $\bar{\kappa}_t$ is determined by the BoP constraint (44) rather than by the first-order condition (23), the growth path $d\bar{\kappa}_t/dk_t^{NRC}$ differs from the unconstrained path $d\kappa_t^*/dk_t^{NRC}$. The FX-conversion ratio $\phi_t = \bar{\kappa}_t/\kappa_t^* < 1$ compresses the intensive margin, and the elasticity under the binding regime is strictly lower:

$$\theta_t^B < \theta_t^A \tag{55}$$

Proposition 5. *Regime-shifting elasticity.* The peripheral aggregate transformation elasticity is a regime-dependent parameter:

$$\theta_t^{periphery} = \theta_t^A \cdot \mathbf{1}\{\kappa_t^* \leq \bar{\kappa}_t\} + \theta_t^B \cdot \mathbf{1}\{\kappa_t^* > \bar{\kappa}_t\} \tag{56}$$

The elasticity shifts discretely when the BoP constraint becomes binding. The regime switch is governed by the interaction between the desired mechanization gap $\kappa_t^*(\omega_{t-1}, p_t^\kappa)$ and the maximum feasible gap $\bar{\kappa}_t(\omega_{t-1})$ determined by export capacity, conspicuous consumption, and wage-goods imports. An increase in the wage share ω_{t-1} simultaneously raises κ_t^* (via capital-using labor-saving choice of technique) and can contract $\bar{\kappa}_t$ (via the wage-goods import channel when $m_w > m_c c_k$), making the regime switch more likely. This is what Prebisch (1981) coined as the strangulation of economic development in the capitalist periphery: distributive pressures that would induce productive mechanization in the center instead triggers the BoP constraint, which in this framework explain the truncated or regime shifting behavior of the transformation elasticity.

3.3.7 The Failure of the Viability Conditions for Technological Change and Structural Inflation

Proposition 6. *Blocked viability condition.* At the boundary $\kappa_t = \bar{\kappa}_t$, the implementation of the viability condition is blocked, i.e. technical change is not net-surplus increasing or conversely, inefficient because it fails to achieve a cost-minimizing choice of technique. The firm satisfies the viability criterion—the technique would reduce unit costs—but cannot deploy it because the required machinery cannot be imported due lack of foreign currency. Hence, the capitalist enterprise in the capitalist periphery suffers a persistent shortfall in its cost-surplus relative to its unconstrained state. *Proof.* At $\kappa_t = \bar{\kappa}_t$, the Kuhn–Tucker multiplier on (46) is strictly positive, so the first-order condition $\partial c/\partial \kappa_t = 0$ does not hold. The marginal viability yield exceeds the marginal cost at the boundary, but the firm cannot expand further. ■

Corollary 1. *Structural Inflation* Unable to extract the distributive dividend via mechanization, capitalists attempt to recover the blocked surplus through the only remaining channel: price markups. When the wage-goods bottleneck forces nominal wage adjustments and the external BoP bottleneck blocks defensive mechanization, any further increase in the wage share directly crushes profitability. The resulting defensive pricing behavior ignites structural inflation dynamic (Kalecki, 1955; Sunkel, 1958; Prebisch, 1981). This is the peripheral realization trap: nominal profitability coexists with truncated mechanization and structurally stalled built-environments that constitute the *"envelope"* of the labor process, permanently destabilizing the long-run relation between investments in plant and machinery.

3.4 Analytical and Theoretical Synthesis

The three models developed in this section formalize a single theoretical claim: that decentralized, individually cost-minimizing decisions generate aggregate disproportionality under the unbalanced growth closure. Table 1 summarizes the articulation between the Okishio and Vidal frameworks:

Table 1: Market Anarchy and the Internally Divided Firm Articulation Across Three Models

Logic	Decentralized Accumulation	Internally Divided Firm	Analytical Insight
Micro logic	Viability condition (cost reduction at prevailing prices)	Internal contradiction (control vs. cooperation)	Optimization problem = cost-minimizing pole; constraints = cooperation limits
Macro logic	Decentralized decisions → disproportionality	Divided firm → contested operating conditions	$\theta \neq 1$ captures gap between accumulation and capacity
Disequilibrium	Unbalanced growth (supply ≠ demand)	"Mediocre sufficiency" (Vidal, 2022) (no coherent optimum)	μ demonstrates gap; θ captures structural inefficiency
Peripheral specificity	BoP constraint blocks *implementation* of viability	External constraint compounds internal division	Corner solution: $\kappa = \bar{\kappa}$ when constraint binds

This analytical framework does not assert equilibrium. It aims demonstrates the opposite: capitalist accumulation, left to its own decentralized logic, does not generate proportional reproduction. The rate of capacity utilization μ is the exhaust valve of this disproportionality: it shows that the normal operating conditions of capitalist firms, determined by their cost-minimizing mechanization choices, do not meet aggregate demand conditions. The transformation elasticity θ captures this

structural gap between accumulation and realized capacity. Whether θ exceeds or falls below unity is an empirical question to be tackled below. In the center, this gap is cyclical—mediated by institutional configurations that can be exhausted triggering processes of capitalist re-structuring. In the periphery, this gap is permanent—locked in by the structural ceiling of external subordination (Kaldor, 1959; Prebisch, 1981).

Three models have now specified what θ depends on, and none of them has said what it equals. The dependence is on objects that no national accounting system reports directly: a capacity ceiling that is never observed, a mechanization gap between two capital stocks that are valued by convention, and, for the periphery, an external state that has to be inferred rather than read off a date. Whether θ can be recovered at all therefore turns on whether those objects can be reconstructed from the historical record without importing the transformation relation into their construction.

4 Data, Measurement, and Empirical Results

4.1 Data and Measurement

4.1.1 A Dual Valuation Distinction

This section constructs the capital stock series for the choice-of-technique models of section 3. The problem with the standard treatment of capital stocks series, is that they treat the physical capacity of installed capital and the monetary value of capital advanced as a single magnitude. This problem introduces a risk of spurious correlation between capital intensity and wage share measures as proxies for distributive conflict: as mechanization raises capital intensity, Consumption of Fixed Capital (CFC) rises in the measurement of Gross Value Added (GVA) denominator, depressing GVA-basis wage shares even when the rate of surplus-value extraction is unchanged. To deal with this bias, I reconstruct the stock account-by-account using the Generalized Perpetual Inventory Method (GPIM) proposed by Shaikh (2016, App. 6.5). This methodology constructs gross and net capital stock measures that maintain stock-flow consistency from direct nominal investment flows and asset-specific price indexes.

Estimates of capacity utilization under the unbalanced growth framework require distinct capital stock variables. The capital value initially invested in plant and equipment returns gradually to its money form as fixed assets depreciate; the accumulated depreciation allowances remain part of total capital advanced regardless of their monetary form, because their recovery is what permits the reproduction of the capitalist enterprise I therefore distinguish two types of measurement of capital stocks:

1. **Capital stock for capacity estimation: gross real GPIM stock.** Productive capacity depends on capital goods still physically in operation. The capital stock measure adequate for estimating capacity is the gross real stock $K_{j,t}^{G,R}$, weighted exclusively by a physical survival schedule. Productive capacity Y_t^p is itself a latent variable, recovered through the estimation of the transformation elasticity θ (section 3); the gross real GPIM stock is the accumulation variable k_t included in my estimation equation, not a measure of capacity itself. A machine whose book value has fallen to zero, but remains in service until it is physically retired, still should be counted as physical capital in operation. The OECD manual confirms that the gross stock, valued at "as new" prices, "is widely used as a broad indicator of the productive capacity of a country" (OECD, 2001, p.31).

2. **Profitability measure: net current-cost stock.** Profitability is a monetary ratio between a profit flow and the value of capital advanced. The relevant measure is the net current-cost stock $K^{N,V}_{j,t}$, which tracks the monetary value of capital after financial depreciation and write-downs. This measure serves as the anchor for the estimation of an implicit price of capital goods under a stock-flow-consistent framework but is not included within the structural estimation of non-observed productive capacities.

A single aggregate net stock serving both purposes is common in empirical macroeconomics. This practice introduces a mechanical correlation between the capital stock and the depreciation flow, introducing potential biases to identify distributive effects on productive capacities formation.

Valuation choice within the GPIM. The dual-valuation distinction does not require imposing a single valuation convention throughout. Shaikh (2016, App. 6.3) advocates the gross stock as the denominator for both the profit rate and the capacity ratio. The GPIM he develops, however, constructs the survival schedule $S_j(v)$, the age-price schedule $(1-d_j)^v$, and the SFC implicit price index $P_{j,t}$ as distinct components within a single stock-flow-consistent recursion. This structure permits the researcher to select the valuation for each analytical purpose without abandoning accounting coherence.

For the profit rate, the denominator must measure the monetary value of capital advanced. The net current-cost stock serves this role: it tracks the book value of capital that the firm must recover to continue production. A machine that has depreciated to a low book value may still operate at full physical capacity, but the firm's cost-recovery problem is defined by the value it must recoup, not by the physical state of the equipment (Basu and Vasudevan, 2013). For estimates related to capacity utilization, the capital stock input must reflect the physical capital still in operation. The gross real GPIM stock, weighted exclusively by the survival schedule $S_j(v)$ and excluding the age-price factor $(1-d_j)^v$, serves this role.

The quantitative significance of the gross/net distinction is not confined to the center. Ffrench-Davis and Vivanco (2016) show that in Chile depreciation absorbs between 8 and 13 percent of GDP and that net investment is barely half of gross investment in the most recent period. Service lives of machinery have declined from 20 years in the 1940s to 11 years by 2005, reflecting accelerating obsolescence. They conclude that the capital stock entering a production function must be the gross stock of physically surviving assets, not the depreciated monetary residual.

4.1.2 Gross Capital Stock Reconstruction and Measurement in the United States

The Generalized Perpetual Inventory Method. In this part I present the procedures of construction for the capital stock series for the United States. Machinery (ME) and nonresidential structures (NRC) do not share a natural physical unit; they become comparable only through a monetary-mediated accounting procedure. I reconstruct the stock account-by-account rather than aggregating nominal values and deflating with a generic index. The GPIM accumulation rules for historical-cost, current-cost, and constant-dollar stocks (Shaikh, 2016, App. 6.5) are derived in detail in Appendix A; here I state the construction.

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage surviving after v years is:

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{57}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1+1/\alpha_j)$, derived from the mean service life L_j. Unlike geometric decay models that continuously erode the physical stock, the Weibull curve ensures that capital goods remain fully operative until they are physically removed from service. Empirical studies of directly observed asset discards confirm that the Weibull family provides a flexible approximation of retirement patterns. Nomura (2005), using Japanese micro-data on 66 asset classes, finds that almost half of assets exhibit a progressively increasing hazard rate ($\alpha > 1$). The calibrated parameters $L_{\text{ME}} = 14$, $\alpha_{\text{ME}} = 1.7$ and $L_{\text{NRC}} = 30$, $\alpha_{\text{NRC}} = 1.6$ follow the BEA Fixed Assets methodology (Fraumeni et al., 1997).

Physical survival alone cannot identify the SFC implicit price index. The BEA current-cost net stock anchors reflect both physical retirement and age-price decay. To recover the asset-specific price index, I impose an externally documented geometric declining-balance age-price profile:[3]

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v \tag{58}$$

where d_j is the asset-specific age-price decay rate. The declining-balance formula applied to the calibrated service lives yields $d_{\text{ME}} = 0.110$ and $d_{\text{NRC}} = 0.024$, consistent with the geometric rates documented by the BEA (Fraumeni et al., 1997).

The SFC implicit price index $P_{j,t}$ is the price level that closes the stock-flow identity for the net current-cost stock. In each period, the BEA anchor must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}} \tag{59}$$

where $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}}/(P_{j,t-v}/100)$ is real investment deflated by the previously recovered SFC price. The first term on the right-hand side is current-year nominal investment; the second is the net-value-weighted sum of surviving real vintages revalued at the current price. The price index is "implicit" because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA's reported current-cost net stock. Solving (59) for $P_{j,t}$:

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\displaystyle\sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}} \tag{60}$$

The forward recursion proceeds period by period: given $P_{j,t-1}, P_{j,t-2}, \ldots$, the current price $P_{j,t}$ is uniquely determined by the requirement that the constructed stock reproduces the BEA anchor. Appendix A documents the recursion, the initial conditions, and the base-year normalization. The BEA's Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline.

The capital stock used to estimate productive capacity is the gross real GPIM stock, constructed

[3]The BEA's current methodology employs geometric depreciation, which assumes that the efficiency pattern and the depreciation pattern have the same form (Fraumeni et al., 1997). The GPIM construction separates these two dimensions: physical survival follows a Weibull retirement distribution, while financial age-price decay follows a geometric profile. This separation is necessary to construct distinct capacity and profitability measures.

as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v}^{\text{REAL}} \tag{61}$$

where $I_{j,t-v}^{\text{REAL}}$ is real investment deflated by the previously recovered SFC price. Because eq. (61) relies exclusively on the survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$, it measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter.

Validation and sample scope. The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors. All GPIM series are normalized to 2017 = 100. The capital stock series has a 30-year warm-up period (1901–1930). The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for the longest-lived asset class (nonresidential structures, $L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. I restrict the estimation sample to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing. This sectoral restriction isolates the private productive business sector from public infrastructure and imputed residential rents.

4.1.3 U.S. Distributive Variables

The GVA inadequacy. The standard GVA-basis wage share is might become contaminated by Consumption of Fixed Capital. Because GVA = CFC + v + s, periods of intensive mechanization raise CFC/GVA even when the rate of surplus-value extraction (s/v) is unchanged. The GVA-basis wage share therefore might be contaminated of spurious correlation between capital intensity and the measured distributive share during capital-deepening episodes. The NVA basis on the other hand, NVA = $v + s$, could avoid this problem. I estimate a permutation of multiple distributive measures to explorate empirically this potential problem.[4]

The No-Double-Accounting Principle. Standard NIPA corporate aggregates combine nonfinancial and financial corporate sectors into a single total. This aggregation violates the No-Double-Accounting Principle: financial sector profits and net interest receipts are secondary claims extracted from primary nonfinancial surplus, not independently created value. Including them in the denominator of the wage share counts value creation twice. The corrected denominator is nonfinancial corporate net value added (NFC_NVA), which excludes financial double-accounting

[4]Eight wage share variants are available: NFC baseline, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, and Clean Corporate, each on a GVA and an NVA basis. Unadjusted Shaikh-Tonak Tier 1 NVA is arithmetically identical to the NFC baseline NVA share, so the non-redundant specifications reduce to the Clean Corporate and Tier 2 supervisory adjustments discussed in section 4.4.

transfers.[5]

Supervisory compensation adjustment. Standard NIPA compensation includes managerial and executive salaries, which under the Shaikh-Tonak classification function as a distribution of surplus value rather than productive variable labor cost (Shaikh and Tonak, 1994). To isolate productive variable capital, total compensation is adjusted by the supervisory compensation ratio ϕ_t^{sup}:

$$v_t^{\text{prod}} = \text{NFC_COMP}_t \times (1 - \phi_t^{\text{sup}}) \,. \tag{62}$$

Two tiers of wage share are defined. Tier 1 uses total compensation without supervisory adjustment; Tier 2 applies the adjustment in (62). The supervisory compensation ratio ϕ_t^{sup} is constructed from Table S512 of the *Measuring the Wealth of Nations* dataset (Shaikh and Tonak, 1994), as reconstructed and extended to 2024 by Anderson (Anderson, 2024).[6].

All measures are constructed in nominal terms (W/NVA and W/GVA), guaranteeing that relative capital-asset price dynamics and depreciation deflators do not distort the physical vs. monetary valuation of surplus extraction.

4.1.4 Productive output and aggregate demand for the United States

The dependent variable in the cointegrating regressions of section 3 is the logarithm of real gross value added of nonfinancial corporate business, $y_t = \ln Y_t^{\text{NFC}}$, measured in chained 2017 dollars and sourced from BEA NIPA Table 1.14, Line 41.[7] The nonfinancial corporate boundary isolates productive value creation within the business sector, excluding financial intermediaries whose income constitutes secondary claims on nonfinancial surplus rather than independently created value. The robustness battery of section 4.1.3 audits eight alternative deflator proxies—including the official NFC implicit deflator (Table 1.15), the GDP implicit deflator, and the nonfarm business output deflator—and confirms that coefficient estimates are invariant to deflator choice.

The numerator of aggregate capacity utilization $\mu_t = Y_t/Y_t^p$ is economy-wide real GDP, drawn from BEA NIPA Table 1.1.9, Line 1, in chained 2017 dollars.[8] The denominator Y_t^p is not directly observed but reconstructed from the estimated transformation elasticity, as detailed in section 4.4.2. The distinction between the two output measures is deliberate: the transformation elasticity θ is estimated from NFC output, which captures the capital-output relationship within the productive corporate sector; the utilization ratio uses aggregate GDP because capacity utilization measures the engagement of total productive capacity by effective demand, including government expenditure, household consumption, and net exports. This separation ensures that the estimation of θ remains uncontaminated by demand-side fluctuations that drive aggregate GDP but do not reflect the structural capital-output relationship.

[5]The implied financial sector surplus is recovered as $\text{FIN_NOS}^{\text{implied}} = \text{CORP_NOS} - \text{NFC_NOS}$. Clean corporate denominators purge both $\text{FIN_NOS}^{\text{implied}}$ and NFC_NET_INT.

[6]The full dataset is available at `https://shaikh.heterodata.org/`

[7]BEA series code `B455RX`; nominal counterpart `A455RC`, Line 17. The implicit output deflator is $P_{y,t}^{\text{NFC}} = 100 \times \text{NFC_GVA}_t^{\text{nom}}/Y_t^{\text{NFC}}$. Coverage: 1929–2025, annual.

[8]BEA series code `A191RL`; coverage 1929–2025, annual. The GDP implicit price deflator (`A191RD`) serves as the economy-wide deflator for the utilization reconstruction.

4.1.5 Capital Series Harmonization and Historical Sources for Chile

This subsection describes the stock-flow consistent (SFC) methodology used to reconstruct gross and net capital stocks for Chile, 1900–1994, in constant 2003 Chilean pesos (CLP). The procedure enforces the accumulation identities for gross and net stocks jointly, yields implicit depreciation and retirement rates for each asset class, and delivers coherent series for machinery and equipment (ME), nonresidential construction (NRC), residential construction (RC), and their aggregates. The construction proceeds from a single investment control total, disaggregated by asset, accumulated via Weibull survival functions, and linked to net stocks through a gap-ratio approach anchored to the Hofman (2000b) overlap window.

The stock-flow identities. The reconstruction is governed by two laws of motion. The gross stock evolves according to

$$\Delta K^g_{j,t} = I_{j,t} - R_{j,t} \tag{63}$$

where $I_{j,t}$ is gross fixed investment and $R_{j,t}$ denotes physical retirements (scrapping of obsolete vintages, demolition, and technological withdrawal). The net stock evolves according to

$$\Delta K^n_{j,t} = I_{j,t} - D_{j,t} \tag{64}$$

where $D_{j,t}$ is depreciation (the monetary consumption of fixed capital). Combining (63) and (64) yields the joint SFC identity

$$\Delta K^g_{j,t} + R_{j,t} = \Delta K^n_{j,t} + D_{j,t} \tag{65}$$

which constrains the combination of depreciation and retirements at every t and for every asset j. Given observed stocks and investment, the implied flows are recovered as $D_{j,t} = I_{j,t} - \Delta K^n_{j,t}$ and $R_{j,t} = I_{j,t} - \Delta K^g_{j,t}$. The capital gap $B_{j,t} \equiv K^g_{j,t} - K^n_{j,t}$ measures the cumulative difference between installed capacity and book value.

The investment control total and the lambda bridge. The aggregate investment control total is drawn from the Pérez-Eyzaguirre (2019) reconstruction of gross fixed capital formation (GFCF), expressed in constant 2003 CLP. The control concept is machinery plus construction, explicitly excluding inventory variation.[9]

Pérez-Eyzaguirre report investment in a price space and asset aggregation that differ from the Hofman (2000a) benchmark. The two sources are linked by a level-normalization factor λ, which corrects for definitional differences between the two series rather than converting between price bases. The factor is estimated as the median of the annual ratio of Hofman to Pérez investment over the 95-year estimation window:

$$\lambda = \text{median}\left(\frac{I^{\text{Hofman}}_t}{I^{\text{Pérez}}_t} \right)_{t=1900}^{1994} = 4.1054 \tag{66}$$

The median rule is chosen for robustness to outliers in the early decades, where source quality is weaker. The scaled aggregate is then disaggregated into ME, NRC, and RC using normalized

[9] Inventories are excluded from fixed-capital accumulation because the reconstruction models fixed capital only. Inventory investment does not enter the perpetual inventory recursion.

asset-specific investment shares from Hofman (2000a):

$$I_{j,t} = s_{j,t} \cdot \lambda \cdot I_t^{\text{Pérez}}, \qquad \sum_{j \in \{\text{ME, NRC, RC}\}} s_{j,t} = 1 \tag{67}$$

for all t.

Weibull survival and gross stock accumulation. Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}, \text{RC}\}$, the proportion of a vintage surviving after v years is

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{68}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j. The survival parameters are estimated from the Hofman (2000b) overlap window (1950–1994), in which both gross and net stocks are observed, by fitting Weibull curves to the implied retirement patterns recovered from the stock-flow identities. Table 2 reports the calibrated parameters.

Table 2: Weibull survival parameters, Chile (1900–1994)

Asset	Scale (λ_j)	Shape (α_j)	Mean life (L_j, years)
ME	15.90	2.42	14.1
NRC	41.30	5.00	37.9
RC	51.16	5.00	47.0

All shape parameters exceed unity, indicating progressively increasing hazard rates: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. The shape parameters for NRC and RC are constrained at 5.0, reflecting the long and relatively uniform service lives of structures. The gross stock is accumulated as the survival-weighted sum of real investment vintages:

$$K_{j,t}^g = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v} \tag{69}$$

Pre-1950 stocks are anchored to the observed Hofman levels at 1950 via a multiplicative factor that matches $K_{j,1950}^g$ by asset.

The gap-ratio approach and net stock construction. Unlike the U.S. GPIM, which recovers the net stock via a separate geometric age-price decay rate and an SFC implicit price recursion, the Chile reconstruction derives the net stock from the observed gross-net relationship in the Hofman (2000b) overlap window. The capital gap ratio

$$\gamma_j \equiv \frac{B_j}{K_j^g} = \frac{K_j^g - K_j^n}{K_j^g} \tag{70}$$

is computed by asset from the Hofman (2000b) data over the overlap period (1950–1994). Two regimes are distinguished: a "pre" ratio (median of years ≤ 1954) and a "post" ratio (median of years ≥ 1990), allowing for structural shifts in the gross-net relationship over the century. Outside the overlap window, the gap ratio is extended at the regime-specific median. The net stock is then recovered as

$$K^n_{j,t} = K^g_{j,t} \cdot (1 - \gamma_j) \tag{71}$$

This approach inherits the gross-net relationship observed in the Hofman accounting reference rather than imposing an external depreciation convention. It is appropriate for Chile because no independent age-price decay data are available for the historical period; the Hofman (2000b) overlap provides the only direct evidence on the joint behavior of gross and net stocks.

Implicit depreciation and retirement rates. Given the reconstructed stocks and investment flows, the implicit annual rates are defined as

$$z_{j,t} = \frac{R_{j,t}}{K^g_{j,t-1}}, \qquad \delta_{j,t} = \frac{D_{j,t}}{K^n_{j,t-1}} \tag{72}$$

where $z_{j,t}$ is the retirement (depletion) rate and $\delta_{j,t}$ is the depreciation rate. These rates are not imposed as parameters; they are derived from the data via the stock-flow identities (63)–(64). They serve as diagnostic indicators of capital durability and replacement dynamics, and are used in section 4.6 to validate the internal consistency of the reconstruction.

Validation and estimation sample. The SFC construction is validated through four checks. First, additivity: all composite stocks and flows satisfy $K_{\text{NR}} = K_{\text{ME}} + K_{\text{NRC}}$ and $K_{\text{T}} = K_{\text{ME}} + K_{\text{NRC}} + K_{\text{RC}}$ at every t. Second, non-negativity: $K^g_{j,t} \geq 0$, $K^n_{j,t} \geq 0$, and $B_{j,t} \geq 0$ for all assets and years. Third, rate bounds: $z_{j,t} \in [0, 1]$ and $\delta_{j,t} \in [0, 1]$ for all observations from 1950 onward. Fourth, identity residuals: $\Delta K^g_{j,t} - (I_{j,t} - R_{j,t}) = 0$ and $\Delta K^n_{j,t} - (I_{j,t} - D_{j,t}) = 0$ to numerical precision. The estimation sample spans 1900–1994, yielding $T = 95$ annual observations. The overlap window (1950–1994) provides 45 years of direct Hofman (2000b) observations against which the reconstructed series are audited.

The BCCh forward extension (1995–2024). Extending the reconstruction beyond 1994 requires integrating the official national accounts published by the Central Bank of Chile (BCCh). A direct chain-linking of the BCCh series to the historical baseline would introduce a level discontinuity, because the two sources differ in base year, valuation conventions, and asset classification. To eliminate this discontinuity while preserving the validated 1900–1994 history, I employ a multiplicative splicing procedure that fixes the legacy panel in place.

For the overlap year 1994, an asset-specific bridge factor is computed as the ratio of the historical net stock to the BCCh reported net stock:

$$\beta_j = \frac{K^{n,\text{hist}}_{j,1994}}{K^{n,\text{BCCh}}_{j,1994}}, \quad j \in \{\text{ME, NRC, RC}\} \tag{73}$$

The same factor β_j is applied to both the BCCh net stock and the BCCh depreciation series for all years 1995–2024. This preserves the depreciation rates and asset dynamics implied by the BCCh

while maintaining continuity with the constant 2003 CLP historical baseline. Table 3 reports the estimated bridge factors.

Table 3: Asset-specific bridge factors, 1994

Asset	Bridge factor (β_j)	Interpretation
ME	5.888	BCCh understates historical ME stock
NRC	1.033	Near parity
RC	1.050	Near parity

The large bridge factor for machinery and equipment reflects the substantially lower level at which the BCCh reports ME stocks relative to the Pérez-based historical reconstruction, a divergence driven by differences in asset coverage and valuation methodology. For nonresidential and residential construction, the bridge factors are close to unity, indicating that the two sources agree substantially on the level of structures.

Post-1994 real investment is inferred via the stock-flow identity:

$$I_{j,t} = \Delta K^{n}_{j,t} + D_{j,t}, \quad t \geq 1995 \tag{74}$$

where $D_{j,t}$ is the bridge-adjusted BCCh depreciation. The gross stock and physical retirements for the extended period are then reconstructed by applying the survival parameters from Table 2 to the complete 1900–2024 investment series via eq. (69). This ensures that the gross stock remains internally consistent across the splice boundary.

4.1.6 Wage Share Data Sources for Chile

The distributive regressor in the transformation elasticity is the wage share, ω_t. For Chile, the wage share is drawn from the revised dataset of real wages in Latin America constructed by Astorga (2017) and updated by Astorga (2023). The dataset provides yearly series of real wages by skill level—unskilled (w_4), semi-skilled (w_3), and skilled (w_2)—together with an economically-active-population (EAP) weighted average (w), expressed in monthly purchasing-power-parity dollars at 1970 prices.[10]

The wage share is computed as the wage-to-income ratio $\omega_t = w_t/y_t$, where y_t is real income per worker derived from national income estimates. The Astorga (2023) series covers 1920–2011. To extend coverage to 2024, I splice Astorga's series with the institutional sector accounts of Chile's Central Bank. The BCCh National Accounts report production income by institutional sector—financial corporations, non-financial corporations, households and NPISH, and general government—in nominal terms at 2018 reference prices. From these, I construct the private non-financial wage share as

$$\omega_t^{\mathrm{BDE}} = \frac{Y_t^{\mathrm{HH}}}{Y_t^{\mathrm{TOT}} - Y_t^{\mathrm{FIN}} - Y_t^{\mathrm{GOV,CFC}}} \tag{75}$$

where Y_t^{HH} is household income (compensation of employees plus mixed income), Y_t^{TOT} is total production income, Y_t^{FIN} is the financial sector's operating surplus, and $Y_t^{\mathrm{GOV,CFC}}$ is general

[10]The revised dataset is publicly available as supplementary material to Astorga (2023).

government consumption of fixed capital. The exclusion of the financial sector from the denominator is consistent with the No-Double-Accounting Principle applied to the U.S. series in section 4.1.3: financial profits are recognized as secondary claims on primary non-financial surplus and do not enter the distributive denominator.[11]

The BCCh series covers 2003–2024, yielding a nine-year overlap (2003–2011) with the Astorga (2023) panel. The extension is anchored at 2011, the final year of the Astorga coverage, by constructing the index

$$\mathcal{I}_t = \frac{\omega_t^{\text{BDE}}}{\omega_{2011}^{\text{BDE}}}, \quad t \geq 2012 \tag{76}$$

and applying it to the Astorga terminal value:

$$\omega_t = \omega_{2011}^{\text{Astorga}} \cdot \mathcal{I}_t, \quad t \geq 2012. \tag{77}$$

The splice assumes that the growth rate of the national-accounts wage share is a valid proxy for the growth rate of the labor-market wage-to-income ratio beyond the Astorga coverage. The nine-year overlap (2003–2011) supports this assumption: both series exhibit the same cyclical pattern and comparable magnitudes of distributive fluctuation. The resulting ω_t series spans 1920–2024, yielding $T = 105$ annual observations.

4.1.7 Aggregate Demand, Price Indices, and External Constraint variables for Chile

The aggregate demand variables—real GDP (Y_t), exports FOB (X_t^{FOB}), and gross machinery investment ($I_{\text{ME},t}$)—are drawn from the aggregate demand reconstruction of Pérez-Eyzaguirre (2019), expressed in constant 2003 Chilean pesos. This series revises and extends the historical national accounts backbone of La República en Cifras (Lüders et al., 2016), correcting additivity and valuation problems in the original Clio-Lab series while preserving the sectoral GDP decomposition. Coverage spans 1860–2024.

The price and trade indices used to construct the relative capital price proxies (p_t^{κ}), the capacity to import (M_t^{cap}), are drawn from *La República en Cifras* (Lüders et al., 2016): the real exchange rate index, the US wholesale capital price index, and the terms of trade index. All three series are rebased to 2003 = 100 for internal consistency with the capital stock series.[12]

The capital import demand imbalance combines sources from *La República en Cifras* with my data set of harmonized capital stocks through a log difference of $ln(I_{MEt})$ - $ln(M_t^{\text{cap}})$, which reconciles the supply export-side identified by the index of imports capacities (M_t^{cap}) of the external constraint, with the internal demand for foreign capital goods providing a both a more accurate variable for the regime shifting empirical strategy detailed below, while also a variant that allows me to check for robustness of my estimates.

A complementary data source for external sector variables are drawn from the World Historical Balance of Payments Database (WBOP) developed by Nievas and Piketty (2025). The WBOP data set provides annual series on global trade flows and the world balance of payments—including goods, services, income, and transfers—covering 48 countries and 9 regions over the 1800–2025

[11] The underlying series are drawn from the Base de Datos Estadísticos (BDE) of the Banco Central de Chile, Cuentas Nacionales, sectoral income accounts, reference year 2018, nominal quarterly flows. Annual values are obtained by simple summation of quarterly observations.

[12] The terms of trade index extends back to 1810; the real exchange rate and US wholesale price index cover 1900–2024.

period. Three series are used for Chile: the ratio of manufactured goods imports to GDP (a proxy for capital goods import dependence), net foreign income outflows as a share of GDP (a proxy for profit remittance drain), and the net current account balance as a share of GDP (an external balance indicator).[13]

4.2 Stylized Facts of Accumulation, Distribution, and External Constraints

This subsection presents stylized facts and descriptive statistics of my data set for the United States (1931–2024) and Chile (1940–2010). First, the data presented for output correspond always to Non Financial Corporate Sector, while the measurement of capital stock correspond to gross measures following my own implementation of Shaikh (2016) *Generalized Perpetual Inventory Method* (GPIM) as presented above. Second, output and capital stocks drift together over the long run in both economies, motivating a cointegrating approach. Third, the level of the output–capital ratio drifts through time in both countries of interest: in the United States presents a declining trend in the long-run, suggesting that their long-run relation might be governed by a transformation elasticity below unity; while in the case in Chile the relation is unclear, while its composition when distinguishing between non-residential construction and machinery and equipments accounts discretely shifts across historical regimes, this motivates a formulation of heterogeneous capital. Third, in the US capital accumulates faster than output on average. Fourth, the state variables that condition the transformation—the distributive state in both economies and the external bottleneck components in the periphery—shift discretely across institutional regimes, motivating a state-dependent and threshold co-integration empirical strategies presented in the next Section 4.3.

4.2.1 Stylized Facts of the U.S. Non-Financial Corporate Economy (1931–2024)

To ground the subsequent econometric analysis in empirical reality, this subsection documents the key stylized facts of the U.S. non-financial corporate economy over the 94-year sample period from 1931 to 2024. Table 4 provides the summary statistics for the core macroeconomic and distributive variables, capturing the long-run means and volatilities of real output, capital accumulation, functional income distribution, and the evolving composition of the capital stock. While these aggregate moments establish the foundational structural characteristics of the economy—such as the persistent divergence between capital and output growth and the shifting dynamics of the wage share—the temporal evolution and historical regime shifts underlying these averages require a detailed chronological examination. The following figures therefore trace the secular trends, cyclical fluctuations, and structural breaks across these dimensions, providing the empirical motivation for the theoretical and econometric specifications developed in the subsequent sections.

[13] All WBOP series are available at `wbop.world` and are also hosted in the World Inequality Database (`wid.world`), together with a detailed replication package including raw data sources, methods, and codes.

Table 4: United States Macroeconomic and Distributive Summary Statistics (1931–2024)

Variable	Mean	Std. Dev.
Log Real Output (y_t)	14.734	1.107
Log Core Capital Stock (k_t^{cap})	14.158	1.810
Log Output–Capital Ratio ($y_t - k_t^{cap}$)	0.576	0.731
Mechanization Gap ($\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$)	-0.637	0.278
Annual Output Growth (Δy_t)	0.039	0.051
Annual Capital Growth (Δk_t^{cap})	0.059	0.045
Net Wage Share ($\omega_{t-1}^{\mathrm{net}}$)	0.707	0.029
Gross Wage Share ($\omega_{t-1}^{\mathrm{gross}}$)	0.623	0.025
IPP Capital Share (ι_{t-1})	0.084	0.043
Effective Wage Pressure (C_t)	0.647	0.041

Note: Sample window spans 1931–2024 ($N = 94$). Output y_t is non-financial corporate real gross value added. Core capital k_t^{cap} comprises machinery, equipment, and non-residential structures ($K_{\mathrm{ME}}+K_{\mathrm{NRC}}$). Mechanization gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$ measures the capital composition ratio. Net wage share (ω^{net} = Compensation/NVA) and Gross wage share (ω^{gross} = Compensation/GVA) are both evaluated in nominal terms. IPP capital share $\iota_{t-1} = K_{\mathrm{IPP}}/(K_{\mathrm{cap}} + K_{\mathrm{IPP}})$. Effective wage pressure $C_t = \omega_{t-1}(1 - \iota_{t-1})$. Data sources: BEA NIPA Tables and Fixed Assets Accounts.

Figure 1 plots log real non-financial corporate output (y_t) and the log of aggregate gross capital stock $k_t^{cap} = \ln(K_t^{\mathrm{ME}} + K_t^{\mathrm{NRC}})$. Over the 94-year sample, the two levels drift together along a common upward trend. From the early 1980s onward, the capital stock runs systematically above output, generating a persistent overaccumulation gap. Estimating in first differences would discard exactly this shared long-run cointegrating trajectory.

Figure 2 plots the corresponding log output–capital ratio ($y_t - k_t^{cap}$). The ratio spikes to its historical maximum under wartime mobilization (1944), falls through the early Cold War, recovers part of the decline in a Fordist hump around 1966, and declines secularly thereafter. The LOESS trend traces a downward drift from 1950 onward, steepening after 1970 and again after 2000. A ratio that falls for decades while capital accumulates provides an empirical suggestion of an over-accumulation tendency $\theta < 1$: capacity formation persistently lags capital accumulation. The balanced-growth knife-edge $\theta = 1$ does not renders as a long-run tendency, once the measurement of capital stock and the dual valuation distinction and the purpose of measurement is considered.

Figure 3 plots the mechanization gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$ (mean -0.637, SD 0.278, Table 4). The gap remains depressed throughout the Great Depression and World War II (≈ -1.15), embarks on a sustained upward march across the Golden Age of Atlantic Fordism as machinery and equipment outpaced industrial structures, and peaks in the late 1990s (≈ -0.313) during the information-technology equipment boom. From 2000 onward, κ_t declines as investment shifted toward structures and commercial real estate while Intellectual Property Products absorbed intangible accumulation, leaving direct physical machinery accumulation subdued. The capital composition ratio is therefore a regime-shifting series rather than a smooth trend, motivating the heterogeneous-capital specification in which the transformation elasticity θ depends dynamically on κ_t.

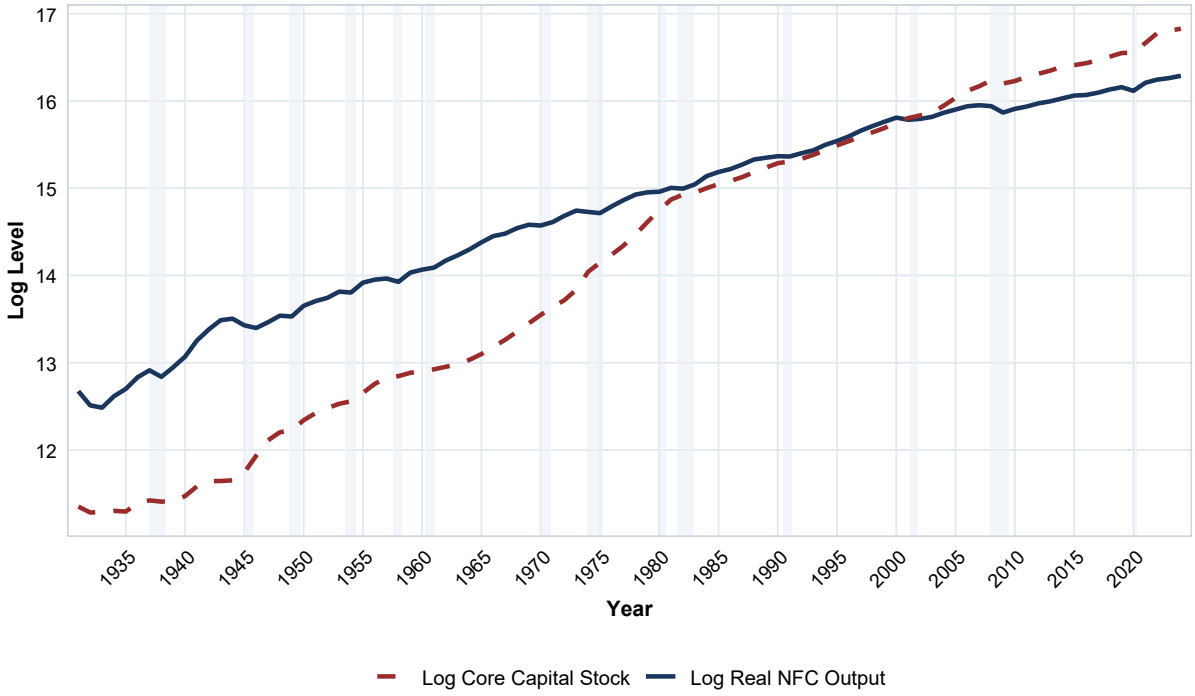
Figure 4 contrasts annual growth rates. Over the full sample, aggregate gross capital accumulation grows faster than output on average (5.9% against 3.9% per year, Table 4), a long-run relation consistent with overaccumulation under an unbalanced growth closure. The two series have comparable standard deviations (0.051 against 0.045) because the pre-1950 window contains large swings in both, including capital-growth spikes around 1946 and 1974. The asymmetry that matters for estimation appears after 1950: output growth repeatedly turns negative in recessions while capital accumultion remains positive, reflecting the durability of physical capiatl in operation

reflected by gross measurements. From the mid-1960s, the capital-growth series sits mostly above the output-growth line.

Figure 5 displays the set of wage share permutations evaluated across our econometric specifications. These capture three dimensions: (1) Corporate boundary cleaning to purge financial sector double-counting; (2) Value-Added denominators (Net Value Added vs. Gross Value Added); and (3) Shaikh and Tonak (1994) supervisory labor adjustments to account for variants of productive vs non-productive labor.

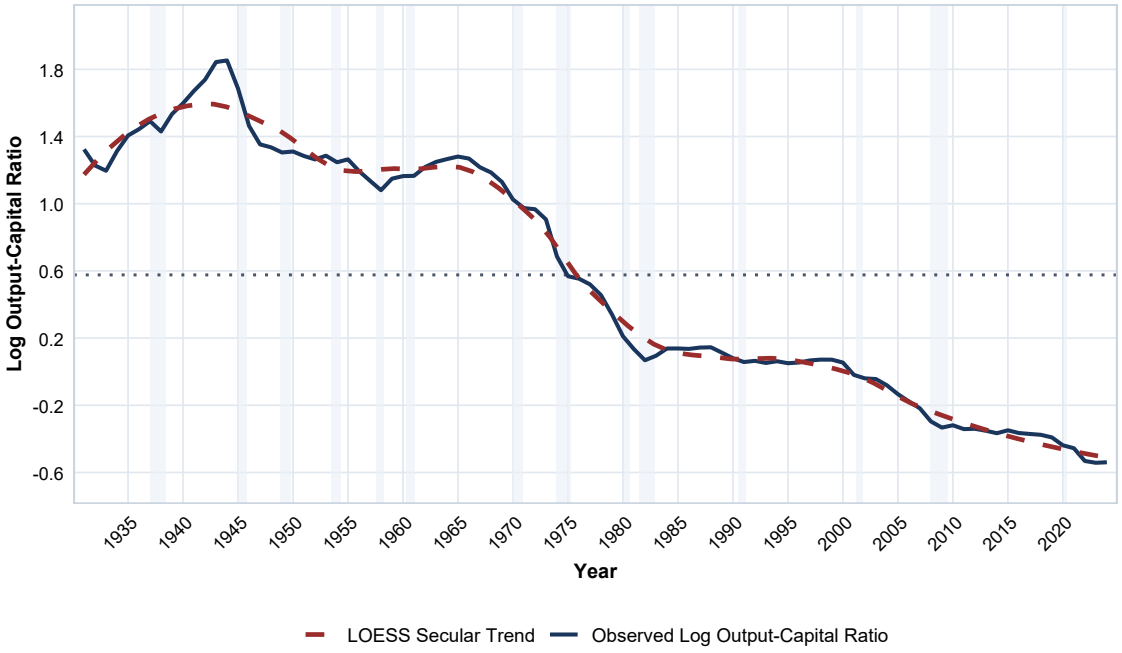
Figure 6 plots the corresponding Effective Wage Pressure permutations, where each wage share series is scaled by the physical capital fraction $(1 - \iota_{t-1})$. Across all specifications, effective wage pressure remains elevated under Atlantic Fordism (peaking around 1981) and trends downward secularly under post-Fordist accumulation as Intellectual Property Products capture an increasing fraction of total capital advanced ($\iota_{t-1} > 0.12$).

Figure 1: U.S. Non-Financial Corporate Real Output and Aggregate Gross Capital Stock (1931–2024)



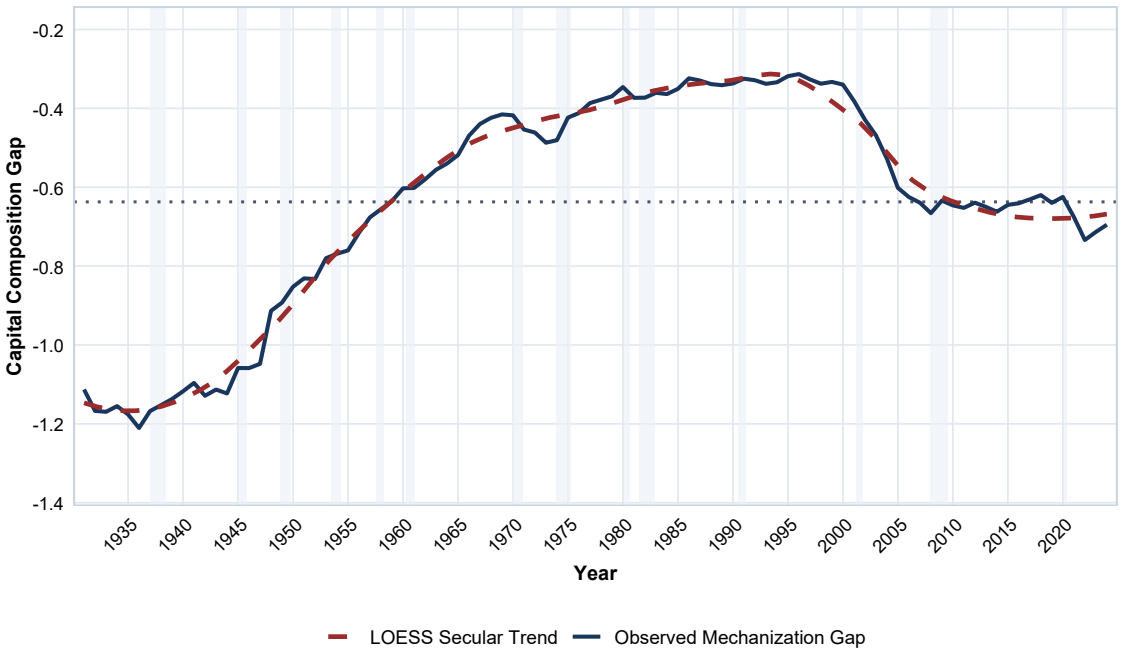
Note: Log real non-financial corporate gross value added (y_t, solid navy line) and log core capital stock ($k_t^{cap} = \ln[K_t^{\text{ME}} + K_t^{\text{NRC}}]$, dashed crimson line). Light gray shaded bands denote official NBER recession dates.

Figure 2: U.S. Log Output–Capital Ratio ($y_t - k_t^{cap}$)



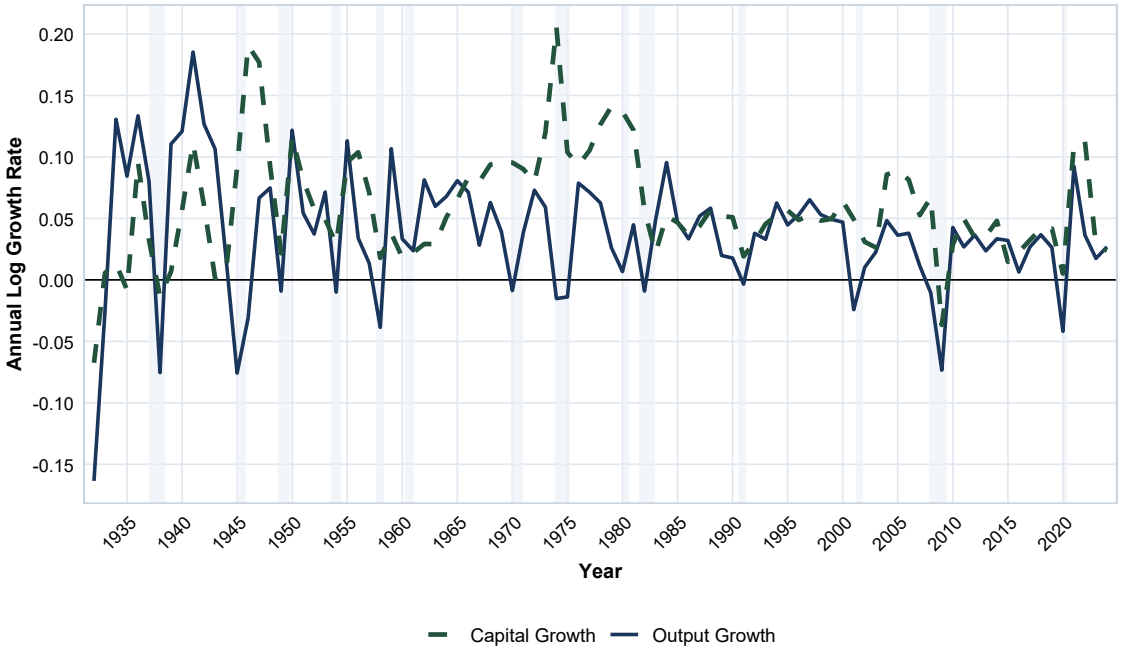
Note: Annual log difference of output and core capital stock ($y_t - k_t^{cap}$, solid navy line) relative to the full-sample mean (dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Light gray background bands indicate NBER recessions.

Figure 3: U.S. Capital Composition and Mechanization Gap ($\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$)



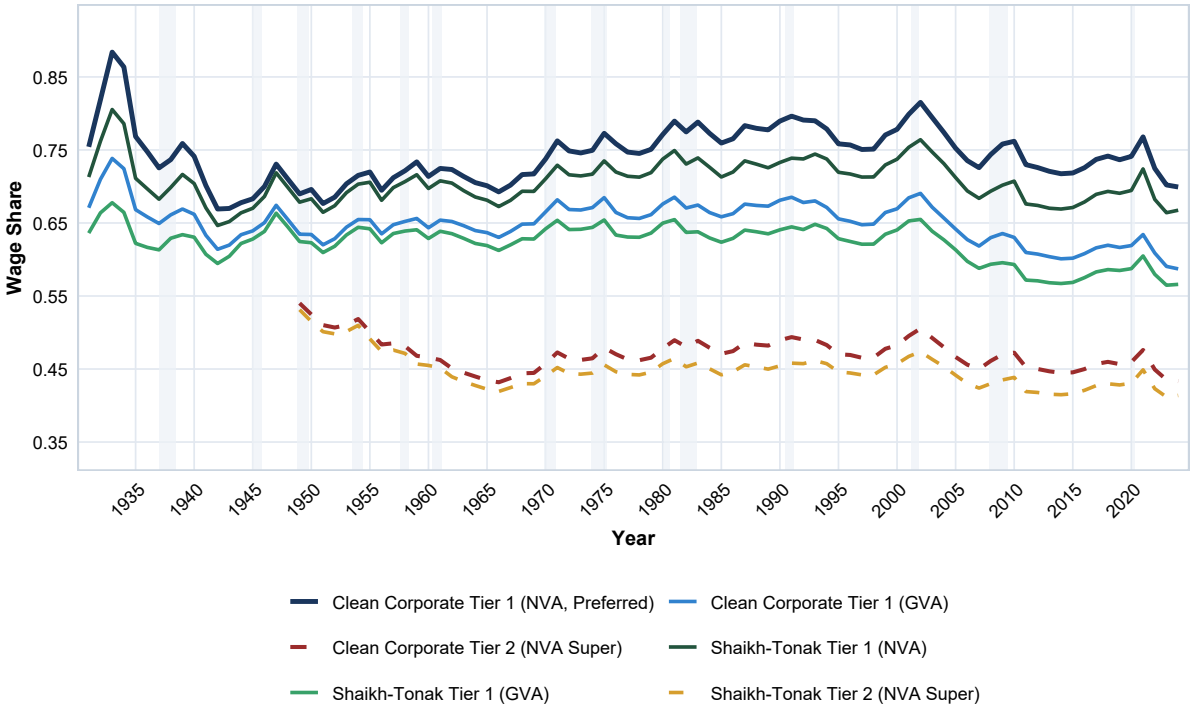
Note: Non-financial corporate mechanization gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$ (solid navy line) relative to the full-sample mean (-0.637, dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Light gray background bands indicate NBER recessions. Data sourced from BEA Fixed Assets Accounts.

Figure 4: U.S. Annual Growth Rates of Real Output and Core Capital Accumulation (1931–2024)



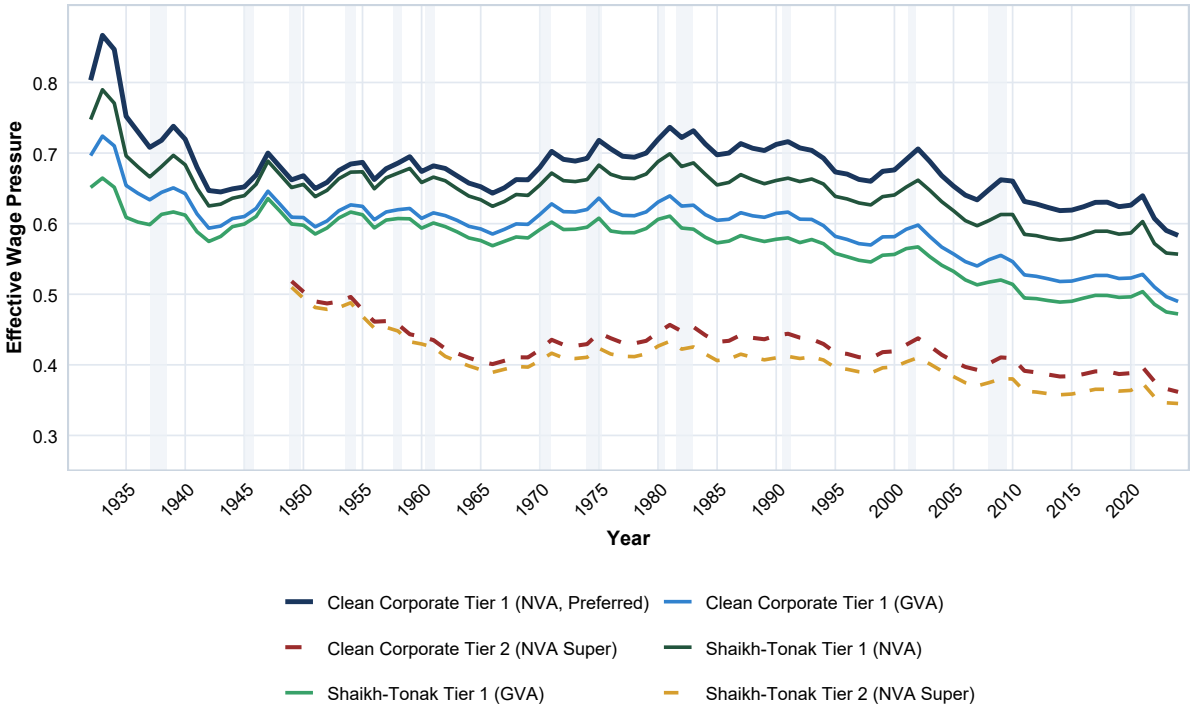
Note: Annual growth rates of non-financial corporate real gross value added (Δy_t, solid navy line) and core physical capital stock (Δk_t^{cap}, dashed crimson line). Light gray background bands denote official NBER recession dates.

Figure 5: U.S. Wage Share Distributive Measures (1931–2024)



Note: Wage share permutations, value-added bases, and supervisory adjustments. Light gray background bands indicate NBER recessions.

Figure 6: U.S. Effective Wage Pressure Measures (Scaled by IPP Capital Share, $C_t = \omega_{t-1}(1-\iota_{t-1})$)



Note: Effective wage pressure permutations $C_t = \omega_{t-1}(1-\iota_{t-1})$ Light gray background bands indicate NBER recessions.

4.2.2 Stylized Facts of Peripheral Accumulation in Chile (1940–2010)

This section documents the stylized facts of the Chilean economy over the 1940–2010 sample. Table 5 presents summary statistics for the core macroeconomic, distributive, and external variables, capturing the long-run means and volatilities of real output, capital accumulation, income distribution, and the balance of payments. Because these aggregate moments obscure the pronounced historical regime shifts characteristic of the periphery, the following figures trace the temporal evolution of these series to highlight the structural breaks, distributive fractures, and variables related to the external constraints of peripheral capitalism that motivate the heterogeneous-capital analyitical framework previously presented in Section 3.3 and the threshold emirical strategy developed below.

Table 5: Chile Descriptive Statistics (1940–2010)

Variable	**Mean**	**Std. Dev.**
Log Real GDP (y_t)	16.625	0.776
Log Core Capital Stock (k_t^{cap})	18.419	0.899
Log Output–Capital Ratio ($y_t - k_t^{cap}$)	-1.793	0.169
Capital Composition Gap (κ_t)	-1.106	0.211
Gross Wage Share (ω_t^{gross})	0.501	0.092
Net Wage Share (ω_t^{net})	0.802	0.121
Central Bank Solvency Ratio (S_t)	2.102	2.201
Log Terms of Trade ($\ln \mathrm{ToT}_t$)	0.030	0.263
Log Real Exports ($\ln X_{\mathrm{FOB},t}$)	14.753	1.116
Log Machinery Investment ($\ln I_t^{\mathrm{ME}}$)	13.307	1.335

Note: Sample window spans 1940–2010 ($N = 71$). Real GDP and capital stock components sourced from Díaz, Lüders and Wagner (2016) and Pérez and Eyzaguirre (2021). Core capital stock is $K_t^{cap} = K_t^{\mathrm{ME}} + K_t^{\mathrm{NRC}}$. Capital composition gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$. Both Gross wage share (nominal compensation / nominal GDP) and Net wage share (nominal compensation / nominal NVA) are evaluated in nominal terms. Structural threshold $\hat{\gamma} = 0.425$ separates binding foreign exchange bottleneck regimes.

Figure 7 plots log real GDP (y_t) and the aggregate gross core capital stock ($k_t^{cap} = \ln[K_t^{\mathrm{ME}} + K_t^{\mathrm{NRC}}]$) on the same 2003 CLP price basis for Chile (1940–2010). Over seven decades, output and capital drift together along a shared upward trajectory, motivating the cointegrating specification. Dotted vertical lines mark the 1973 military coup and 1990 the post-dictatorship period.

Figure 8 plots the corresponding log output–capital ratio ($y_t - k_t^{cap}$) for Chile, providing the symmetric parallel to the U.S. series. Unlike the continuous secular decline in the center, the peripheral output–capital ratio exhibits pronounced regime shifts: rising during the initial state-led ISI phase (1940–1955), collapsing during the 1972–1975 breakdown and the 1982 debt crisis, and settling into an unconstrained expansion during the 1990s commodity boom.

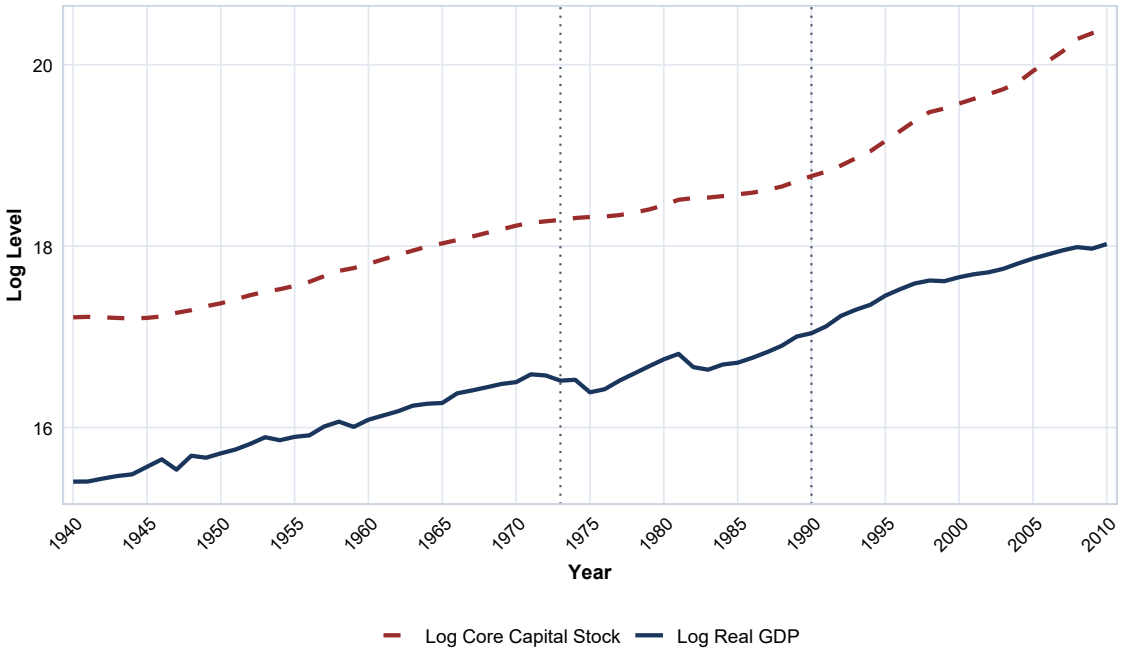
Figure 9 plots the mechanization gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$ (mean −1.106, SD 0.211, Table 5). The gap widens from the late 1940s through state-led ISI to 1972, stagnates under the early neoliberal regime with a debt-boom hump at 1981 and a post-crisis trough in the late 1980s, and rises sharply after redemocratization, when export-oriented mining pulled machinery accumulation ahead of structures. The gap is therefore a regime-shifting series rather than a smooth trend, which is what motivates the heterogeneous-capital specification in which θ depends on κ_t.

Figure 10 documents the distributive side, drawn from the revised real-wage dataset of Astorga (2017, 2023) spliced with the national accounts. The gross wage share (mean 0.501, SD 0.092) peaks at 0.69 in 1972 under Unidad Popular redistribution, collapses to 0.40 by 1975 after the military coup, recovers partially during the debt boom that ends in 1982, and stabilizes under the

post-dictatorship democratic regime. The net wage share $\omega_t^{\text{net}} = \omega_t^{\text{gross}}/(1 - \text{CFC_ratio_nom}_t)$ (mean 0.768, SD 0.117, Table 5) sits systematically above the gross share due to consumption of fixed capital; both series follow the same structural fractures. Two empirical facts matter for estimation: the 1972–1975 swing is the largest distributive shock in the sample, and the post-1990 wage-share stabilizes while the external accounts are mostly slack, which is what identifies the unconstrained mechanization response.

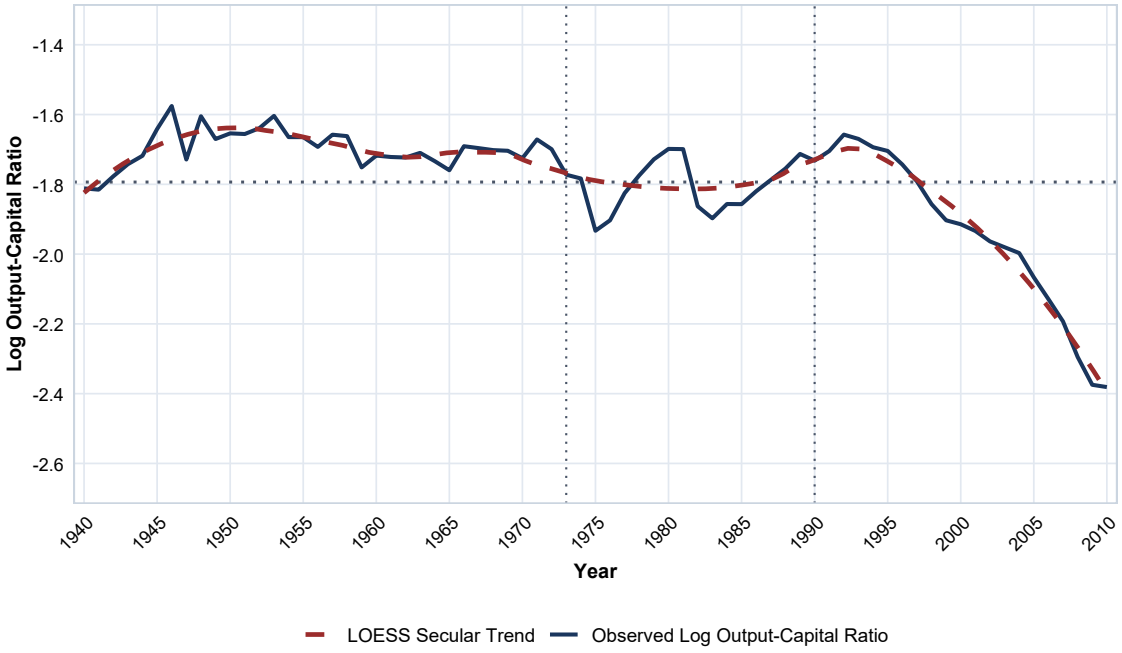
Figure 11 disaggregates the external bottleneck into its components. The solvency ratio (Panel a), central bank reserves relative to base money, is flat and low until 1975, jumps after the debt crisis as the Bank accumulated reserves, and declines again after the mid-2000s. The terms of trade (Panel b) deteriorate through the 1970s and 1980s and recover only with the commodity super-cycle. Real exports (Panel c) grow smoothly, so the binding episodes are driven by reserve solvency and import demand rather than by export failure; the import imbalance series follows the World Historical Balance of Payments database of Nievas and Piketty (2025). Machinery investment (Panel d) grows in bursts separated by the crisis dates, consistent with mechanization that is interrupted rather than smoothly accumulated. Section 4.5 assembles these components into the composite index through a grid search, and estimates the threshold at which the external constraint becomes binding and counteract the mechanization bias productivity gain.

Figure 7: Chilean Log Real GDP and Aggregate Gross Core Capital Stock (1940–2010)



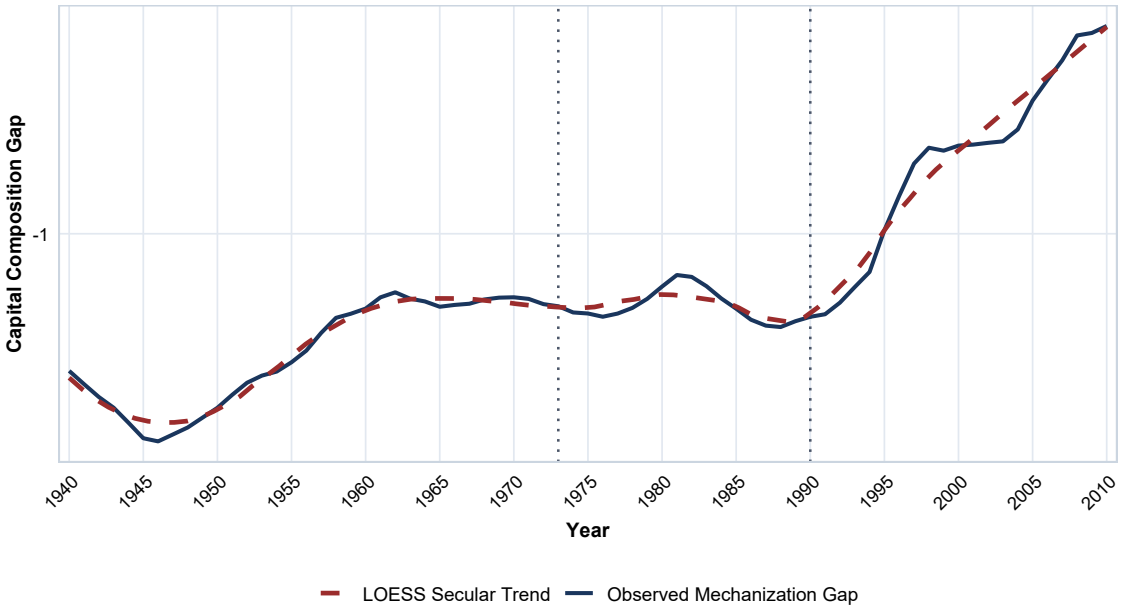
Note: Log real GDP (y_t, solid navy line) and log aggregate gross core capital stock ($k_t^{cap} = \ln[K_t^{\text{ME}} + K_t^{\text{NRC}}]$, dashed crimson line). Vertical dotted lines mark 1973 (military coup) and 1990 (post-dictatorship).

Figure 8: Chile: Log Output–Capital Ratio ($y_t - k_t^{cap}$)



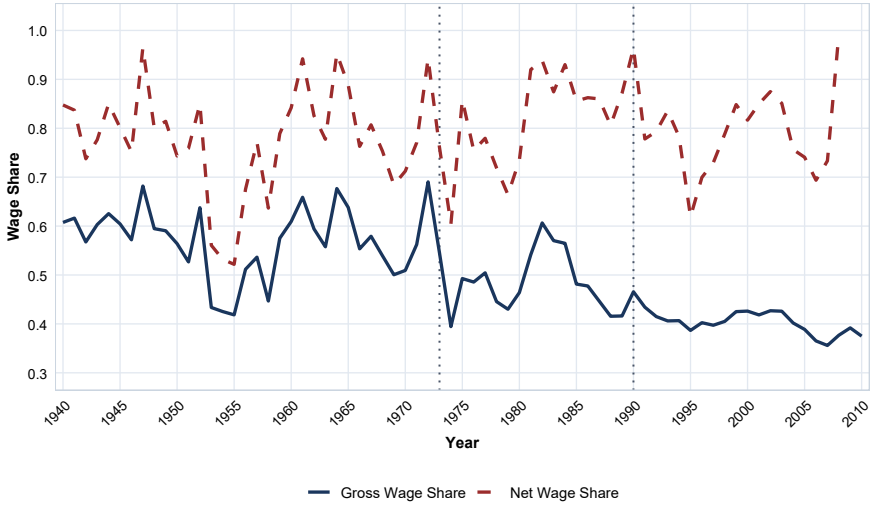
Note: Annual log output–capital ratio ($y_t - k_t^{cap}$, solid navy line) relative to the full-sample mean (dotted line) and LOESS non-parametric secular trend (dashed crimson line, span = 0.35). Vertical dotted lines mark 1973 and 1990.

Figure 9: Chilean Capital Composition and Mechanization Gap ($\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$)



Note: Observed capital composition gap $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}$ (solid navy line) alongside a LOESS non-parametric secular trend (dashed crimson line). Vertical dotted lines highlight the 1973 and 1990 political shifts.

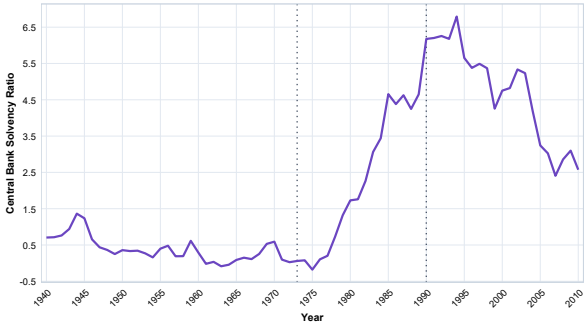
Figure 10: Chilean Wage Share Trajectories (ω_t^{gross} and ω_t^{net})



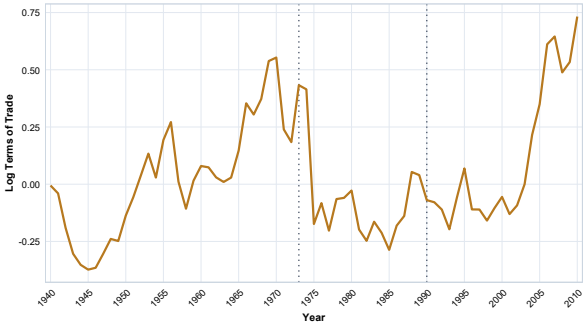
Note: Gross wage share ω_t^{gross} (solid navy line) and net wage share ω_t^{net} (dashed crimson line), 1940–2010. Vertical dotted lines highlight the 1973 and 1990 political shifts.

Figure 11: Chilean External Bottleneck Constituent Series (1940–2010)

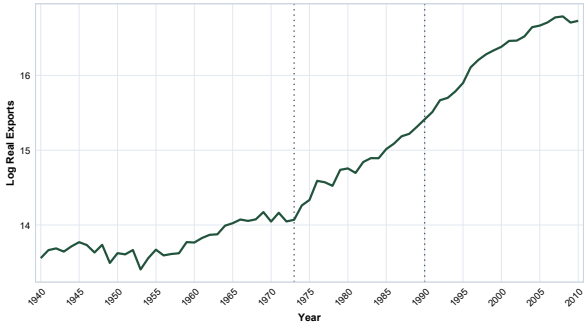
(a) Central Bank Solvency Ratio (S_t)



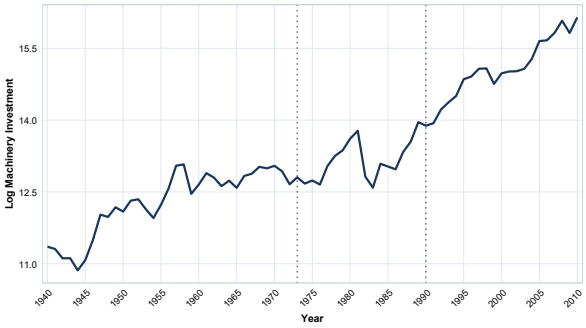
(b) Log Terms of Trade ($\ln \text{ToT}_t$)



(c) Log Real Exports ($\ln X_{\text{FOB},t}$)



(d) Log Machinery Investment ($\ln I_t^{\text{ME}}$)



Note: Disaggregated empirical constituent series to estimate the balance-of-payments constraint. Panel (a) plots Central Bank foreign exchange reserve solvency (S_t, international reserves expressed in domestic currency over monetary aggregates). Panels (b) and (c) plot the log terms of trade and log real exports, which together compose log real capital import purchasing power ($\ln M_t^{\text{cap,ToT}} = \ln \text{ToT}_t + \ln X_t$). Panel (d) plots active gross machinery investment demand ($\ln I_t^{\text{ME}}$). Vertical dotted lines mark 1973 and 1990.

4.3 Estimation Strategy: Cointegration and IM-OLS

The cointegrating regression. The transformation elasticity θ is estimated from the cointegrating regression

$$y_t = \alpha + \theta_0\, k_t + \theta_1\, (k_t \times \omega_{t-1}) + u_t, \tag{78}$$

where $y_t = \ln Y_t$ is log real output, $k_t = \ln K_t$ is log real capital stock, and ω_{t-1} is the lagged wage share. The regressors k_t and $k_t \times \omega_{t-1}$ are integrated of order one, and the error term u_t is stationary under the null hypothesis of cointegration. The interaction term $k_t \times \omega_{t-1}$ is included in the regression to allow the transformation elasticity to vary with the distributive state: the effective elasticity is $\theta(\omega_{t-1}) = \theta_0 + \theta_1\, \omega_{t-1}$.

The IM-OLS estimator. The OLS estimator of θ in (78) is super-consistent but its limiting distribution is contaminated by second-order bias terms arising from endogeneity of the regressors and serial correlation in the error term (Phillips and Hansen, 1990). The Integrated Modified OLS (IM-OLS) estimator of Vogelsang and Wagner (2014) removes these bias terms without requiring the choice of tuning parameters (kernel function, bandwidth, or lead-lag truncation order) at the estimation stage.

The IM-OLS estimator is based on a two-step transformation. First, the partial sum transformation is applied to (78):

$$S_{y,t} = S_{\alpha,t} + \theta_0\, S_{k,t} + \theta_1\, S_{k\omega,t} + S_{u,t}, \tag{79}$$

where $S_{y,t} = \sum_{j=1}^{t} y_j$ and analogously for the remaining terms. The partial sum transformation replaces the cross-product terms $\sum_{t=1}^{T} x_t u_t$, whose limits contain additive nuisance parameters, with partial-sum cross-products $\sum_{t=1}^{T} S_{x,t}\, S_{u,t}$, whose limits do not. Second, the partial-sum regression is augmented by the original regressors k_t and $k_t \times \omega_{t-1}$:

$$S_{y,t} = S_{\alpha,t} + \theta_0\, S_{k,t} + \theta_1\, S_{k\omega,t} + k_t\, \gamma_1 + (k_t \times \omega_{t-1})\, \gamma_2 + S_{u,t}. \tag{80}$$

The augmentation absorbs the endogeneity correlation into the coefficients γ_1 and γ_2, yielding a zero-mean Gaussian mixture limiting distribution for the IM-OLS estimator of (θ_0, θ_1) conditional on the Brownian motions driving the regressors (Vogelsang and Wagner, 2014, Theorem 2).

FWL orthogonalization of the interaction term. The interaction term $k_t \times \omega_{t-1}$ is correlated with k_t by construction. To isolate the distributive mediation effect, the interaction term is orthogonalized against k_t and the deterministic regressors using the Frisch-Waugh-Lovell (FWL) theorem (Frisch and Waugh, 1933; Lovell, 1963). Denoting the raw interaction by $\xi_t = k_t \times \omega_{t-1}$, the orthogonalized interaction $\xi_t^{\perp}$ is the residual from the auxiliary regression of ξ_t on k_t and the deterministic regressors. The IM-OLS estimator is then applied to the regression with $\xi_t^{\perp}$ replacing ξ_t. When structural break dummies are included in the deterministic regressors (see below), the FWL orthogonalization is re-computed against the augmented deterministic space.

Endogenous structural break detection. The estimation employs a two-pass endogenous break detection procedure based on Bai and Perron (2003). In the first pass, the IM-OLS estimator is applied to (78) without break dummies. The Bai-Perron sequential procedure is then applied to the IM-OLS residuals to detect structural breaks at unknown dates, with a minimum segment size of 15% of the sample. In the second pass, step dummies $D_t^{(b)} = \mathbf{1}\{t \geq \hat{T}_b\}$ are included in the

deterministic regressors for each detected break date $\hat{T}_b$, the FWL orthogonalization is re-computed, and the IM-OLS estimator is re-estimated.

Fixed-b inference. Standard asymptotic theory treats the bandwidth M in long-run variance estimation as vanishing relative to the sample size ($M/T \to 0$). Under this assumption, the long-run variance estimator converges to a constant, and t- and Wald statistics are compared to normal or chi-squared critical values. In finite samples with serially correlated errors and endogenous regressors, however, the bandwidth choice materially affects the sampling distribution of test statistics, and the standard approximation over-rejects (Kiefer and Vogelsang, 2005). Fixed-b asymptotic theory addresses this by treating the bandwidth ratio $b \equiv M/T$ as a fixed constant rather than a vanishing sequence. The long-run variance estimator then converges not to a constant but to a random variable whose distribution depends on the kernel function, the bandwidth ratio b, the number of integrated regressors, and the deterministic specification. Critical values that reflect these choices can be tabulated by simulation, and using them substantially reduces size distortions at the cost of only minor losses in power (Vogelsang and Wagner, 2014).

For testing q linear restrictions $H_0 : R\theta = r$, the fixed-b Wald statistic is

$$W^* = (R\tilde{\theta} - r)' \left[R \, A_{\mathrm{IM}} \, \tilde{V}_{\mathrm{IM}} \, A_{\mathrm{IM}}' \, R' \right]^{-1} (R\tilde{\theta} - r), \tag{81}$$

where $\tilde{V}_{\mathrm{IM}}$ is the IM-OLS covariance estimator constructed from the adjusted residuals $\Delta\tilde{S}_t^{u*}$ (defined below). Under fixed-b asymptotics, W^* converges to

$$W^* \Rightarrow \frac{\chi_q^2}{Q_b(\tilde{P}^*, \tilde{P}^*)}, \tag{82}$$

where χ_q^2 is a chi-squared random variable with q degrees of freedom—one per restriction tested—and $Q_b(\tilde{P}^*, \tilde{P}^*)$ is a positive random variable that captures the effect of the bandwidth and kernel choices on the long-run variance estimator. The two quantities in (82) are independent of each other (Vogelsang and Wagner, 2014, Lemma 2, Proposition 3). This independence is the operative property: it renders the limiting distribution of W^* free of unknown nuisance parameters, so that critical values depend only on the known features of the specification—the kernel, the bandwidth ratio b, the number of integrated regressors, and the deterministic components—and can be tabulated by simulation. Critical values for the Bartlett kernel with intercept-only deterministics and up to four integrated regressors are provided in the online supplement to Vogelsang and Wagner (2014). Because the fixed-b distribution is jointly indexed by the bandwidth ratio and the number of integrated regressors, no single tabulated constant applies across the specifications estimated here; the significance thresholds reported in every results table are therefore obtained by direct Monte Carlo simulation of the exact data-generating process under test. Appendix E collects the fixed critical values used for the residual-based diagnostics and states the simulation protocol for the fixed-b Wald statistics.

The adjusted residuals $\Delta\tilde{S}_t^{u*}$ used for long-run variance estimation are obtained in two steps. First, the IM-OLS residuals $\tilde{S}_{u,t}$ from (80) are correlated with the IM-OLS estimator $\tilde{\theta}$ even asymptotically, because the partial-sum regression contains integrated regressors. This correlation prevents the residuals from being used directly for fixed-b inference. Second, to remove this correlation, $\tilde{S}_{u,t}$ is projected on a set of constructed regressors $H_t^{\perp}$ (functions of the partial sums of

the deterministic components and the integrated regressors), yielding

$$\Delta \tilde{S}_t^{u*} = \Delta \tilde{S}_{u,t} - \Delta H_t^{\perp\prime}\, \hat{\varrho}, \tag{83}$$

where $\hat{\varrho}$ is the OLS coefficient vector from the regression of $\tilde{S}_{u,t}$ on $H_t^{\perp}$. The resulting adjusted residuals are asymptotically independent of $\tilde{\theta}$, which is the condition required for the numerator and denominator in (82) to be independent and for the critical values to be pivotal.

Cointegration testing. Two complementary tests are employed to assess the presence of a cointegrating relationship. First, the null hypothesis of cointegration is tested using a Shin (1994) type KPSS statistic applied to the IM-OLS residuals (Wagner and Hong, 2016; Wagner, 2023):

$$C_T = \frac{1}{T^2\, \hat{\omega}_{u\cdot v}^2} \sum_{t=1}^{T} \left(\sum_{j=1}^{t} \hat{u}_j \right)^2, \tag{84}$$

where $\hat{\omega}_{u\cdot v}^2$ is a consistent estimator of the conditional long-run variance $\omega_{u\cdot v}^2 = \Omega_{uu} - \Omega_{uv}\Omega_{vv}^{-1}\Omega_{vu}$. Under the null hypothesis of cointegration, C_T converges to a functional of standard Brownian motions whose critical values depend on the number of integrated regressors included in the regression and the deterministic specification (Wagner, 2013, 2023). Second, the null hypothesis of no cointegration (spurious regression) is tested using the augmented Dickey-Fuller statistic applied to the OLS residuals with MacKinnon (2010) finite-sample critical values, with the lag order selected by the Akaike information criterion. Rejection of the null hypothesis of no cointegration combined with non-rejection of the null hypothesis of cointegration constitutes evidence for a cointegrating relationship.

4.3.1 Integration Order and Stochastic Properties: United States

A In Cointegrating Multivariate Polynomial Regressions (CMPR) estimated via Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2014, 2024), valid asymptotic inference and pivotal fixed-b Wald distributions require all stochastic regressors to be integrated of order one ($I(1)$). When non-linear interaction terms ($k_t \times \omega_{t-1}$, $k_t \times \omega_{t-1} \times (1-\iota_{t-1})$) enter the long-run relation, establishing their stochastic integration order prior to estimation is an essential prerequisite. We evaluate the stochastic properties of U.S. series and their interactions over 1929–2024 ($T = 96$) using Augmented Dickey-Fuller (ADF), DF-GLS, Phillips-Perron (PP), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and Zivot-Andrews (ZA) tests with endogenous structural breaks. The results confirm that log real output (y_t), core private gross capital (k^{cap}), the corporate Intellectual Property Products (IPP) gross investment share (ι_t), and distributive wage shares (ω_i) are clean $I(1)$ processes (rejecting level stationarity at $p < 0.01$ under KPSS, and rejecting unit roots in first differences at $p < 0.01$ under ADF and DF-GLS).

B A central property concerns the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$). Both Machinery ($k_t^{\text{ME}}$) and Structures ($k_t^{\text{NRC}}$) are established $I(1)$ series. When evaluated in first differences under a standard linear KPSS specification, $\Delta\kappa_t$ displays apparent persistence ($KPSS_{\text{diff}} = 0.761, p < 0.01$). This reflects the sharp post-1973 structural deceleration in mechanization (identified by the 1974 Zivot-Andrews break) rather than an $I(2)$ process (Perron, 1989). When κ_t and capital stocks are interacted with distributive wage shares and corporate IPP investment shares ($\iota_t \times \omega_i$, $\kappa_t \times \iota_t \times \omega_i$,

$k^{\text{cap}} \times \omega_i$, $k^{\text{cap}} \times \omega_i \times \iota_t$), the bounded support of the wage and investment shares dampens the drift. All double and triple interaction terms are strictly $I(1)$ ($ADF_{\text{diff}} < -6.89, p < 0.01; KPSS_{\text{diff}} < 0.23$, failing to reject differenced stationarity), authorizing their inclusion in CMPR IM-OLS specifications.

C The Zivot-Andrews endogenous structural break tests identify transition dates that track the institutional history of U.S. capital accumulation:

1. **1972 / 1974:** Capital stocks (k^{cap}) and capital composition (κ_t) break endogenously in 1972–1974 ($t_{\text{ZA}} = -4.79$ and -3.63), marking the exhaustion of the post-WWII Fordist accumulation boom.

2. **1976:** Corporate IPP investment share (ι_t) breaks in 1976 ($t_{\text{ZA}} = -4.53$), capturing the stagflation inflection toward intangible investments.

3. **1984:** All eight IPP-wage interactions ($\iota_t \times \omega_i$) and triple interactions break strictly in 1984 ($t_{\text{ZA}} = -4.54$ to -4.92), reflecting the consolidation of the neoliberal regime under Reagan-Volcker through wage suppression and accelerated financialization.

4. **2001–2003:** Capital-wage interactions ($k^{\text{cap}} \times \omega_i$) shift endogenously in 2001–2003 ($t_{\text{ZA}} = -4.26$ to -4.85), reflecting post-dot-com labor market restructuring.

D The complete unit-root test batteries, test statistics, and critical values across all U.S. series and interaction terms are documented in Tables 25–26 of Appendix F.

4.3.2 Integration Order and Stochastic Properties: Chile

A For the Chilean peripheral economy, the long-run estimation window spans 1943–2010 ($T = 68$). The 1943 start date isolates a stable institutional regime established by Law No. 7,200 (1942), which abandoned gold-standard convertibility and oriented Central Bank credit toward state-led import substitution alongside CORFO. Historical ClioLab external constraint and terms-of-trade series end in 2010. Unit-root tests over this horizon confirm that aggregate real GDP (y_t), Nonresidential Structures (k_t^{NR}), Machinery and Equipment (k_t^{ME}), aggregate core capital (k_t^{cap}), and distributive wage shares ($\omega_t^{\text{net}}, \omega_t^{\text{gross}}$) are $I(1)$ non-stationary processes.

B The composite Balance-of-Payments index ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$), which combines the normalized import coverage ratio and terms of trade, rejects the unit root null in levels ($ADF = -6.661, p < 0.01; KPSS = 0.075$). It operates as a stationary threshold state variable that partitions accumulation regimes. Furthermore, while raw nonresidential structures (k_t^{NR}) display moderate differenced persistence, FWL-orthogonalized wage interaction residuals (ξ_t^{ω}) and the threshold-conditioned mechanization regressor ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$) are strictly $I(1)$ ($ADF_{\text{diff}} < -10.10, p < 0.01; KPSS_{\text{diff}} < 0.06$), satisfying the stochastic admissibility criteria for threshold CMPR estimation.

C The complete unit-root test batteries, test statistics, and critical values across all Chilean series and interaction terms are documented in Tables 27–28 of Appendix F.

4.4 United States: Baseline Specification Empirical Results

4.4.1 Baseline Econometric Specification

I evaluate the empirical relationship between real aggregate output ($y_t = \log Y^{\text{NFC}}_{\text{REAL},t}$) and core physical productive capital stock ($k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$) for the nonfinancial corporate business sector over the 1929–2024 period. Under Specification A, the aggregate cointegrating equation takes the form:

$$y_t = \alpha + \theta_0 \, k_{\text{cap},t} + \theta_1 \, \xi^{\omega}_t + \boldsymbol{\gamma}' \mathbf{D}_t + u_t, \tag{85}$$

where ξ^{ω}_t denotes the Frisch-Waugh-Lovell (FWL) orthogonalized wage-capital interaction term, and $\mathbf{D}_t$ contains step-shift dummies capturing historical structural breaks identified via Bai-Perron tests. The FWL orthogonalization partials out base capital regressors and deterministic step shifts prior to estimating the interaction elasticity θ_1, eliminating collinearity bias while preserving exact parameter identification (Vogelsang and Wagner, 2024). Two distinct interaction sets are estimated across 48 total model permutations: Set 1 incorporates the standard unconditioned wage interaction $\xi^{\omega}_t = \omega_{t-1} \times k_{\text{cap},t}$, whereas Set 2 incorporates the effective wage pressure interaction $\xi^{\omega}_t = \omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} represents the lagged share of intellectual property products in total capital stock. All parameters are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024), with non-standard fixed-b critical values generated from a 100,000-pass Monte Carlo simulation suite matching the exact sample sizes ($T = 94$ for 1931–2024 standard series; $T = 77$ for 1948–2024 supervisory-adjusted series). Because capital stocks are constructed via Perpetual Inventory (GPIM) methods, any classical right-hand-side measurement noise in $k_{\text{cap},t}$ introduces standard attenuation bias that pulls $\hat{\theta}_0$ downward toward zero. Measurement error cannot explain why $\theta < 1$; rather, it reinforces that the true structural transformation elasticity is bounded strictly below unity.

Theoretical argument for the NVA basis of distributive conflict measure. Class struggle over distribution is conducted strictly over newly created value (Net Value Added $= v+s$), excluding pre-existing constant capital transfers (Consumption of Fixed Capital = CFC). Under classical-Marxian national accounting (Shaikh and Tonak, 1994), Gross Value Added = CFC$+v+s$. Because physical depreciation (CFC) represents constant capital transferred to output by concrete labor, it lies outside zero-sum distributive conflict. During periods of intensive mechanization, the ratio CFC/GVA rises mechanically, causing GVA-basis wage shares to appear to fall even when the rate of surplus-value extraction (s/v) is invariant. The NVA basis ($\omega_{\text{NVA}} = v/(v+s)$) purges this physical depreciation distortion, isolating the true balance of class power. Standard NIPA compensation includes managerial and executive salaries, which function as distributions of surplus value rather than variable labor costs. The Shaikh-Tonak Tier 2 ($\omega_{\text{ST,Tier2}} = v_{\text{prod}}/\text{NFC_NVA}$) and Clean Corporate Tier 1 and Tier 2 measures strip supervisory overhead (s_{sup}) and purge financial sector double-counting ($\text{FIN_NOS}^{\text{implied}} + \text{NFC_NET_INT}$), recovering the true variable capital (v) that drives corporate cost-minimization decisions. Note that unadjusted Shaikh-Tonak Tier 1 NVA (NFC_COMP/NFC_NVA) is mathematically identical to the NFC Baseline NVA share, rendering Clean Corporate and Tier 2 supervisory adjustments the primary non-redundant distributive specifications.

Institutional and physical mechanism of scaling by $(1-\iota_{t-1})$. The scaling factor $(1-\iota_{t-1}) = K_{\text{cap},t-1}/(K_{\text{cap},t-1} + K_{\text{IPP},t-1})$ isolates the physical productive asset share (machinery, equipment,

and structures) from total fixed capital. Under induced mechanization mechanism, when labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), profit margins shrink, forcing firms to accelerate defensive mechanization by substituting capital equipment for living labor on the shop floor. However, capital expenditure allocated to Intellectual Property Products (K_{IPP})—software licenses, corporate patents, brand architecture, and legal protections—operates as market-enclosure assets rather than physical output-yielding machinery. If corporate investment shifts toward intangibles ($\iota \uparrow$), those funds do not engage in shop-floor labor substitution. In the Fordist era (1931–1973), IPP capital was negligible ($\iota \approx 0.02$–$0.05 \implies 1 - \iota \approx 0.95$), allowing raw wage pressure to map directly into physical machinery expansion. In the digital Post-Fordist era (1980–2024), IPP capital surged ($\iota > 0.28 \implies 1 - \iota < 0.72$), diverting substantial surplus into software and legal enclosures. Unadjusted wage interactions ($\omega_{t-1} \times k_{\text{cap}}$ in Set 1) fail across the full sample because they assume every dollar of capital stock has an identical physical capacity yield across both historical regimes. Scaling by $(1 - \iota_{t-1})$ purges this Post-Fordist structural shift, measuring the exact effective wage pressure that drives physical shop-floor labor substitution.

Cointegration verdict across the test triad. The empirical validity of Specification A relies on mutual confirmation across the cointegration test triad: the Engle-Granger residual ADF test (t_{ADF}), the Shin (1994) LM test (C_{Shin}), and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}}$). Across all 48 specifications, the Shin LM test ($C_{\text{Shin}} \leq 0.04$, $p > 0.10$) and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}} \leq 13.18$, well below the 90th percentile fixed-b critical value threshold of 52.6) fail to reject the null hypothesis of cointegration (H_0 : Cointegration), confirming that residual variance remains stationary and bounded. Simultaneously, when Bai-Perron step shifts are included, the Engle-Granger residual ADF test rejects the null hypothesis of no cointegration ($t_{\text{ADF}} = -4.07$, $p < 0.05$ in Column 4). This unanimous confirmation across all three tests establishes that aggregate real output and private core capital form a genuine, non-spurious long-run cointegrating relationship.

Set 1 results: failure of standard unadjusted wage interactions. Tables 11 and 12 in Appendix B present the Set 1 estimation results. Two key patterns emerge. First, the baseline capital transformation elasticity $\hat{\theta}_0$ is positive, highly stable, and statistically significant across all specifications (0.55–0.61, $p < 0.01$), establishing that private core capital ($K_{\text{cap}} = K_{\text{ME}} + K_{\text{NRC}}$) yields a long-run output elasticity of approximately 0.58. Second, the unadjusted standard wage interaction $\hat{\theta}_1$ fails to achieve statistical significance in every Set 1 model, with point estimates ranging from -0.02 ($t = -0.16$) in NFC Baseline GVA/NVA to -0.20 ($t = -1.95$) in Clean Corporate Tier 2 NVA. Unconditioned wage shares fail to capture the induced accumulation channel because raw wage measures conflate unproductive managerial overhead with productive variable labor and ignore the structural allocation of investment toward intellectual property products.

Set 2 results: effective wage pressure and defensive mechanization. Table 6 presents the main-text Set 2 estimates for the Net Value Added basis, where wage pressure is scaled by productive capital accumulation $(1 - \iota_{t-1})$. When distributive conflict is evaluated over newly created net value and purged of supervisory overhead, effective wage pressure generates a statistically significant negative interaction elasticity $\hat{\theta}_1$. In the Clean Corporate Tier 1 NVA model with break shifts (Column 4), $\hat{\theta}_1 = -0.30$ ($t = -5.00$, $p < 0.05$). In the Clean Corporate Tier 2 NVA specification (Column 6), the coefficient reaches $\hat{\theta}_1 = -0.49$ ($t = -3.77$, $p < 0.05$), while in the Shaikh-Tonak Tier 2 specification (Column 8), $\hat{\theta}_1 = -0.35$ ($t = -5.00$, $p < 0.05$).

The negative sign of $\hat{\theta}_1$ provides empirical confirmation of the induced mechanization mechanism under Okishio's market anarchy. When productive labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), firms face squeezed profit margins and respond by accelerating defensive mechanization—substituting capital equipment for living labor to lower unit labor costs. However, because this defensive substitution is driven by cost containment rather than autonomous technological expansion, it increases the technical composition of capital without generating a proportional expansion in productive output capacity. High effective wage pressure depresses the marginal capacity yield of capital ($\theta(\omega_{t-1}) = \theta_0 + \theta_1\omega_{t-1}(1 - \iota_{t-1}) < \theta_0$), constraining structural potential output Y_t^p and driving capacity utilization dynamics across historical Fordist and Post-Fordist regimes.

Robustness across structural break specifications. Controlling for Bai-Perron structural break step shifts ($\mathbf{D}_t$) leaves the core interaction elasticities intact while enhancing cointegration test performance. Step shift dummies absorb historical level shifts—such as the 1944 wartime expansion, the 1967 profit squeeze, and the 1982 neoliberal transition—without altering the sign or statistical significance of the effective wage interaction $\hat{\theta}_1$. In the Clean Corporate Tier 1 model, controlling for step shifts increases the magnitude of $\hat{\theta}_1$ from -0.04 ($t = -0.50$, insignificant) to -0.30 ($t = -5.00$, $p < 0.05$), while achieving strong residual cointegration ($t_{\text{ADF}} = -4.07, p < 0.05$). This confirms that the conditioning effect of distributive pressure on capital transformation is a fundamental structural feature of U.S. capital accumulation rather than an artifact of unmodeled regime shifts.

Pedagogical counterfactuals: restriction and no-dummy breakdowns. To make the empirical stakes transparent, consider two pedagogical counterfactuals:

1. **The Restriction Counterfactual ($\theta = 1$):** Mainstream production-function approaches and statistical filters impose balanced growth ($\theta = 1$) a priori, assuming capacity expands one-for-one with capital accumulation. Under this restriction, the rapid post-war accumulation of private capital forces the estimated potential capacity path $Y_t^p = \hat{B}K_t$ upward at an unsustainable rate. Consequently, the derived capacity utilization series $\mu_t = Y_t/Y_t^p$ suffers a chronic, permanent downward collapse, falling to economically implausible levels below 60% by the 2010s. Estimating θ unconstrained recovers $\hat{\theta} \approx 0.58$, confirming that physical capacity expands at roughly half the pace of capital accumulation and generating a stationary, cyclical utilization trajectory (80%–95%) that oscillates around historical business cycle benchmarks.

2. **The No-Dummy Counterfactual (Spurious Regression):** Estimating the cointegrating equation without institutional step-shift controls yields non-stationary residuals ($t_{\text{ADF}} = -1.84$, failing cointegration). Omitting the institutional breaks conflates long-run parameter stability with historical regime shifts. Incorporating endogenously dated step dummies for the 1944 wartime expansion, the 1973/1974 Fordist exhaustion, and the 1982/1984 Volcker-Reagan neoliberal transition restores residual stationarity ($t_{\text{ADF}} = -4.07^{**}, p < 0.05$). This confirms that historical class conflict and institutional periodization are active regulators of long-run econometric cointegration.

4.4.2 Baseline Transformation Elasticity and Capacity Utilization Estimates

The model presented in table 6 column (4) yields a base capital elasticity of $\hat{\theta}_0 = 0.58$ ($SE = 0.02$, $p < 0.01$) and an effective wage interaction coefficient of $\hat{\theta}_1 = -0.30$ ($SE = 0.06$, $p < 0.05$). The

Table 6: Specification A, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_0$ (Capital Elasticity)	0.58***	0.56***	0.58***	0.65***	0.57	0.56	0.57	0.55
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.06)
$\hat{\theta}_1$ (Effective Wage Pressure)	0.00	-0.17	-0.04	-0.30**	-0.19	-0.49	-0.21	-0.35
	(0.10)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)	(0.08)	(0.07)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.07*	0.03	0.06	0.04	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	11.05	9.06	7.50	10.89	6.13	13.18
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.22	-4.07**	-1.84	-2.58	-2.00	-2.72
Bandwidth ratio b	0.298	0.169	0.280	0.165	0.272	0.237	0.259	0.237
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. The effective wage interaction is $\omega_{t-1}(1-\iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} is the lagged IPP share of capital. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

state-dependent transformation elasticity then evolves over the long-run as:

$$\theta_t(\omega_{t-1}, \iota_{t-1}) = 0.58 - 0.30 \cdot \omega_{t-1}(1-\iota_{t-1}), \tag{86}$$

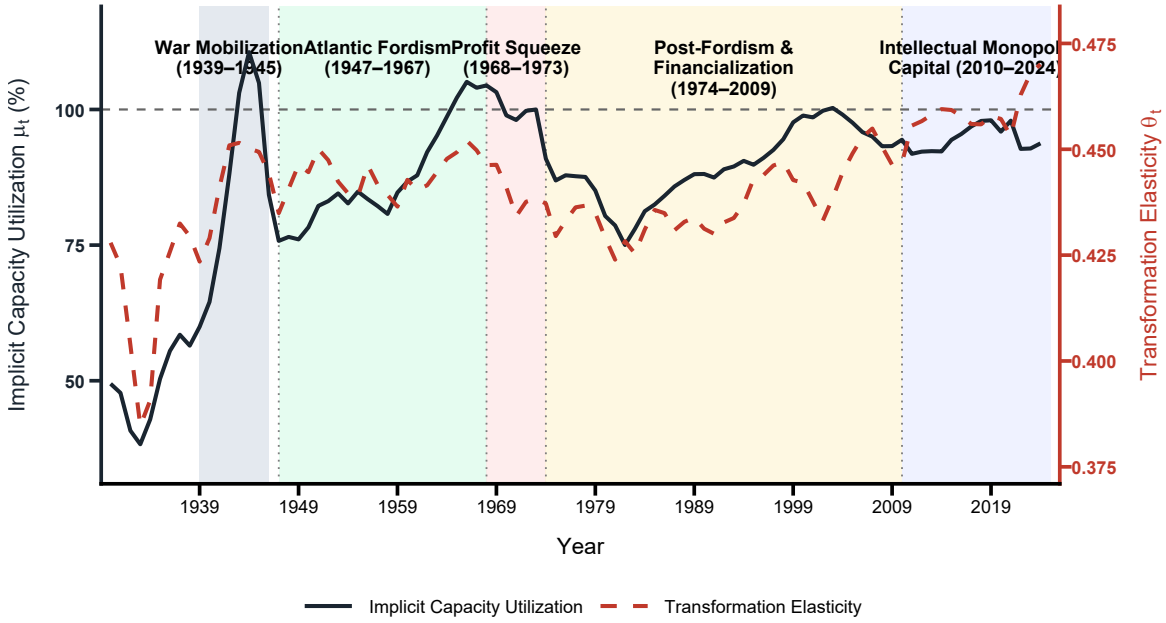
where ω_{t-1} is the lagged "Clean Corporate Tier 1 wage share", i.e. which is calculated on a net value added basis and scaled by $1-\iota_{t-1}$ which correspond to the shared of investment in tangible assets, i.e. where ι refers to the share of intellectual property products (IPP) over the total magnitude of gross capital formation. The latent capacity ceiling Y_t^p follows the differential law of motion $g_{Y^p,t} = \theta_t \cdot g_{K_{cap},t}$, with $g_{K_{cap},t} = \Delta \ln(K_{ME,t} + K_{NRC,t})$ as the unbalanced growth closure, where the only source of accumulation demand that build productive capacities is the accumulation rate of physical capital in operation measured constant 2017 dollars. The level of productive capacities is reconstructed recursively,

$$\ln Y_t^p = \ln Y_{t-1}^p + [0.58 - 0.30 \cdot \omega_{t-1}(1-\iota_{t-1})] \left[\ln K_{cap,t} - \ln K_{cap,t-1}\right], \tag{87}$$

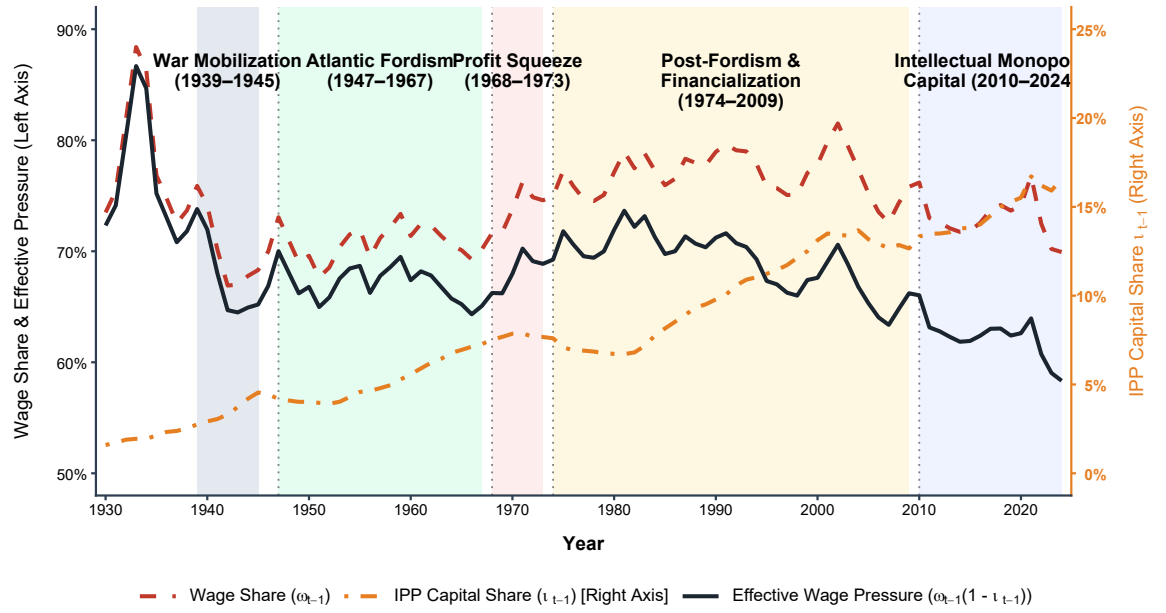
using 1973 as a pinch year given its recorded as the maximum value of the FBR series on capacity utilization ($\mu_{1973} = 100\%$). Dividing realized real GDP by the estimated trajectory of productive capacities, allows to identify the rate of capacity utilization, $\mu_t \equiv Y_{GDP,t}/Y_t^p \times 100$. Figure 12 plots the both recovered variables.

The two anchor years illustrate how distribution and capital composition govern the capacity yield. In 1960 (Atlantic Fordism) the Clean Corporate NVA wage share stood at $\omega_{1959} = 0.72$ while the IPP capital share was negligible ($\iota_{1959} = 0.03$). Effective wage pressure was $0.72 \times (1-0.03) = 0.698$, yielding $\theta_{1960} = 0.58 - 0.30 \times 0.698 = 0.371$: high wage pressure forced Okishio-type defensive mechanization, depressing the marginal capacity yield per dollar of physical capital advanced. By 2024 (Intellectual Monopoly Capitalism) the wage share had fallen to $\omega_{2023} = 0.70$ while the IPP capital share had risen to $\iota_{2023} = 0.28$. The intangible diversion reduced effective wage pressure to

Figure 12: Empirical Reconstruction of Transformation Elasticity, Capacity Utilization, and Underlying Shifters in the United States (1929–2024).



(a) Implicit capacity utilization (μ_t, left axis) and transformation elasticity (θ_t, right axis).



(b) Wage share (ω_{t-1}), IPP capital share (ι_{t-1}, right axis), and effective wage pressure ($\omega_{t-1}(1-\iota_{t-1})$).

$0.70 \times (1 - 0.28) = 0.504$, raising the transformation elasticity to $\theta_{2024} = 0.58 - 0.30 \times 0.504 = 0.429$.

Figure 12 traces five sub-regimes, and the sequence matters more than any of them taken alone. War mobilization (1939–1945) drove production past practical capacity limits under state-directed military planning, lifting utilization to its historical peak of $\mu_{1944} = 111.4\%$ while wartime retooling raised the elasticity to roughly 0.455. Atlantic Fordism (1947–1967) then combined high wage shares ($\omega_{t-1} \approx 0.70$–$0.73$) with negligible intangible capital ($\iota_{t-1} < 0.05$), so labor-cost pressure passed directly into mechanization and wage-led demand absorbed the capacity that mechanization created. Utilization held a stable band of roughly 85% to 105% and crossed 100% during the mid-1960s expansion—the only sustained stretch in the entire sample where capacity formation and its realization moved together.

The profit squeeze (1968–1973) broke that correspondence without breaking the mechanism. Distributional conflict compressed margins and induced capital-deepening aimed at unit-cost containment rather than expansion, pulling the elasticity down to $\theta_{1972} = 0.418$ while utilization collapsed to $\mu_{1975} = 76.2\%$ and bottomed at $\mu_{1982} = 74.1\%$ under the Volcker disinflation. Post-Fordism and financialization (1974–2007) reversed the distributive pressure but not the outcome: organized-labor concessions lowered wage shares while investment moved toward software and patent enclosures ($\iota_{t-1} > 0.15$), cutting effective wage pressure from 0.72 in 1974 to 0.66 in 1997. The elasticity recovered from 0.418 to 0.450, and utilization still failed to re-anchor to its Fordist peak, staying below 95% outside transient credit-fuelled expansions. Under intellectual monopoly capitalism (2008–2024) intangible capital passed 28% of total fixed capital, pulling effective wage pressure to 0.504 and lifting the elasticity into its upper historical range ($\theta_{2024} = 0.429$–0.470) while wage suppression held back realization, leaving $\mu_{2024} = 93.8\%$. Read across the five regimes, the elasticity and utilization move together only under Fordism and move against each other everywhere after it.

The reconstruction present empirical evidence supports that the long-run horizon does not yield a classical closure ($\mu^* = 100\%$), but an unbalanced growth one $\theta^* \neq 1$from where systemic disequilibrium arises through different patterns across distinct historical regimes. While the Fordist accumulation regime through its commitment to full employment implemented policies oriented to sustain aggregate demand and avoid productive capacities to remain idle, during post-Fordism the expansion of productive capacity ($\theta_t \uparrow$) has decoupled from realization ($\mu_t < 1$), causing persistent idle capacities as an institutional feature.

4.5 Empirical Estimates Under a Heterogeneous Capital Specification

4.5.1 Econometric Specification of Mechanization Bias and Heterogenous Capital

Specification A treats aggregate core capital $k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$ as a single regressor. This collapses the physical structure of the workplace into a scalar, obscuring whether wage pressure operates through the *level* of capital or through its *composition*—the relative weight of machinery and equipment (K^{ME}) versus nonresidential structures (K^{NRC}). Specification B decomposes the aggregate into two components: a structural base ($k_t^{\text{NRC}} = \log K_t^{\text{NRC}}$) and the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), which measures the mechanization gap between equipment and structures in log-levels. The disaggregated cointegrating equation takes the form:

$$y_t = \alpha + \theta_{\text{NRC}}\, k_t^{\text{NRC}} + \theta_\kappa\, \kappa_t + \theta_{\omega\kappa}\, \xi_t^{\omega\kappa} + \theta_{\kappa p}\, \xi_t^{p\kappa} + \boldsymbol{\gamma}'\mathbf{D}_t + u_t, \tag{88}$$

where $\xi_t^{\omega\kappa}$ denotes the FWL-orthogonalized effective wage–mechanization interaction and $\xi_t^{p\kappa}$ denotes the FWL-orthogonalized relative price–mechanization interaction. Both interaction terms

are residualized against the two base capital regressors $(k_t^{\text{NRC}}, \kappa_t)$ and deterministic step shifts $(\mathbf{D}_t)$ prior to estimation. The effective wage interaction under Set 2 is $\xi_t^{\omega\kappa} = \omega_{t-1}(1-\iota_{t-1}) \times \kappa_t$; the relative price interaction is $\xi_t^{p\kappa} = p_t^{\kappa} \times \kappa_t$, where p_t^{κ} denotes the relative asset price ratio. Non-standard fixed-b critical values are generated from a dedicated 100,000-pass Monte Carlo simulation engine that replicates the exact Specification B integration geometry—five independent $I(1)$ stochast ic trends, two simultaneous $I(1) \times I(1)$ interaction products, and joint FWL orthogonalization—matching the empirical sample sizes ($T = 94$ and $T = 77$).

Economic intuition of the decomposition. The parameter θ_{NRC} captures the long-run capacity yield of structures—factories, warehouses, commercial buildings—which embody extensive accumulation: expanding the physical envelope within which production occurs. The parameter θ_κ captures the additional capacity yield of machinery and equipment *above and beyond* what structures deliver. A positive θ_κ means that a wider mechanization gap—more equipment per dollar of structures—raises productive capacity independently of the total capital stock. The interaction $\theta_{\omega\kappa}$ asks whether effective wage pressure conditions this compositional margin. Under the Okishio-induced mechanization mechanism, rising wages force firms to substitute equipment for labor on the shop floor. If the resulting machinery expansion is defensive—driven by unit-cost containment rather than autonomous technical expansion—it raises the equipment-to-structures ratio without proportionally expanding output capacity, yielding $\theta_{\omega\kappa} < 0$.

Standard wage interactions fail at the intensive margin. Tables 16 and 17 in Appendix C present the estimation results using the unadjusted wage share across all 24 specifications. The structural base elasticity $\hat{\theta}_{\text{NRC}}$ remains stable and highly significant (0.47–0.61, $p < 0.01$), confirming that nonresidential structures anchor the cointegrating relationship regardless of wage measure or break treatment. The standalone mechanization gap $\hat{\theta}_\kappa$ is positive in nearly all models (0.34–1.11), indicating that equipment contributes to capacity above structures. The wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$, however, fails to achieve statistical significance in any $T = 94$ model, with point estimates ranging from -4.89 ($t = -1.58$) to $+7.95$ ($t = 1.33$). The standard wage interaction fails at the intensive margin for the same reason it fails at the aggregate level in Specification A: unadjusted wage shares conflate supervisory overhead with productive variable capital and ignore the post-Fordist diversion of investment toward intellectual property products.

Effective wage pressure and defensive mechanization at the intensive margin. Table 7 presents the main estimates using the effective wage pressure measure for the NVA basis. When effective wage pressure is scaled by $(1 - \iota_{t-1})$ and interacted with the mechanization gap κ_t, the wage–mechanization interaction $\hat{\theta}_{\omega\kappa}$ is negative and statistically significant across all eight models. In the preferred Clean Corporate Tier 1 NVA specification with Bai-Perron step shifts (Column 4), $\hat{\theta}_{\omega\kappa} = -7.93$ ($SE = 2.96$, $p < 0.05$), with a structural base elasticity of $\hat{\theta}_{\text{NRC}} = 0.60$ ($SE = 0.08$, $p < 0.01$) and a standalone mechanization gap of $\hat{\theta}_\kappa = 0.39$ ($SE = 0.29$). In the Shaikh-Tonak Tier 2 NVA specification without breaks (Column 7), the interaction strengthens to $\hat{\theta}_{\omega\kappa} = -17.05$ ($SE = 2.34$, $p < 0.01$), with $\hat{\theta}_{\text{NRC}} = 0.54$ ($SE = 0.01$) and $\hat{\theta}_\kappa = 0.33$ ($SE = 0.06$, $p < 0.01$). The Clean Corporate Tier 2 NVA models (Columns 5–6) yield $\hat{\theta}_{\omega\kappa} = -14.97$ ($SE = 2.01$, $p < 0.01$), showing the tightest standard errors in the table.

The negative sign of $\hat{\theta}_{\omega\kappa}$ provides direct evidence of defensive mechanization operating at the *intensive margin* of capital composition. Specification A established that effective wage pressure

depresses the aggregate capacity yield of total capital ($\hat{\theta}_1 = -0.30$). Specification B isolates the channel: wage pressure forces firms to widen the mechanization gap—substituting equipment for structures on the shop floor—but the capacity yield of that compositional shift is negative. The firm accumulates more machinery per dollar of structures, yet productive capacity does not expand proportionally, because the substitution is driven by cost containment rather than by genuine technical complementarity between new equipment and existing plant.[14]

Relative price interactions. The relative asset price interaction $\hat{\theta}_{\kappa p}$ is generally small and statistically insignificant. In the NFC Baseline and Clean Corporate Tier 1 models (Columns 1–4), it is statistically indistinguishable from zero. In the Shaikh-Tonak Tier 2 specification (Columns 7–8), $\hat{\theta}_{\kappa p} = -0.79$ ($SE = 0.32$, $p < 0.05$), suggesting that when equipment becomes relatively more expensive, the capacity yield of the mechanization gap declines. However, this result is fragile and does not hold across wage share measures. The Okishio mechanism operates through distributional conflict over newly created value, not through relative asset prices.

Cointegration inference. The effective wage models show strong evidence of cointegration. The Shin LM test never rejects the null of cointegration ($C_{\text{Shin}} \leq 0.03$, $p > 0.10$), and the Vogelsang-Wagner Type 2 Wald statistic remains well below critical thresholds ($S_{\text{Type 2}} \leq 15.37$). The Engle-Granger residual ADF test rejects the null of no cointegration at the 5% level in every break-adjusted model: $t_{\text{ADF}} = -4.58$ (Column 2), -4.60 (Column 4), -4.71 (Column 7), and -4.57 (Column 8). Without step shifts, the ADF test rejects at the 10% level across all $T = 94$ models ($t_{\text{ADF}} \leq -4.33$) and at the 5% level for the $T = 76$ Shaikh-Tonak Tier 2 model ($t_{\text{ADF}} = -4.71$, Column 7). The full cointegration triad—Shin, Vogelsang-Wagner, and ADF—confirms that the disaggregated capital composition, real output, and effective wage pressure form a stable long-run relationship.

Robustness. Including Bai-Perron step shifts absorbs historical level discontinuities, but leaves the sign and significance of the wage–mechanization interaction unchanged. In the Clean Corporate Tier 1 NVA model, controlling for breaks at 1944, 1977, and 2001 sharpens the ADF rejection from -4.34 ($p < 0.10$) to -4.60 ($p < 0.05$) while reducing the Vogelsang-Wagner statistic from 15.37 to 6.37. The wage interaction $\hat{\theta}_{\omega\kappa}$ moves from -11.08 (Column 3) to -7.93 (Column 4). This modest attenuation indicates that the step dummies absorb part of the interaction's raw variance. The interaction remains negative and significant at the 5% level, confirming that the compositional conditioning channel survives institutional break controls.

The structural base elasticity $\hat{\theta}_{\text{NRC}}$ falls consistently within the 0.51–0.61 band across wage share measures, closely tracking the aggregate $\hat{\theta}_0 = 0.58$ from Specification A. This implies the capacity yield of structures is a stable parameter of U.S. capital accumulation. The wage interaction $\hat{\theta}_{\omega\kappa}$ grows larger in absolute value as I remove supervisory overhead from the wage share: from -10.13 in the NFC Baseline (Column 2) to -7.93 in Clean Corporate Tier 1 (Column 4) to -16.76 in Shaikh-Tonak Tier 2 (Column 8). Removing managerial salaries isolates the productive variable

[14]The apparent difference in magnitude between Specification A ($\hat{\theta}_1 \approx -0.30$) and Specification B ($\hat{\theta}_{\omega\kappa} \approx -7.93$ to -17.05) reflects regressor scaling, not economic inconsistency. In Specification A, θ_1 multiplies $\omega_{t-1}(1-\iota_{t-1}) \times k_{\text{cap},t}$, where aggregate capital $k_{\text{cap},t} \approx 16$–$17$ in log-levels. In Specification B, $\theta_{\omega\kappa}$ multiplies $\omega_{t-1}(1-\iota_{t-1}) \times \kappa_t$, where the mechanization gap $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}} \approx 0.3$–$0.8$. The total marginal effect (coefficient $\times$ regressor scale) is comparable across both specifications.

capital driving the firm's cost-minimization decision, which sharpens the defensive mechanization effect.

Table 7: Specification B, Set 2: Effective Wage Interaction (NVA Basis)

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{\text{NRC}}$ (Structures k^{NRC})	0.61***	0.58***	0.60***	0.60***	0.54***	0.54***	0.54***	0.51***
	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.32*	0.39	0.32*	0.39	0.31***	0.31***	0.33***	0.06
	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.06)	(0.06)	(0.09)
$\hat{\theta}_{\omega\kappa}$ (Wage Interaction)	-12.77**	-10.13**	-11.08**	-7.93**	-14.97***	-14.97***	-17.05***	-16.76***
	(3.81)	(3.43)	(3.48)	(2.96)	(2.01)	(2.01)	(2.34)	(2.55)
$\hat{\theta}_{\kappa p}$ (Price Interaction)	0.18	0.92	0.69	1.41*	-0.31	-0.31	-0.79**	-0.71**
	(0.51)	(0.69)	(0.63)	(0.75)	(0.33)	(0.33)	(0.32)	(0.31)
Step Shifts ($D^{\text{step}}_{T_b}$)	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.03	0.03	0.02	0.02	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	15.37	6.37	8.96	8.96	10.28	8.70
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.34*	-4.60**	-4.24*	-4.24*	-4.71**	-4.57**
Bandwidth ratio b	0.185	0.173	0.193	0.181	0.208	0.208	0.208	0.208
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors ($k_t^{\text{NRC}}, \kappa_t$) and step dummies. The effective wage interaction is $\xi_t^{\omega} = \omega_{t-1}(1-\iota_{t-1}) \times \kappa_t$, where $\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ is the capital composition ratio (mechanization gap) and ι_{t-1} is the lagged IPP share of capital. Significance: $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

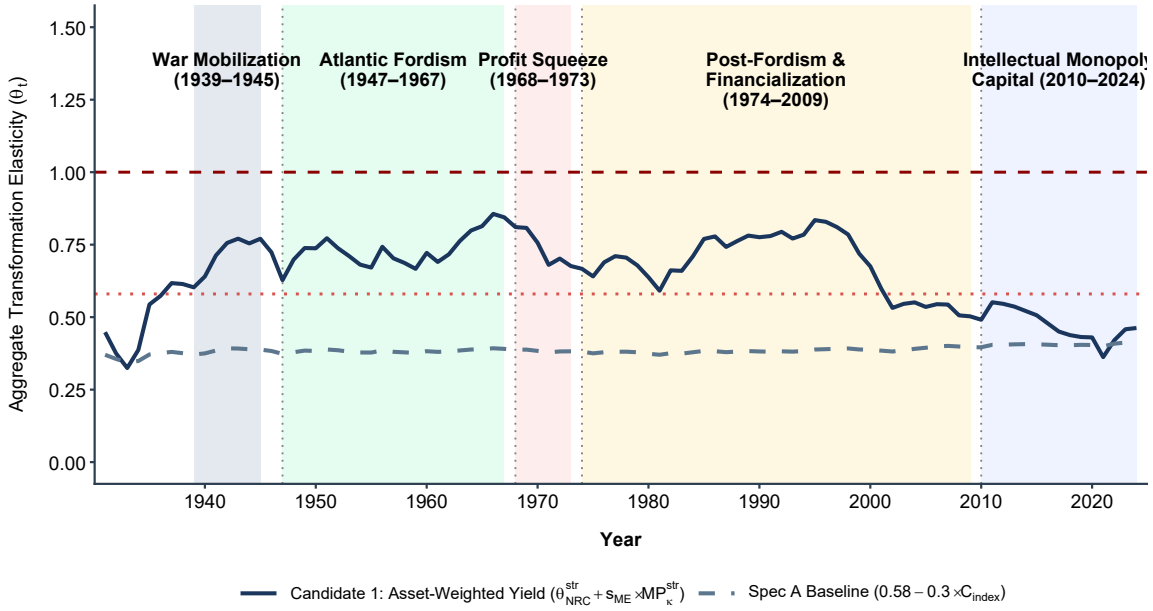
4.5.2 Transformation Elasticity from Heterogeneous Capital Specification and Capacity Utilization Estimates

The transformation elasticity along the historical path. Having established the structural asset-weighted transformation elasticity θ_t^{SpecB} in Equation (37) (Section 3.2), I evaluate its empirical trajectory using the estimated parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 7). The structural marginal capacity yield of mechanization $MP_{\kappa,t}^{\text{str}} = \beta_\kappa + \theta_{\omega\kappa} C_{\text{index},t} + \theta_{\kappa p} p_t^{\kappa}$ is driven directly by the state-dependent levels of effective wage pressure ($C_{\text{index},t}$) and relative asset prices (p_t^{κ}).[15]

Figure 13 plots $\theta_t^{\text{Aggregate}}$ against the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30 \cdot C_{\text{index},t}$) and the Harrodian knife-edge threshold ($\theta = 1.00$) under standard structural regime shading.

[15] Initial-observation centering ($\theta_\kappa^{\text{str}} = 0.39$) is an anchoring convention that identifies the baseline dimensionless yield at $t = 1931$.

Figure 13: Aggregate Transformation Elasticity ($\theta_t^{\text{Aggregate}}$), Specification B vs. Baseline

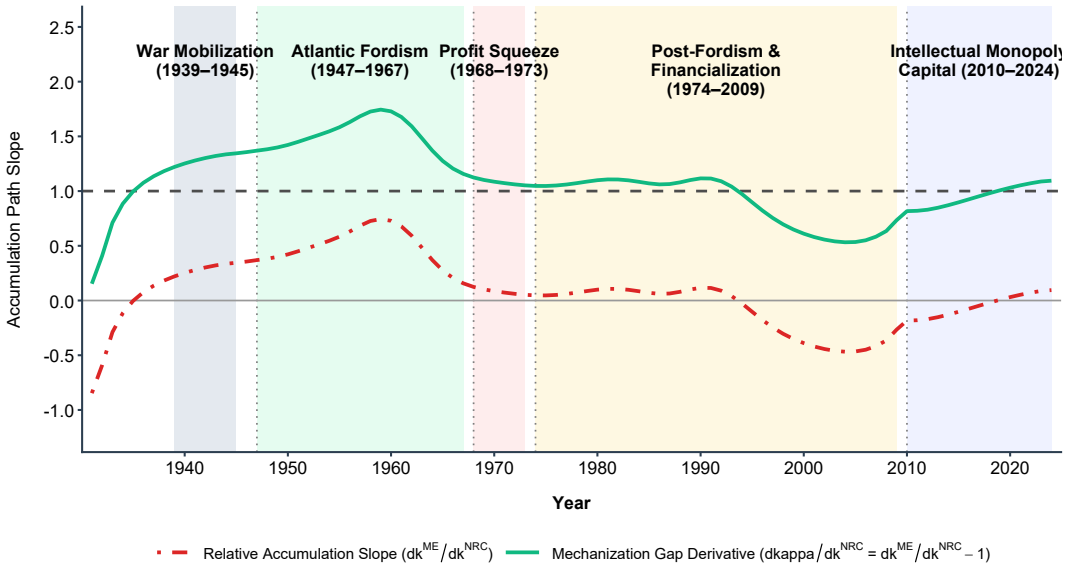


Note: Aggregate transformation elasticity $\theta_t^{\text{Aggregate}}$ (Navy line) computed according to Equation (37) using FWL structural parameters from the preferred Specification B model (Clean Corporate Tier 1 NVA, Set 2, Column 4 of Table 7). The Slate Blue line traces the dynamic Specification A baseline ($\theta_t^{\text{SpecA}} = 0.58 - 0.30\, C_{\text{index},t}$). The dotted horizontal line at $\theta = 0.58$ marks the constant base from Specification A. The dashed line at $\theta = 1.00$ marks the Harrodian knife-edge. Background shading indicates structural accumulation regimes.

Empirical comparison: Specification B disciplines the Specification A baseline. Figure 13 delivers two core empirical findings. First, the structural elasticity θ_t^{SpecB} tracks the dynamic Specification A path closely but swings across a wider range—$[0.32, 0.86]$ against $[0.37, 0.75]$. The two paths move together because both register the same underlying force: effective wage pressure depresses the marginal capacity yield of capital under either specification, and the estimated base elasticity of structures ($\hat{\theta}_{\text{NRC}} \approx 0.60$) sits close to Specification A's aggregate base ($\hat{\theta}_0 = 0.58$). The wider swing is the point of the disaggregation rather than a cost of it: Specification A averages the composition effect into a single scalar, while Specification B lets the equipment share move the elasticity directly, so the same distributive shock registers with more amplitude. Second, the Specification B trajectory remains strictly below the Harrodian knife-edge ($\theta < 1$) across the entire 1931–2024 sample, so aggregate capacity formation in the United States never leaves the sub-Harrodian regime.

The disaggregated capital structure explains what Specification A's aggregate elasticity averages over. During the Fordist era ($C_{\text{index},t} > 0.65$), high effective wage pressure drove negative marginal mechanization yields ($MP^{\text{str}}_{\kappa,t} < 0$). Firms substituted equipment for labor to contain unit labor costs, but because this expansion was defensive rather than technically autonomous, it depressed the aggregate capacity yield of total capital. In the Neoliberal period ($C_{\text{index},t} < 0.50$), declining wage pressure restored $MP^{\text{str}}_{\kappa,t}$ to positive territory, pulling $\theta_t^{\text{Aggregate}}$ back toward the structural base. Specification A collapses this compositional mechanism into a scalar interaction ($\hat{\theta}_1 = -0.30$); Specification B opens the intensive margin and confirms the Okishio mechanism empirically.

Figure 14: Empirical Accumulation Path Derivatives ($dk^{\text{ME}}/dk^{\text{NRC}}$ and $d\kappa/dk^{\text{NRC}}$)



Note: Empirical accumulation path derivatives estimated via non-parametric LOESS time-domain chain rule (span = 0.35). The green solid line measures the relative accumulation slope $dk_t^{\text{ME}}/dk_t^{\text{NRC}}$. The red dot-dashed line measures the mechanization gap derivative $d\kappa_t/dk_t^{\text{NRC}} = (dk_t^{\text{ME}}/dk_t^{\text{NRC}}) - 1$. The horizontal dashed line at 1.00 indicates balanced expansion where equipment and structures grow at identical rates.

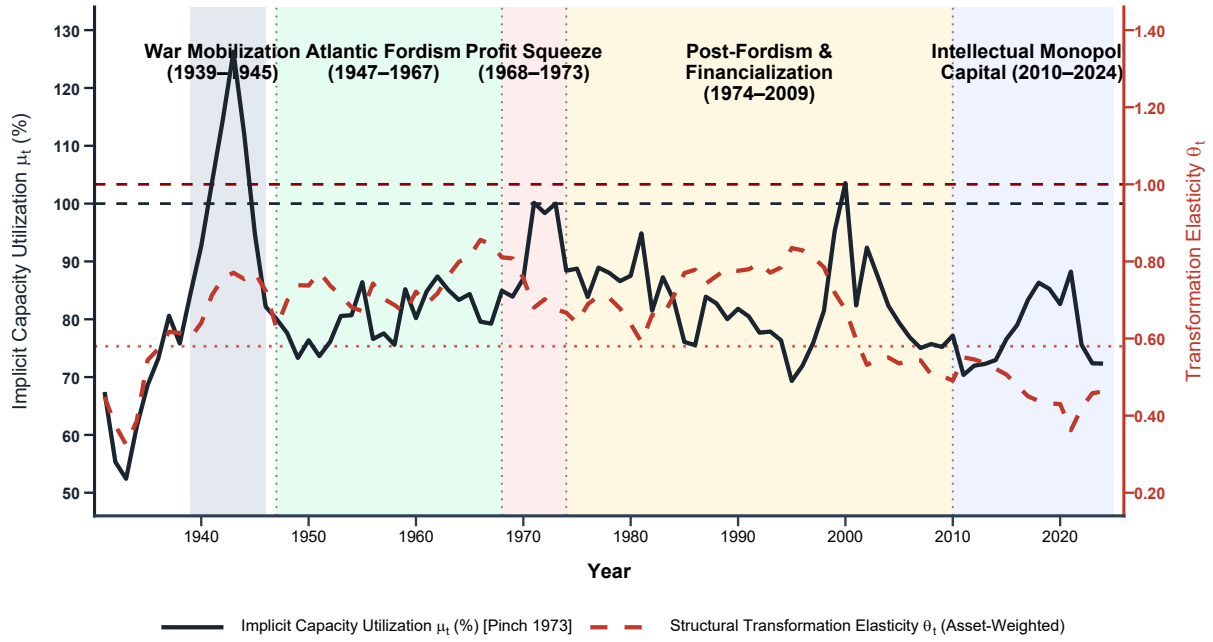
Empirical failure of the conventional Divisia derivative path. Figure 14 illustrates why the empirical total-derivative path $\theta_{\text{total},t}^{\text{Aggregate}}$ (Equation (36)) fails in practice. In the mid-1990s, corporate accumulation shifted away from physical machinery toward intellectual property products and financial assets, causing physical equipment growth to slow relative to structures ($dk_t^{\text{ME}}/dk_t^{\text{NRC}} < 1.00$). As $d\kappa_t/dk_t^{\text{NRC}}$ crossed zero, the Divisia denominator in Equation (36) approached zero, sending the raw total-derivative elasticity to spurious infinite spikes. As established in Section 3.2, this instability reflects the theoretical pathology of differentiating a value aggregate under structural accumulation shifts. The structural asset-weighted elasticity $\theta_t^{\text{Aggregate}}$ in Equation (37) purges these Divisia singularities, yielding a continuous, empirically well-behaved trajectory.

Driving Forces The inclusion of heterogeneous capital reveals a structural paradox in the post-war US economy: while the relative cheapening of machinery drove a massive shift in the physical composition of capital, the capacity-generating power of this mechanization was severely degraded by the distributive conflicts of the late 1960s and early 1970s. Figure 15 plots the implicit capacity utilization rate (μ_t) alongside the asset-weighted aggregate transformation elasticity (θ_t^{SpecB}). Unlike the static baseline elasticity of Specification A, the transformation elasticity here is not solely driven by the evolving mix of machinery and structures, and the macroeconomic environment mediating their installation. As the physical accumulation of machinery and equipment accelerated relative to nonresidential structures, the aggregate transformation elasticity collapsed, severing the Harrodian link between accumulation and potential output. This structural degradation is not an artifact of index-number aggregation but the direct result of shifting marginal capacity yields. To unpack the mechanics of this collapse, Figure 16 illustrates the two primary institutional driving forces

behind this choice of technique: the secular decline in relative asset prices ($p_t^\kappa = P_t^{\text{ME}}/P_t^{\text{NRC}}$) and the episodic spikes in effective distributive pressure ($C_t = \omega_{t-1}(1-\iota_{t-1})$).

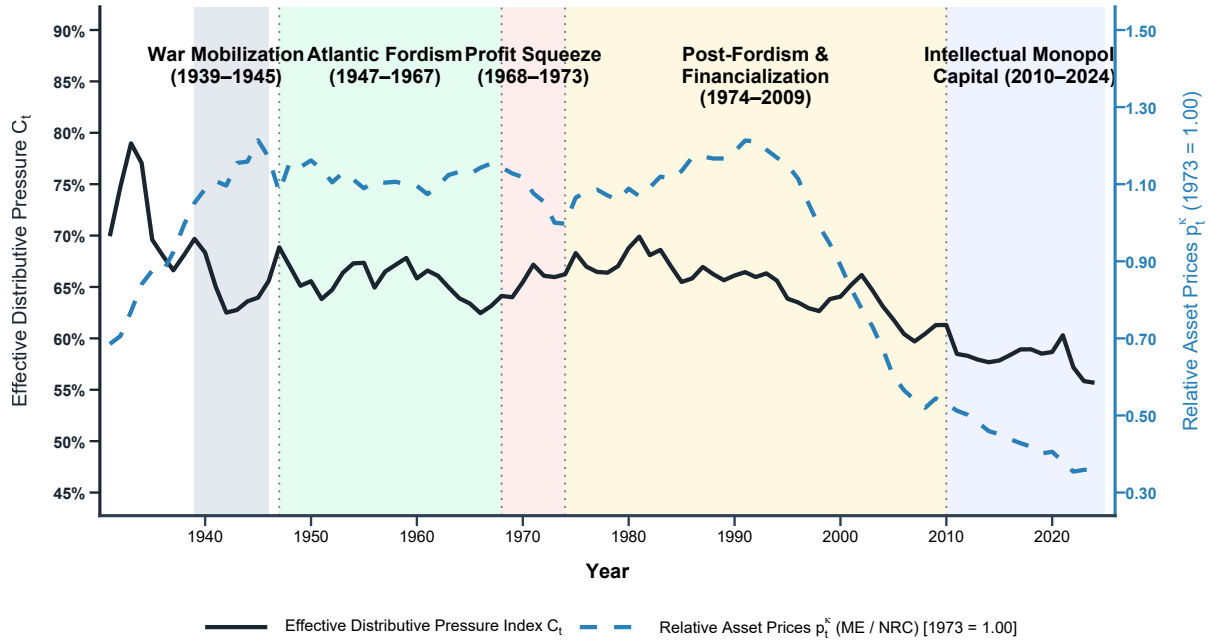
Consider a given period during the Golden Age of American Capitalism (e.g., the 1950s) where the structural yield of nonresidential structures is $\theta_{\text{NRC}}^{\text{str}} = 0.60$ and the equipment share is $s_{\text{ME}} = 0.30$. Under a stable distributive environment, if the marginal capacity yield of mechanization is $MP_\kappa = 0.40$, the transformation elasticity yields an order of magnitude around 0.72 ($\theta^{\text{SpecB}} = 0.60 + 0.30 \times 0.40$). However, as the post-war era progressed, machinery became structurally cheaper, as can be appreciated on the downswing tendency of p_t^κ in Figure 16, pushing the physical equipment share up toward 0.50. Likewise, the Profit Squeeze (1968–1973) produced a sharp spike in effective distributive pressure (C_t). Because the interaction term $\theta_{\omega\kappa}$ is negative, this heightened wage pressure pushing firms into a defensive tactical choice of technique, increasing their efforts for capital-using labor-saving mechanization, i.e. Marx-biased technical change (Basu, 2022). Consequently, the marginal capacity yield of this machinery dropped (say, to $MP_\kappa = 0.20$). In the post-Fordist era the aggregate elasticity then stands at 0.70 ($\theta^{\text{SpecB}} = 0.60 + 0.50 \times 0.20$). The arithmetic is the point: a 20-percentage-point rise in the physical share of machinery leaves the aggregate elasticity essentially where it started, because the yield per unit of that machinery fell by half. Cheapening delivered the equipment; defensive investment under falling profitability drained its capacity-generating power. This mechanism explicitly limits accumulation through stagnation tendencies towards over-accumulation due firms over-shooting their investments biased towards mechanization not to expand capacity, but to replace increasingly expensive labor, resulting into an over-accumulation regime crisis due the decreasing returns to mechanization and increasingly harder to transform capital accumulation to productive capacities within the political economy of full employment with intense labor militancy (Kalecki, 1943).

Figure 15: Implicit Capacity Utilization (μ_t) and Asset-Weighted Transformation Elasticity ($\theta_t^{\text{Aggregate}}$)



Note: Reconstructed capacity utilization rate anchored at the 1973 pinch year alongside the dynamic aggregate transformation elasticity derived from Specification B. The Harrodian knife-edge ($\theta = 1.00$) and the Specification A baseline ($\theta = 0.58$) are provided as reference lines.

Figure 16: Effective Distributive Pressure (C_t) and Relative Asset Prices (p_t^{κ})



Note: The figure illustrates the institutional driving forces behind the choice of technique. The left axis tracks the effective distributive pressure to mechanize (C_t), while the right axis tracks the relative price of machinery and equipment to nonresidential structures (p_t^{κ}, indexed to 1973 = 1.00).

4.6 Chile's Empirical Evidence: Peripheral Caiptalism under External Constraint

4.6.1 Econometric Strategy: Threshold Cointegrating Multivariate Polynomial Regression

In core economies, capital accumulation converts into productive capacity through domestic technical choice. Rising wages induce capitalists to substitute labor with machinery, expanding capacity unless surplus is diverted into unproductive accumulation (Rotta, 2018). In peripheral economies like Chile, this mechanism depends on international liquidity. Because capital equipment (K_t^{ME}) cannot be produced domestically, labor-saving technical change requires imported machinery. When foreign-exchange reserves collapse, peripheral firms hit a structural bottleneck. Rather than adjusting smoothly through domestic prices, the capacity-generating effect of capital accumulation breaks down: wage pressure cannot induce mechanization because firms cannot acquire the foreign exchange needed to import machinery.

To evaluate this state-dependent breakdown, I estimate a Cointegrating Multivariate Polynomial Regression (CMPR) using Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2024) with an endogenously identified threshold regime. The estimating equation models log real GDP (y_t) as a cointegrating function of non-residential construction (k_t^{NR}), the capital composition gap (κ_t), distributional conflict (dist_{t-1}), the state-dependent external constraint (D_t^{γ}), and relative price shocks ($\tilde{p}_t^{\perp}$):

$$y_t = \alpha + \theta_1 k_t^{\text{NR}} + \theta_2 \kappa_t + \theta_3(\text{dist}_{t-1} \cdot \kappa_t) + \theta_4(\kappa_t \cdot D_t^{\gamma}) + \theta_5(\text{dist}_{t-1} \cdot \kappa_t \cdot D_t^{\gamma}) + \beta_p \tilde{p}_t^{\perp} + u_t \tag{89}$$

where y_t is log real GDP (2003 CLP); $k_t^{\text{NR}} = \ln K_t^{\text{NR}}$ is the log net stock of non-residential structures; $\kappa_t = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$ is the capital composition gap between intensive machinery and extensive structures; dist_{t-1} is the lagged distributive variable, evaluated across four permutations: net wage share ($\omega_{t-1}^{\text{net}}$), gross wage share ($\omega_{t-1}^{\text{gross}}$), net rate of exploitation (e_{t-1}^{net}), and gross rate of exploitation (e_{t-1}^{gross}); $D_t^{\gamma} = \mathbb{I}(Z_t \leq \gamma)$ is a binary regime indicator equal to 1 when the composite balance-of-payments index (Z_t) falls below threshold γ; $\tilde{p}_t^{\perp}$ is the standardized relative capital price, orthogonalized against physical capital stocks via Frisch-Waugh-Lovell (FWL) regression; and u_t is a stationary $I(0)$ error.

The partial derivative of output y_t with respect to the capital composition gap κ_t yields the marginal transformation elasticity $\theta(\text{dist}, D_t)$:

$$\frac{\partial y_t}{\partial \kappa_t} = \theta_2 + \theta_3 \text{dist}_{t-1} + \theta_4 D_t^{\gamma} + \theta_5(\text{dist}_{t-1} \cdot D_t^{\gamma}) \tag{90}$$

The coefficients carry direct political-economy interpretations:

1. **Extensive Construction Elasticity (θ_1)**: Measures capacity generated by non-residential construction.

2. **Unconstrained Mechanization Elasticity (θ_2)**: Measures capacity generated by intensive capital composition under normal liquidity ($D_t^{\gamma} = 0$).

3. **Domestic Induced Innovation (θ_3)**: Captures wage-led induced technical change under unconstrained liquidity, where worker bargaining power forces capitalists to substitute labor with machinery.

4. **Balance-of-Payments Structural Shift (θ_4)**: Measures the direct collapse in mechanization capacity when external constraints bind ($D_t^{\gamma} = 1$).

5. **Peripheral Distributional Inversion (θ_5)**: Captures the breakdown in wage-led technical change under foreign-exchange starvation. A negative estimate ($\theta_5 < 0$) demonstrates that when reserves collapse, wage pressure cannot induce mechanization because imported machinery is rationed.

Constructing the Balance-of-Payments Threshold Index (Z_t) I define the composite balance-of-payments index $Z_t(p)$ as a convex linear combination of two standardized macroeconomic indicators:

$$Z_t(p) = p\,\tilde{M}_t + (1-p)\,\tilde{S}_t, \quad p \in [0,1] \tag{91}$$

where $\tilde{S}_t = \text{scale}(F_t/M0_t)$ measures central bank foreign reserves ($F_t = e_t \cdot \text{IR}_t$) relative to base money ($M0_t$) (Nievas and Piketty, 2025), and $\tilde{M}_t = \text{scale}(\text{Imbalance}_t)$ measures the capital-goods import demand imbalance.

Because the structural weight p and threshold level γ are unknown *a priori*, I estimate $(\hat{p}, \hat{\gamma})$ via a joint profile Sum of Squared Residuals (SSR) search:

$$(\hat{p}, \hat{\gamma}) = \arg \min_{p \in \mathcal{P},\, \gamma \in \Gamma(p)} \mathrm{SSR}(p, \gamma) = \arg \min_{p \in \mathcal{P},\, \gamma \in \Gamma(p)} \sum_{t=1}^{T} \hat{u}_t(p, \gamma)^2 \tag{92}$$

The weighting grid $\mathcal{P} = \{0.0, 0.1, 0.2, \ldots, 1.0\}$ steps across the unit interval in increments of 0.1. For each candidate weight p, the threshold grid $\Gamma(p)$ evaluates trimmed quantiles $\Gamma(p) = \{\gamma_q(p) : q \in [0.15, 0.85],\, \Delta q = 0.05\}$. The global minimizer selects $\hat{p} = 0.2$ ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$) and $\hat{\gamma} = 0.425$ (or $\hat{\gamma} = 0.379$ in parsimonious specifications), confirming that international reserve cover ($\tilde{S}_t$) dominates the empirical weighting in isolating the structural bottleneck.

Bai-Perron Structural Break Benchmark To confirm that $D_t^{\hat{\gamma}}$ captures a genuine balance-of-payments constraint rather than a deterministic calendar trend, I compare the threshold model against a benchmark specification using Bai and Perron (2003) endogenous structural breaks:

$$y_t = \alpha + \theta_1 k_t^{\mathrm{NR}} + \theta_2 \kappa_t + \theta_3(\mathrm{dist}_{t-1} \cdot \kappa_t) + \theta_4(\kappa_t \cdot D_t^{\mathrm{BP}}) + \theta_5(\mathrm{dist}_{t-1} \cdot \kappa_t \cdot D_t^{\mathrm{BP}}) + \beta_p \tilde{p}_t^{\perp} + u_t \tag{93}$$

where $D_t^{\mathrm{BP}} = \mathbb{I}(t \geq T_b)$. Dynamic search over candidate break dates $T_b \in [0.15T, 0.85T]$ identifies two historical break points:

- **Net Distribution ($\omega^{\mathbf{net}}, e^{\mathbf{net}}$)**: Identifies $T_b = 1987$, marking structural consolidation under the Büchi reforms following the 1982 crisis, where wage suppression and export promotion restored financial stability.
- **Gross Distribution ($\omega^{\mathbf{gross}}, e^{\mathbf{gross}}$)**: Identifies $T_b = 1999$, marking the Asian Financial Crisis in Chile, when the Central Bank floated the Chilean peso and abandoned exchange-rate crawling bands.

Achieving lower SSR and stronger cointegration statistics under $Z_t \leq 0.425$ rules out deterministic calendar trends in favor of true balance-of-payments state dependence.

Relative Price Controls and Asset Deflators The relative price control accounts for international purchasing power shifts affecting imported capital equipment. In Chilean historical national accounts, separate long-run deflators for structures (P_t^{NRC}) and machinery (P_t^{ME}) do not exist back to 1940. Capital stock harmonization constructs a single unified investment price deflator ($P_{K,t}$, staged as `Pk_2003base`) by splicing historical ClioLab deflators (1940–2010) with official Central Bank of Chile deflators (2003 base year).

I evaluate two complementary proxies:

1. **Baseline Real Exchange Rate Control ($\tilde{p}_t^{\kappa,\mathbf{RER}}$)**: The standardized log Real Effective Exchange Rate ($\ln \mathrm{TCREAL}_t$, rebased to 2003 = 1.0), capturing external purchasing power shocks that alter the domestic price of imported machinery (K_t^{ME}).
2. **Robustness Asset-to-Output Ratio ($p_t^{\kappa,P_Y/P_K}$)**: The log ratio of aggregate GDP deflator ($P_{Y,t}$) to unified capital price index ($P_{K,t}$):

$$p_t^{\kappa,P_Y/P_K} = \ln\left(\frac{P_{Y,t}}{P_{K,t}}\right) \tag{94}$$

measuring relative shifts between aggregate output prices and capital goods prices.

To prevent relative price volatility from collinearizing with physical capital stocks $(k_t^{\mathrm{NR}}, \kappa_t)$, I extract the orthogonal price component $\tilde{p}_t^{\perp}$ via a Frisch-Waugh-Lovell (FWL) auxiliary regression prior to estimation:

$$\tilde{p}_t^{\kappa,\mathrm{RER}} = c_0 + c_1 k_t^{\mathrm{NR}} + c_2 \kappa_t + \tilde{p}_t^{\perp} \tag{95}$$

The residual vector $\tilde{p}_t^{\perp}$ enters the estimating equation additively. Preliminary grid tests introducing interacted price terms $(\kappa_t \cdot \tilde{p}_t)$ caused severe multicollinearity (VIF > 88), destroying matrix invertibility $(X'X)^{-1}$ and inflating Monte Carlo critical bounds ($|t^*| > 19$). Keeping $\tilde{p}_t^{\perp}$ as an additive control isolates physical capital composition without polluting the structural elasticity.

The estimation sample and its limitations. Two data limitations define the estimation sample window, and I acknowledge them here rather than resolve them. First, the pre-1942 monetary and external series do not provide a coherent basis for the threshold variables: under the gold-exchange standard the reserve-cover ratio records a legal requirement rather than a binding external constraint, and the monetary base responds to convertibility rules and fiscal pressure rather than to the balance of payments. Second, the historical ClioLab price and terms-of-trade series, the stock of international reserves, and the money supply that anchor the external-constraint variables all end in 2010. Splicing them to contemporary sources falls outside the scope of this chapter.[16]

The Kemmerer regime, 1925–1931. Decree-Law No. 486 (21 August 1925) created the Central Bank of Chile (BCCh), and the Organic Law and Monetary Bill of 14 October 1925 (Decree-Law No. 606) placed it under a gold-exchange standard: the peso was pegged to sterling and the Bank was required to hold a minimum gold cover of 50 percent against note issue, with the remainder of the reserve requirement met in foreign exchange. Reserve coverage of the money supply rose from 27 percent at the start of 1925 to exactly 50 percent by year-end. The system's credibility was anchored abroad: it depended on sterling stability and steady copper revenues, functioning only as long as export receipts replenished central bank reserves.

Collapse and fiscal dominance, 1931–1941. The external anchor failed during the Great Depression. Between January 1930 and July 1931, the Bank lost more than half of its gold holdings defending the peg, suspending convertibility on 21 July 1931; the United Kingdom's departure from gold two months later erased the value of remaining sterling reserves. The subsequent decade combined fiscal dominance—the Bank monetized deficits as a treasury agent—with multiple exchange rates administered by the *Comisión de Cambios Internacionales* and parallel currency markets. Once convertibility was suspended, the cover ratio ceased to operate as a statutory discipline and became instead an indicator of systemic fragility. During a currency panic, commercial banks hoard base money at the central bank while the money multiplier contracts, making the reserves-to-M_0 ratio the true measure of the Bank's capacity to absorb speculative attacks. The denominator follows economic exposure rather than the letter of the 1925 law: while the statute defined requirements against note issue alone, commercial banks in a panic seek to convert their reserve deposits, making total central-bank liabilities (M_0) the relevant exposure. Under fiscal dominance, however, even this fragility indicator no longer anchored the external sector: the monetary base responded to deficit

[16] The capital-stock and distributive panels of sections 4.1.5 and 4.1.6 span 1900–2024, but the threshold estimation discards 1900–1941 and 2011–2024. The institutional history below explains why the first discard separates two regimes rather than truncating a single one.

financing and exchange controls rather than balance-of-payments flows, rendering the 1931–1941 cover series a measure of fiscal distress rather than a stable balance-of-payments state variable.

The 1942 regime shift. Law No. 7,200 (18 July 1942), together with Law No. 7,434, rewrote the Central Bank's Organic Law: the reserve-cover rule was formally abandoned, the mandate shifted from gold convertibility to the promotion of national production, and foreign exchange was allocated through centralized credit ceilings and state budgets. Working alongside CORFO (Law No. 6,640, 1939), which directed credit allocation toward import-substituting industrialization, the Bank financed development planning rather than currency pegs. Macroeconomic stability thereafter derived from internal planning rather than external gold credibility. Econometrically, pooling the pre-1942 data with the post-1942 period violates cointegration: the structural regime shift introduces severe non-stationary drift, generating massive finite-sample size distortions and explosive Monte Carlo t-statistics ($|t| > 15$) that destroy matrix invertibility. Restricting the estimation window to 1943–2010 ($T = 68$) is therefore both institutionally necessary and econometrically sound, isolating a single, stable cointegrating regime across distinct historical regimes: under state-led industrialization and its violent rupture through neoliberalization.

4.6.2 Empirical Findings for Chile: Truncated Mechanization

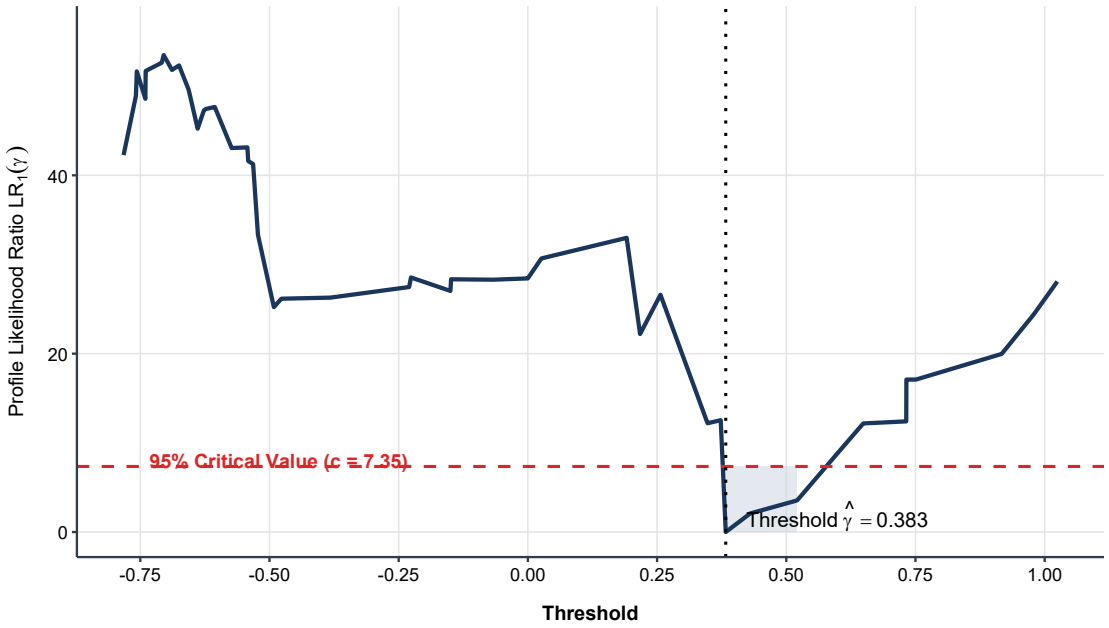
Table 9 compares the parsimonious regime-shifting specification across four distributive permutations: Net Wage Share (ω^{net}), Gross Wage Share (ω^{gross}), Net Rate of Exploitation (e^{net}), and Gross Rate of Exploitation (e^{gross}). Across all four variants, the composite external threshold ($Z_t \leq 0.425$) produces super-consistent long-run cointegration ($t_{\text{ADF}} = -3.81^*$ to -3.89^*, $CT_{\text{IM}} = 8.34$ to 15.20).

Gross vs. Net Measures in Peripheral Capitalism. In the United States (Section 4.3), Net Value Added (NVA) is the necessary theoretical basis for evaluating distribution. Because Consumption of Fixed Capital (CFC) rises mechanically during intensive capital-deepening, evaluating distribution on a Gross Value Added (GVA) basis introduces a spurious negative correlation between capital intensity and wage shares. In the US core, class struggle takes place strictly over newly created net value ($v + s$), and purging CFC isolates a cost-minimizing mechanization response to distributive pressure.

In the capitalist periphery, however, gross distributive measures (Columns 2 and 4) achieve superior empirical fit and cointegration (SSR $= 0.275$, $CT_{\text{IM}} \approx 8.34$) compared to net measures (SSR $= 0.294$, $CT_{\text{IM}} \approx 15.10$). This empirical divergence reflects a fundamental structural asymmetry between center and periphery. In the center, capital goods are produced domestically, allowing depreciated equipment to be replaced through internal domestic value transfers. In the periphery, capital accumulation relies on *imported machinery and equipment* (K^{ME}). Deducting official accounting depreciation (CFC) under linear conventions fails to capture the unencumbered gross cash flow (Π^{gross}) required by peripheral firms to obtain foreign exchange and finance machinery imports. Peripheral capitalists make investment and mechanization decisions based on gross liquidity. Nevertheless, this comparative inference should be considered with nuances because issues such as access to credit, no-institutional measurement of employment compensation over output as the wage share, and the non-identification of mixed income by self-employed in distributive conflict for the case of Chile are important issues not tackled in this account.

Likelihood Ratio and Threshold Identification. Figure 17 plots Hansen (2000) profile Likelihood Ratio statistic $LR_1(\gamma) = [\text{SSR}(\gamma) - \text{SSR}(\hat{\gamma})]/\hat{\sigma}^2$ across candidate threshold values $\gamma \in [\gamma_{0.15}, \gamma_{0.85}]$ for the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. The profile displays a single, sharp minimum at $\hat{\gamma} = 0.383$ (within the $[0.383, 0.521]$ 95% confidence interval encompassing the benchmark threshold $\hat{\gamma} = 0.425$), matching a binding foreign-exchange reserve cover threshold. Inverting the likelihood ratio statistic at the exact 95% analytical critical boundary $c(0.05) = -2\ln(1 - \sqrt{0.95}) = 7.35$ yields a tight non-parametric confidence interval. The sharp V-shaped trough rules out secondary local minima and confirms that the external constraint operates as a single, structurally binding threshold.

Figure 17: Profile Likelihood Ratio Sequence for the Peripheral External Constraint (Chile, 1943–2010)



Note: Profile Likelihood Ratio statistic $LR_1(\gamma)$ calculated for the preferred gross wage share model (ω^{gross}, Column 2 of Table 9) over the trimmed evaluation domain $\Gamma_n = [\gamma_{0.15}, \gamma_{0.85}]$ ($\pi = 0.15$). The horizontal dashed red line marks the exact analytical 95% critical boundary ($c(0.05) = 7.35$). The vertical dotted line marks the identified profile minimum. The shaded region depicts the non-parametric 95% confidence interval $[0.383, 0.521]$ defined by $LR_1(\gamma) \leq 7.35$.

Distributive Conflict Preferred Measure: Center versus Periphery. Comparing the wage share model in Column (2) (ω^{gross}) with the rate of exploitation model in Column (4) (e^{gross}) illuminates how external financial constraints alter the transmission of class conflict into capital composition. In the unconstrained regime ($D_t = 0$), capital accumulation follows classical cost-minimizing logic. Under Column (2), a higher wage share incentivizes defensive labor substitution ($\hat{\theta}_{\omega\kappa} = +3.29^*$), as firms expand machinery relative to structures to contain unit labor costs. Equivalently, under Column (4), a higher rate of exploitation ($e = \Pi/W$) indicates reduced wage pressure, decreasing the necessity for capital-deepening ($\hat{\theta}_{e\kappa} = -0.69^*$).

Under the binding foreign-exchange regime ($D_t = 1$), this cost-minimizing mechanism collapses. When the external constraint becomes binding ($Z_t \leq 0.425$), peripheral capitalists hit a hard

import constraint ($\bar{\kappa}_t$). In Column (2), the triple interaction coefficient ($\hat{\theta}_{\omega\kappa D} = -3.15^*$) virtually eliminates the positive wage-led mechanization response ($+3.29 - 3.15 = +0.14$). Because machinery and equipment (K^{ME}) must be imported, foreign-exchange rationing prevents peripheral firms from translating domestic wage pressures into shop-floor labor substitution. Unable to extract a distributive dividend through technical change, capitalists attempt to protect profit margins through price markups.

State-dependent thresholds versus exogenous historical breaks. To verify that D_t^{γ} captures a structural macroeconomic constraint rather than an arbitrary trend break, Appendix Tables 21 and 22 (Appendix D) benchmark the threshold model against Bai and Perron (2003) endogenous calendar breaks. The Bai-Perron procedure identifies two break dates: 1987 under net wage shares (ω^{net}) and 1999 under gross measures (ω^{gross}, e^{gross}). The 1987 break marks the post-1982 debt-crisis restructuring—fiscal austerity, privatization, and wage suppression. The 1999 break marks a change in monetary regime: the Central Bank abandoned the crawling peg and its accompanying capital controls, moving to a free float with open capital mobility in the wake of the Asian Financial Crisis.

Comparing specifications confirms that the composite threshold $Z_t \leq 0.425$ dominates calendar-time breaks across all metrics, achieving lower residual variance (SSR = 0.275 against 0.292) and stronger cointegration rejections ($t_{\text{ADF}} = -3.89^*$ against -2.98). Two conditions must meet for the constraint to bind: a wide gap between effective imports and the capacity to finance them, and foreign-exchange reserves too thin for the Central Bank to defend the currency. Neither condition is tied to a calendar date, which is why a state-dependent gate outperforms a break at 1987 or 1999.

Table 8 shows that the binding constraint ($D_t = 1$) does not always announce itself the same way. During the ISI era (1943–1984), under fixed or administered pegs, reserve depletion ($S_t \to 0$) forced devaluations and structural inflation (Sunkel, 1958; Prebisch, 1981). During the 2006–2010 commodity super-cycle, under a free float and inflation targeting, the same gate closed through markup expansion instead.

These findings suggest that in financialized peripheral capitalism the external bottleneck shifts its pressure valve from domestic price inflation toward *re-primarization, real exchange rate appreciation (Dutch Disease), and rentier surplus drainage* (Palma, 2019; Ahumada, 2019). Capital goods import requirements (K^{ME}) for extractive mining projects, combined with foreign profit remittances, truncate the translation of accumulation into productive capacity while the Central Bank still meets its inflation target. Disentangling the transmission between inflationary rationing (1943–1984) and rentier-drain re-primarization (2006–2010) would require a monetary-regime-specific specification that this chapter does not estimate.

Model selection. The comparative results in Table 9 establish Column (2)—the gross wage share specification (ω^{gross}) under the composite balance-of-payments threshold ($Z_t \leq 0.425$)—as the primary empirical model for Chile. Column (2) achieves the global minimum residual variance (SSR = 0.275), and its distributive channel is the one that maps onto the United States baseline, which is what makes the center–periphery comparison in Section 5 an argument about the same relation under different constraints rather than a juxtaposition of two unrelated estimates.

4.6.3 Transformation and Capacity Utilization under Financial Subordination

Balance-of-payments subordination inverts the classical choice of technique. When foreign exchange is slack ($Z_t > 0.425$), a higher wage share accelerates mechanization, as cost minimization predicts

Table 8: Institutional Manifestations of External Constraint Regimes (Chile, 1943–2010)

Historical Period	Regime Status (D_t^γ)	Duration	Institutional Manifestation of the External Constraint
1943–1984	Binding ($D_t = 1$)	42 years	ISI Reserve Scarcity & Structural Inflation: Established under Law 7,200 foreign-exchange budgeting and fixed/crawling pegs; reserve depletion ($S_t \to 0$) under wage pressure forced maxidevaluations, cost-plus markups, and structural inflation spirals.
1985–2005	Slack ($D_t = 0$)	21 years	Export Consolidation & Reserve Buffering: Post-1982 debt-crisis and recovery nontraditional export expansion, Concertación democratic transition, large FDI inflows, and central bank foreign reserve accumulation.
2006–2010	Binding ($D_t = 1$)	5 years	Commodity Boom, Rentier Surplus Drain & Reprimarization: Under a free-floating exchange rate and inflation targeting (3%), FX starvation did not trigger inflation; instead, high import demand ($\tilde{M}_t$) and massive MNC profit repatriation (DL 600) manifested as a rentist surplus drain and Dutch Disease re-primarization.

Notes: Classification based on the optimal profile threshold $\hat{\gamma} = 0.425$ evaluated over the standardized composite balance-of-payments index $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$. Regime $D_t = 1$ denotes a binding balance-of-payments constraint where foreign-exchange starvation forces peripheral firms into a corner solution ($\bar{\kappa}_t$). Total sample size $T = 68$ (1943–2010 effective cointegrating window), yielding 47 binding years (69.1%) and 21 slack years (30.9%).

($\hat{\theta}_3 = +0.9583$). When foreign exchange binds, the interaction shift ($\hat{\theta}_5 = -0.9518$) cancels the base sensitivity and leaves a slope of +0.0065. Peripheral capitalists keep the incentive to substitute machinery for labor but lose the means, because the machinery is imported.

I evaluate the aggregate elasticity from the preferred CMPR IM-OLS model (Column 2 of Table 9, gross wage share and composite threshold $Z_t \leq 0.425$) and integrate the capacity law of motion, $g_{Y^p,t} = \theta_t^{\text{Agg}} g_{K,t}$, forward and backward from the 1981 Ffrench-Davis full-capacity peak ($\mu_{1981} = 1.00$). The state-dependent elasticity is

$$\theta_t^{\text{Agg}} = \hat{\theta}_1 + s_{\text{ME},t} \cdot MP_{\kappa,t} = \hat{\theta}_1 + s_{\text{ME},t} \left[\hat{\theta}_2 + \hat{\theta}_3\, \omega_{t-1}^{\text{gross}} + D_t^\gamma (\hat{\theta}_4 + \hat{\theta}_5\, \omega_{t-1}^{\text{gross}})\right] \tag{96}$$

where $\hat{\theta}_1 = +0.6970$ is the base elasticity of non-residential structures, $s_{\text{ME},t} = K_t^{\text{ME}}/(K_t^{\text{ME}} + K_t^{\text{NR}})$ is the current-price share of machinery and equipment in core capital, $\kappa_t = \ln(K_t^{\text{ME}}/K_t^{\text{NR}})$ the composition gap, $\omega_{t-1}^{\text{gross}}$ the lagged gross wage share, and $D_t^\gamma = \mathbb{I}(Z_t \leq 0.425)$ the binding-state indicator. The remaining coefficients are $\hat{\theta}_2 = -0.1293$, $\hat{\theta}_3 = +0.9583$, $\hat{\theta}_4 = +0.6645$, and $\hat{\theta}_5 = -0.9518$, so the marginal product of the composition gap takes one form in each state:

$$MP_{\kappa,t}^{\text{slack}} = -0.1293 + 0.9583\, \omega_{t-1}^{\text{gross}} \quad (D_t^\gamma = 0) \tag{97}$$

$$MP_{\kappa,t}^{\text{bind}} = +0.5352 + 0.0065\, \omega_{t-1}^{\text{gross}} \quad (D_t^\gamma = 1) \tag{98}$$

In slack periods a higher wage share raises the capacity yield of mechanization. In binding periods the slope collapses from +0.9583 to +0.0065, and foreign-exchange rationing replaces the distributive state as the governor of technique choice.

Table 9: Comparative Estimates Across Distributive Variable Permutations (Chile, 1943–2010)

	Wage Share (ω)		Rate of Exploitation (e)	
	Net (ω^{net}) (1)	Gross (ω^{gross}) (2)	Net (e^{net}) (3)	Gross (e^{gross}) (4)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.79***	0.82***	0.80***	0.82***
	(0.07)	(0.08)	(0.07)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.12	-1.48*	0.15	0.85
	(0.39)	(0.54)	(0.25)	(0.38)
$\hat{\theta}_{\text{dist}\cdot\kappa}$ (Distributive Interaction)	-0.01	3.29*	0.02	-0.69*
	(0.22)	(1.07)	(0.09)	(0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	0.34	1.52**	0.15	-0.73
	(0.16)	(0.45)	(0.08)	(0.29)
$\hat{\theta}_{\text{dist}\cdot\kappa D}$ (Triple Interaction)	-0.17	-3.15*	0.12	0.67*
	(0.23)	(1.04)	(0.10)	(0.22)
Regime Threshold (D_t)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	No	No	No	No
CT_{IM} [H_0: Cointegration]	15.20	8.39	14.99	8.34
t_{ADF} [H_0: Unit Root]	-3.81*	-3.81*	-3.84*	-3.89*
Bandwidth ratio b	0.654	0.654	0.654	0.654
Sample Size T	68	68	68	68

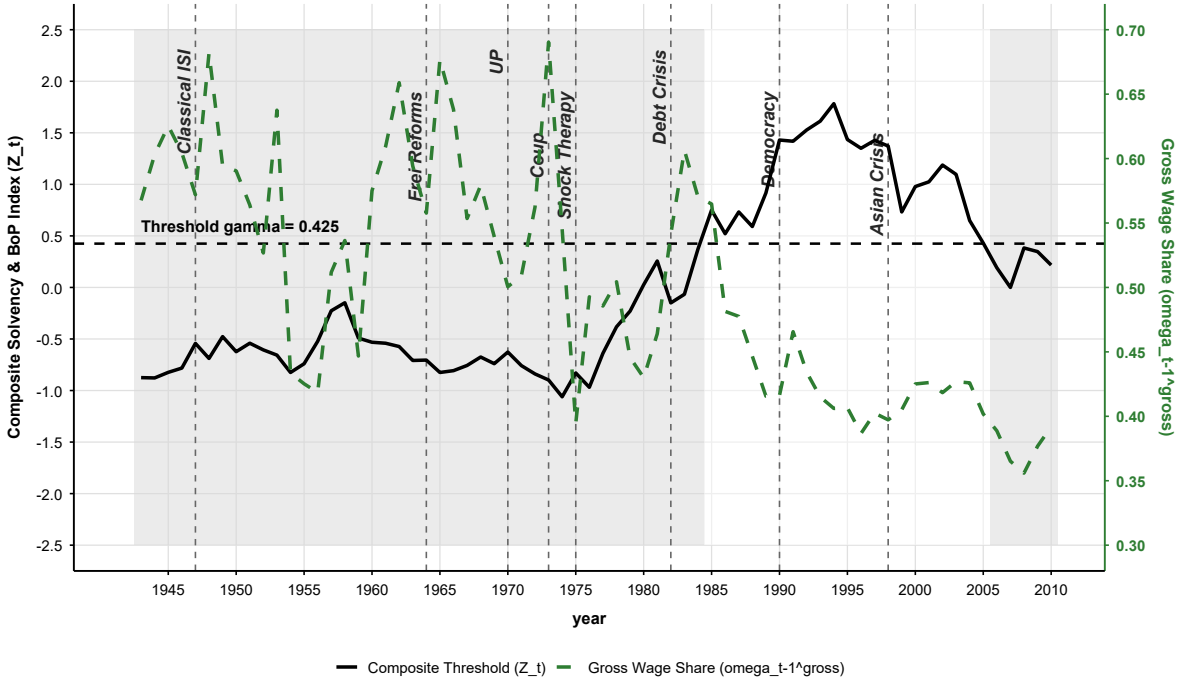
Notes: Dependent variable: y_t (log real GDP). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2024). All models specify the ultra-parsimonious balance-of-payments threshold ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t \leq 0.425$) over the effective sample 1943–2010 ($T = 68$). Columns (1) and (2) evaluate Net (ω^{net}) and Gross (ω^{gross}) wage shares; Columns (3) and (4) evaluate Net (e^{net}) and Gross (e^{gross}) rates of exploitation. Significance stars ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$) are benchmarked against 10,000-pass Monte Carlo simulation critical bounds under profile search.

Section 3.3 predicted exactly this pair of equations. Proposition 4 says that once the constraint binds, the composition gap is pinned at $\bar{\kappa}_t$ and the first-order condition stops governing technique choice; the only distributive channel left is the slope of the feasible gap, $(m_c c_k - m_w)/\nu$. The estimated residual slope of $+0.0065$ is consistent with $m_c c_k \approx m_w$, the conspicuous-consumption drain and the wage-goods import drain roughly offsetting each other. Proposition 6 supplies the second half: at the corner the marginal capacity payoff is strictly positive, because the firm would like to mechanize further and cannot, and the significant intercept shift $\hat{\theta}_4 = +0.6645$ is that payoff entering the capacity equation, the empirical counterpart of a positive Kuhn–Tucker multiplier on the external constraint. Binding is therefore not a state in which mechanization yields nothing; it is a state in which it yields something but responds to nothing domestic. Proposition 5 predicts a discrete shift of the elasticity at the threshold; in the data the shift appears less as a level drop than as the loss of the wage slope, with the binding path running flat through distributive shocks that move the slack path. The gap between nominal profitability and truncated technique is the blocked surplus of Definition 4, and Corollary 1 names the adjustment channel that remains once mechanization is rationed, price markups.

Figure 18 plots the composite index against the lagged gross wage share, and the Unidad Popular episode is the clearest observation in the sample. Between 1970 and 1973, redistribution raised the gross wage share to its historical peak of $\omega_{1972} = 0.6903$, about twenty percentage points up. Over the same years, credit blockade and capital flight drove reserves to $\tilde{S}_t \to -2.5$ and the index

to $Z_t = -1.06$, deep inside the binding domain. Figures 20 and 19 show that the elasticity did not respond: Equation (98) governed technique choice and held θ_t^{Agg} flat near 0.83 through the wage expansion. The wage expansion could not become productive capacity, and per Corollary 1 it became unrealized demand and structural inflation instead (Kalecki, 1955; Sunkel, 1958). Utilization collapsed to 0.72 by 1975.

Figure 18: Chilean Composite External Constraint (Z_t) and Lagged Gross Wage Share ($\omega_{t-1}^{\text{gross}}$)



Note: Primary left axis (solid black line) tracks the composite balance-of-payments threshold variable $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$, with the optimal threshold $\hat{\gamma} = 0.425$ marked by the horizontal dashed line. Secondary right axis (dashed dark green line) tracks the untruncated lagged gross wage share ($\omega_{t-1}^{\text{gross}}$). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark historical institutional breakpoints.

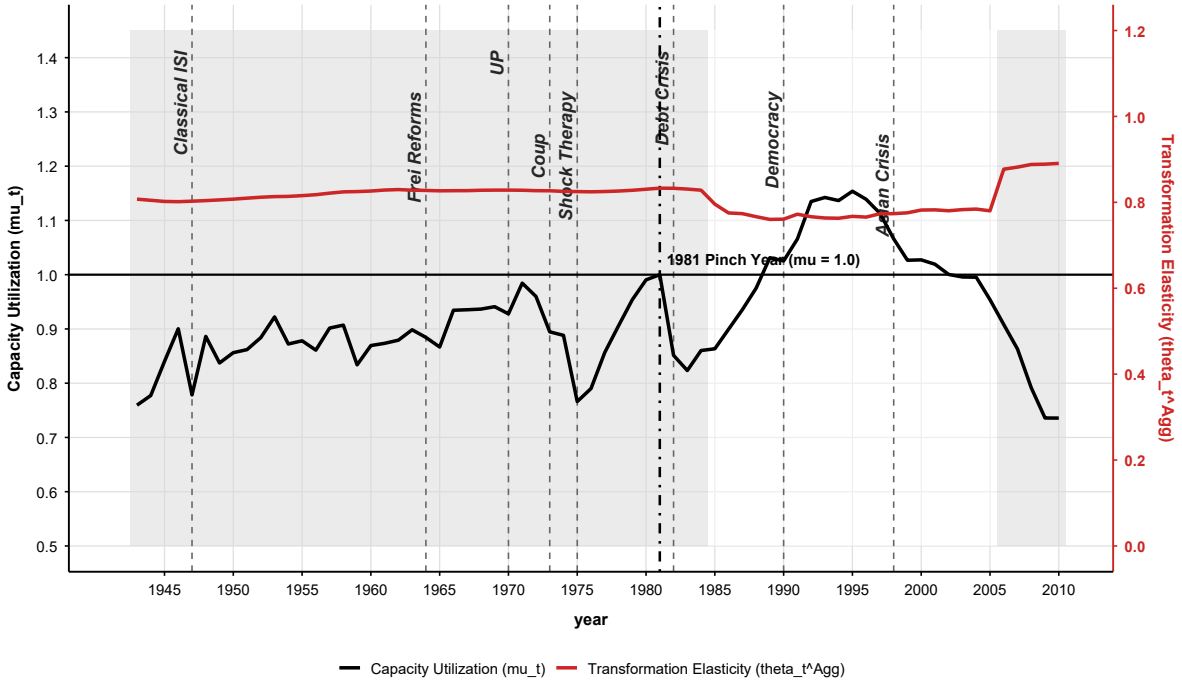
Based on Ffrench-Davis (2018) account on over-heating pre-debt crisis in 1982, I recover the order of magnitude for productive capacitieis from the recursion

$$\ln Y_t^p = \ln Y_{t-1}^p + \theta_t^{\text{Agg}} \left(\ln K_t^{\text{tot}} - \ln K_{t-1}^{\text{tot}} \right), \quad Y_{1981}^p = Y_{1981} \implies \mu_{1981} = 1.00 \tag{99}$$

and divide realized GDP by it to obtain utilization. Figure 19 plots the two series on dual axes and Figure 20 the elasticity alone. Five regimes organize the path, and the external constraint separates them. The early CORFO expansion (1942–1946) combined postwar demand above normal capacity ($\mu_{1946} = 1.12$) with slack foreign exchange and elasticities near 0.85. The ISI stop-go crises (1947–1957) reversed both: recurring shortages forced stabilization cycles that pulled utilization to $\mu_{1955} = 0.84$. The socialist transition (1970–1975) combined redistribution with credit blockade, and utilization bottomed at 0.72 after the 1975 shock therapy. The debt boom and crash (1975–1982) ran the sequence in reverse, with inflows carrying utilization to the 1981 peak and the crisis erasing it within a year ($\mu_{1982} = 0.68$). The recovery and commodity super-cycle (1984–2010) absorbed idle capacity rather than creating new capacity, holding μ_t between 0.85 and 0.98 through 1997. In

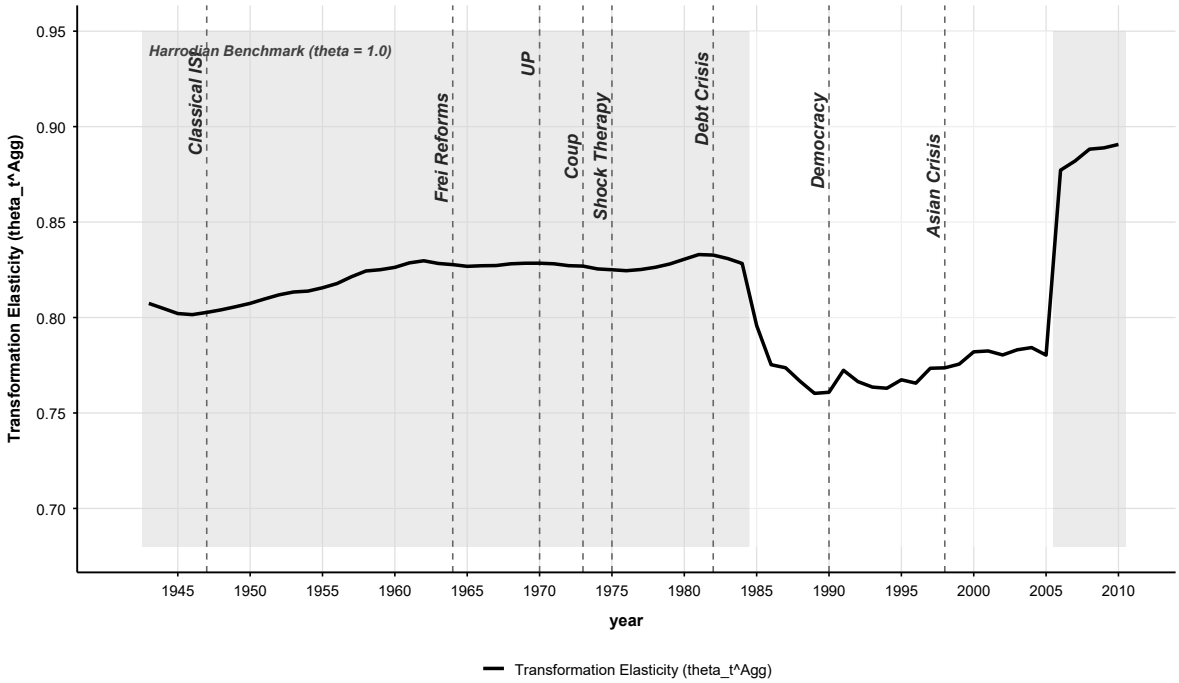
four of the five regimes the turning point is external rather than distributive, the asymmetry the threshold specification was built to detect.

Figure 19: Chilean Capacity Utilization (μ_t) and Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Reconstructed capacity utilization rate μ_t (solid black line, left axis) anchored at the 1981 Ffrench-Davis pinch year ($\mu_{1981} = 1.00$) alongside the state-dependent aggregate transformation elasticity θ_t^{Agg} (solid firebrick line, right axis). Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark institutional breakpoints.

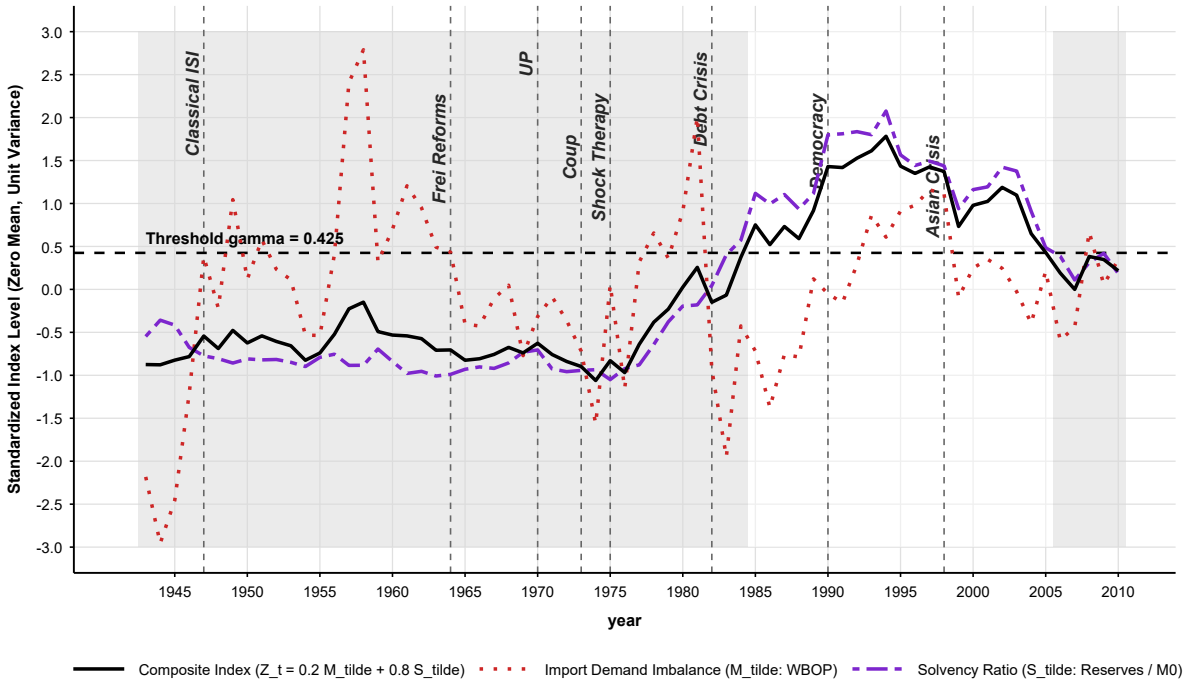
Figure 20: Standalone Chilean Aggregate Transformation Elasticity (θ_t^{Agg})



Note: Standalone trajectory of the aggregate transformation elasticity θ_t^{Agg} (solid firebrick line) on an uncompressed vertical scale (0.68 to 0.95). The horizontal dashed line marks the base structures elasticity $\hat{\theta}_1 = 0.6970$. Background light gray shading marks binding balance-of-payments regimes ($Z_t \leq 0.425$). Vertical dashed lines mark institutional breakpoints.

Figure 21 decomposes the index into reserve cover and import demand imbalances, and the two components show that the constraint changes its institutional form across regimes. Under the fixed and crawling pegs of the ISI period (1943–1973), reserve exhaustion forced sharp devaluations and structural inflation (Sunkel, 1958), in line with the periodization of Ffrench-Davis (2002) and Ahumada (2019). Under the post-2000 free float with inflation targeting, binding conditions produced no runaway inflation; they produced extractivist re-primarization, real appreciation, and profit repatriation by multinational mining (Palma, 2019; Ahumada, 2019). Capital goods imports for extractive projects, together with dividend outflows, truncated the conversion of accumulation into broad-based domestic capacity.

Figure 21: Decomposition of the Chilean External Constraint Index (Z_t Components)



Note: Decomposed ingredients of $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$: standardized Central Bank solvency ratio $\tilde{S}_t$ (reserves to base money $M0$, purple twodash line), WBOP import demand imbalance $\tilde{M}_t$ (red dotted line, thickness 1.3), and composite index Z_t (solid black line). Horizontal dashed line marks the threshold $\hat{\gamma} = 0.425$. Background light gray shading marks binding regimes ($Z_t \leq 0.425$).

Two caveats qualify the reading of these regimes. The first concerns the classifier. States are assigned by a single linear combination, $Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$, with the weight and the threshold chosen by grid search over the profile sum of squared residuals and a hard cutoff at $\hat{\gamma} = 0.425$. This is a deterministic classification in the sense of Hansen (2000), not a probabilistic regime model: there are no transition probabilities and no persistence parameter, and after the search $(\hat{p}, \hat{\gamma})$ enter the elasticity path as if known. The simulation bounds charge for the threshold search but not for the weight grid, so regime uncertainty is confined to the interval $[0.380, 0.472]$ and is not propagated into θ_t^{Agg}. A Markov-switching or smooth-transition extension would assign a binding probability to each year and separate persistent shifts from transient bites; I leave that to future work.

The second caveat concerns what the hard classification nonetheless captures that a periodization cannot. D_t^{γ} does not switch only between the major regimes. Within the ISI era the index crosses the threshold at the frequency of the stop-go cycle: import surges in the expansion phase push it below the cutoff before the crisis, and post-crisis stabilizations (devaluation, import compression, wage suppression) briefly lift it back above. The 1998–99 tightening and the 2006–10 mining boom are intra-cycle bites inside eras the conventional periodization labels slack. The constraint operates at cycle frequency; the five regimes organize the path at the lower frequency at which the institutional form of the constraint changes, from inflation valve under administered exchange rates to re-primarization valve under a free float. That double frequency is also why the state-dependent indicator outperforms calendar breaks at 1987 or 1999: the breaks mark changes in the form, and the threshold tracks the bites.

The center–periphery comparison follows. In the United States, where capital equipment is

produced domestically, wage pressure induces defensive mechanization ($\hat{\theta}_1 < 0$, Section 4.3). In Chile, where machinery must be imported, wage pressure induces mechanization ($\hat{\theta}_3 = +0.9583$) only while foreign exchange is unconstrained. When reserves collapse, the constraint binds and truncates the choice of technique. The distributive dividend that Okishio's viability condition promises is not unavailable in principle in the periphery; it is unavailable whenever the balance of payments binds.

5 Discussion: The Political Economy of Capacity Utilization in the Center and the Periphery

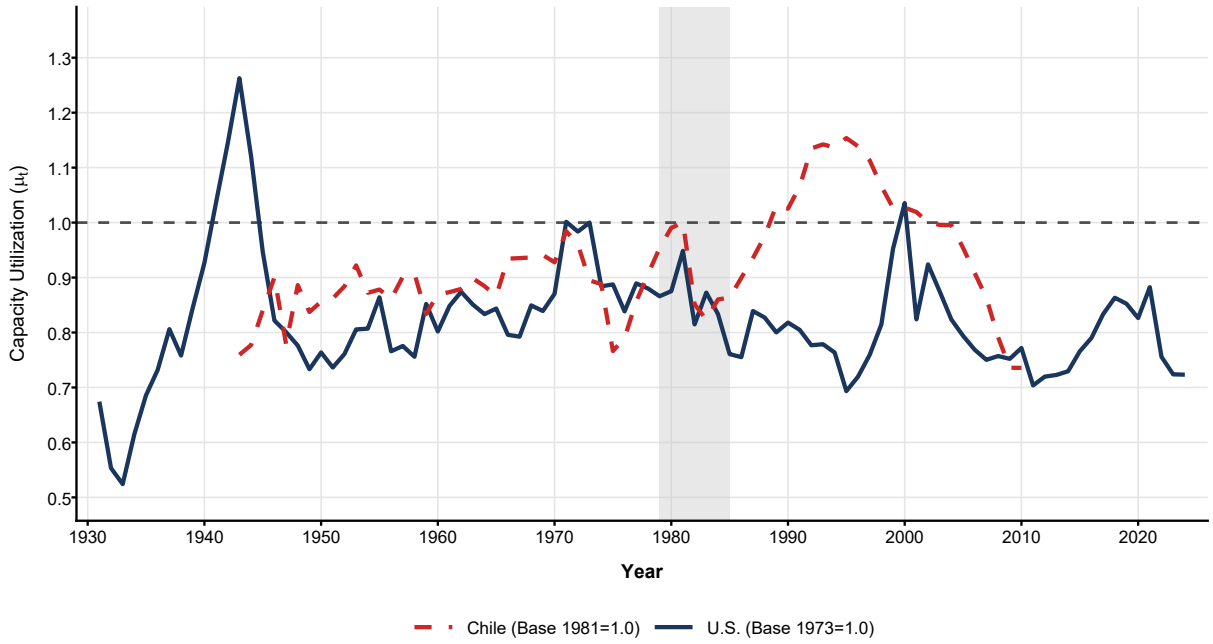
Section 4 established that the transformation elasticity θ_t sits strictly below unity in both the center and the periphery. Within a single-sector framework, the persistent tendency toward overaccumulation is a structural disproportionality between capital accumulation and its transformation into productive capacity, yielding systemic disequilibrium and chronic idle capacity. Capacity formation is the supply-side dimension of this process; its macroeconomic realization remains mediated through the rate of capacity utilization (μ_t).

While $\theta < 1$ is the shared premise of unbalanced growth, the historical trajectories of μ_t reveal fundamentally distinct crisis formations across the global divide. In the United States, utilization crises stem from the exhaustion of institutional class compromises and the failure of effective demand. In Chile, stagnation and crisis originate in the external truncation of capital-import capacity and the physical strangulation of the productive apparatus.

The Relational Dynamics of Idle Capacity. Figure 22 places the reconstructed implicit capacity utilization rates (μ_t) for the U.S. non-financial corporate sector (1931–2024) and the Chilean economy (1940–2010) on a single axis.[17]

[17]The U.S. series is anchored at the 1973 pinch year ($\mu_{1973} = 100\%$), while the Chilean series is anchored at the 1981 full-capacity peak ($\mu_{1981} = 100\%$). Despite differing base years, the absolute levels of μ_t are directly commensurable as measures of the engagement of productive capacity by effective demand and external constraints.

Figure 22: The Political Economy of Idle Capacity: Center vs. Periphery (1931–2024)



Note: Implicit capacity utilization (μ_t) for the U.S. (solid navy) and Chile (dashed crimson). The shaded band highlights the early 1980s global overaccumulation crisis (1979–1985), resolved via financialization in the center and external subordination in the periphery.

Neither economy in Figure 22 converges toward a normal rate of utilization in the long run ($\mu = 100\%$); persistent idle capacity remains the structural norm of decentralized accumulation. The volatility of μ_t in Chile vastly exceeds that of the United States. This higher amplitude directly tracks the structural asymmetry between a diversified, reserve-currency center capable of dampening demand swings and a dependent periphery vulnerable to sudden international liquidity collapses; it is not an econometric anomaly. Installed gross fixed capital (K_t) cannot vanish during downturns, so peripheral output collapses, such as the supply shocks of 1975 and the 1982 debt crisis, translate into abrupt contractions of utilization. The shaded window (1979–1985) captures a global overaccumulation crisis and structural turning point: the Volcker interest-rate shock triggered a severe domestic realization drop in the center ($\mu_{1982} \approx 82\%$) while ending Chile's debt-led boom ($\mu_{1981} = 100\%$) and driving peripheral utilization into a severe macro-historical trough. This common conjuncture was subsequently resolved through radically divergent regime shifts: debt-led financialization in the center, and a violent transition toward neoliberalization and external subordination in the periphery.

The Center: Realization Failure and Intellectual Monopolies. In the United States, the dynamics of μ_t are governed by the continuous mediation of distributive conflict and the shifting composition of capital. Because capital goods are produced domestically, the transformation of accumulation into capacity (θ) is not externally constrained; while stagnation and crisis may have multifaceted causes, they ultimately manifest as an issue of *realization*. Under Atlantic Fordism, the institutionalized class compromise absorbed growing productive capacity via mass consumption (Aglietta, 2000). High wage shares, alongside negligible diversion of investment toward intangible

assets like intellectual property, meant that consumer demand effectively matched expanding physical capacities, sustaining μ_t within a high, stable corridor (85–105%). While a commitment to full employment, Keynesian policymaking, and demand-inducing militarism masked secular overaccumulation tendencies, distributive conflict exerted cost-minimizing pressures that forced capitalists to engage in mechanization, configuring a contradictory utilization regime.

When the profit squeeze emerged in the late 1960s, μ_t fell to 76.2% in 1975 and bottomed at 74.1% in 1982 during the Volcker disinflation. In his decomposition of this postwar U.S. profit rate collapse, Weisskopf (1979) pinned the crisis on the "Rising Strength of Labor" (RSL), treating the "Realization Failure" (RF) variant—a drop in μ_t—as a secondary, survey-driven symptom. The Federal Reserve Board data left the physical conversion of accumulation into capacity unexamined. Here, μ_t is evaluated as an endogenous residual of θ_t. The Fordist profit squeeze was not just a distributional fight over a fixed pie; it was a structural breakdown in θ_t itself. When effective wage pressure peaked, θ_t fell to 0.32. The subsequent fall in μ_t directly reflected that distributive conflict degrading the capacity-generating power of capital. Post-Fordism dismantled the institutional foundations of wage-led growth to resolve this crisis. The capitalist class restored profitability through deunionization and wage suppression. This structural wage suppression nevertheless fundamentally undermined the domestic realization of productive capacity over the long run, an effect only partially offset by debt-led consumption (Stockhammer, 2012; Vidal, 2019). Average utilization failed to return permanently to its Fordist peaks, exceeding the 100% benchmark temporarily only during the speculative tech-investment bubble of 2000 ($\mu_{2000} \approx 103\%$). The center's idle capacity records secular stagnation after financialization and its shift toward the predominance of intangible-asset accumulation (Rikap, 2021; Rotta, 2018).

The Periphery: Dual Strangulation and Export Reprimarization. The terminal crisis of the inward-oriented regime in the peripheral Chilean economy cannot be explained by the balance-of-payments constraint alone, nor by the domestic profit squeeze in isolation. As Prebisch (1981) fundamentally articulated, the structural condition of the periphery is defined by its subordinate integration into the global division of labor. Because technological development inherently necessitates imported capital goods (K^{ME}) for late-industrializing economies, peripheral capitalist formations face a rigid structural ceiling: the transformation of accumulation into productive capacity is perpetually subject to the limits of external constraints. However, this external constraint did not operate in a distributive vacuum. Historical analyses of the Chilean developmental state emphasize that early import-substitution industrialization (ISI) was subsidized by a massive rural-to-urban demographic transition, providing an elastic urban reserve army *à la* Lewis (1954) that depressed real wages and temporarily offset chronic foreign exchange shortages (Polanco, 2021). Yet, as this rural surplus was exhausted by the late 1950s, the structural limits of the ISI model were exposed. The urban working class consolidated and distributive conflict intensified, culminating in the aggressive, wage-led redistribution of the *Unidad Popular* (1970–1973).

In a closed economy, the 20-percentage-point rise in the gross wage share—from roughly 49% in 1970 to a historical peak of 69.03% in 1972—would have simply induced rapid labor-saving mechanization. In peripheral capitalism, however, the internal profit squeeze and the external balance-of-payments constraint locked together. The threshold cointegrating model formalizes why this distributive impulse failed to induce mechanization: as it rose, the composite balance-of-payments index fell deep into the binding domain ($Z_t = -1.06 < \hat{\gamma} = 0.425$). The external constraint truncated the choice of technique: peripheral capitalists retained the microeconomic

incentive to substitute machinery for labor but lost the macroeconomic means. The working class's success in capturing a larger share of the surplus increased import demand for wage-goods, leaving insufficient foreign exchange to import capital goods. The cost of this "dual strangulation" appears in capacity utilization. When the *Unidad Popular* assumed power, μ_t operated at a constrained baseline of 94.1% in 1969. The initial redistributive boom drove μ_t to an unsustainable cyclical peak of 98.4% in 1971. Deprived of the foreign exchange required to import machinery—exacerbated by credit blockades, capital flight, and the 1972 bosses' lockout (de Vylder, 1976)—the economy lost more than 26 percentage points of utilization in just four years, reaching roughly 72% by 1975. The unconverted distributive surplus manifested as a blocked surplus, generating structural inflation and severe supply bottlenecks rather than expanding productive capacity.

The post-1975 resolution did not rebuild domestic manufacturing. While the unconstrained international liquidity conditions and copper booms of the 1990s temporarily pushed utilization to hyper-capacity levels ($\mu > 110\%$ during the 1993–1996 peak), leading to overheating and the Asian Crisis during the 1997-1998 period. In the aftermath, trade liberalization and financial opening ultimately locked the accumulation regime into premature de-industrialization and export reprimarization (Palma, 2019). Under the 2000s commodity super-cycle, external solvency was stabilized through primary extractive rents rather than domestic industrial capacity (Torres and Ahumada, 2022; Ahumada et al., 2026); high import leakages and multinational profit repatriation once again depressed capacity utilization to roughly 73% by 2010. From an institutional design standpoint, this trajectory yields a critical lesson: redistributive programs in the periphery cannot succeed in isolation from strategic foreign-exchange regime and monetary policy articulated with effective cross-borders of capital controls (Ffrench-Davis, 2018).

The composite index Z_t and the threshold $\hat{\gamma} = 0.425$ are estimated via a deterministic hard classification. This binary gate captures the frequency with which the constraint binds, not smooth, probabilistic regime shifts. It captures the "stop-go" shocks of the ISI era and the intra-cycle tightness of the 2000s mining boom rather than positing a single macro-historical break, but does not provide granular identification within regimes. Within these limits, the classification captures the discrete severing of the transformation of accumulation into capacity whenever external solvency collapses in the periphery. Future research requires backward and forward extension of time-series and exploration of other econometric techniques, such as Markov-switching modelling, to assign a time-varying threshold for regime classification. Applying this double-constraint metric to comparative panel datasets across other commodity-dependent or semi-peripheral spaces remains a promising avenue for future research.

Synthesis: Scar Tissue of the Global Divide. This comparative analysis demonstrates that the contemporary political economy of capacity utilization is fundamentally dictated by an economy's position within the global division of labor as originally formulated by Prebisch (1981). In the current stage of *"geriatric capitalism"* (Vidal, 2019), chronic idle capacity is not a mere anomaly, but the structural manifestation of overlapping stagnationist tendencies across the uneven development of the capitalist world system. In the center, this takes the form of an exhausted accumulation regime prone to secular stagnation: unable to mobilize idle physical capacity through wage-led demand, partially offset through debt-led consumption and later diverted to asset inflation and accumulation of intangible unproductive assets (Rotta, 2018; Rikap, 2021). In the periphery, the condition of financial subordination express through differential channels and the specific integration to the world market but with a common ceiling imposed by external constraints, insofar this become

binding due financial imbalanced caused by a low degree of import capacities with respect to the effective demand for imports from domestic markets (McCombie and Thirlwall, 1994).

The dead labour embedded in under-utilized productive capacities is crystallized past labour denied the living labour necessary for its valorization. This idle fixed capital belongs to an accumulation regime unable to sustain its own reproduction. In the center, a domestic social contract has been dismantled in the name of profitability and financialized rent extraction; in the periphery, an asymmetrical global order conditions growth on structural subordination. The same past labour remains idle under these distinct conditions.

6 Conclusion

Capacity utilization has been treated as something an economy reports. This chapter has treated it as something an economy produces. Once the denominator of the ratio is recognized as the outcome of a transformation—accumulated capital becoming usable productive capacity through a choice of technique made under distributive conflict—utilization stops being a primitive and becomes a residual of that transformation. The elasticity θ is what has to be recovered first, and everything the ratio can be made to say depends on recovering it without assuming its value.

The apparatus for that recovery came from two frameworks that had not previously been joined at this point. Okishio's account of decentralized accumulation establishes that private investment commitments generate disproportionality as a structural norm rather than as a friction around proportional reproduction (Okishio, 1961, 2022). Vidal's account of the internally divided firm establishes that the organization converting capital into operating capacity cannot maximize control and cooperation at once, and settles instead into arrangements that sustain production without resolving the conflict inside it (Vidal, 2022, 2019). Together they rule out any automatic passage from accumulation to capacity, which is what makes θ an object with a determinate value worth estimating rather than a parameter to be normalized to one.

The estimates say two things about the center. American capacity formation has stayed strictly below the Harrodian knife-edge across 1931–2024 under both the homogeneous and the heterogeneous capital specification: accumulation persistently outruns the capacity it creates. And the elasticity has drifted upward across the postwar period, from 0.371 in 1960 to 0.429 in 2024, for a reason that does not vindicate the accumulation regime that produced it. Effective wage pressure fell from 0.698 to 0.504 over those decades, partly because the wage share declined and partly because investment migrated into intellectual property, which now exceeds a quarter of fixed capital. A rising transformation elasticity coincided with falling utilization, which stood at 93.8% in 2024 after a trough of 74.1% in 1982. Capital became somewhat better at building capacity precisely as the distributive settlement that would have realized that capacity was dismantled.

The periphery shows the same relation cut short from outside. For Chile over 1943–2010, a composite balance-of-payments index built from reserve cover and the import-financing gap identifies a threshold at $\hat{\gamma} = 0.425$ by profile likelihood ratio search (Hansen, 2000). In the unconstrained state a rising wage share induces mechanization as cost-minimization predicts, with an estimated interaction of $+3.29$. When the threshold binds, a triple interaction of -3.15 removes nearly all of that response. Chilean capitalists under foreign-exchange rationing keep the incentive to substitute machinery for labor and lose the means, because the machinery is imported. The Unidad Popular period is the clearest case: a twenty-point rise in the gross wage share to $\omega_{1972} = 0.6903$ produced almost no mechanization response, and utilization fell to 0.72 by 1975.

What follows for how utilization is used. Any empirical strategy that recovers utilization by fitting output to capital has imposed $\theta = 1$ before measuring anything, and the estimates here put the value near 0.4 for the United States and between 0.68 and 0.95 for Chile. Utilization series built that way do not measure the distance between output and capacity; they measure the residual of an assumption. For the long-running debate over whether firms converge on a normal utilization rate, the finding is more direct still: the denominator that debate presupposes is itself a function of the distributive conflict the debate treats as separate from it.

Two extensions follow with enough specificity to be worth naming. The Chilean series terminate in 2010, which places the unwinding of the commodity super-cycle outside the sample; splicing the historical reserve, terms-of-trade, and money series to contemporary Central Bank sources through 2024 would test whether the external gate has migrated again, from markup expansion toward lithium-era rent capture, and the threshold specification estimated here transfers to that extended sample without modification. Separately, the two economies are identified on different specifications, which limits what the comparison can carry; constructing an intellectual-property capital series for Chile, or an import-biased mechanization gap for the United States, would let a single specification be estimated on both panels and would turn the sign-structure comparison offered here into a quantitative one.

The scope of this chapter is deliberately bounded. It does not present a complete crisis theory, a fully specified model of profitability dynamics, or an exhaustive dependency framework. It reconstructs capacity utilization as a structurally mediated object and identifies the transformation elasticity as the hinge of its empirical recovery in the center and the periphery, and it leaves the broader crisis and profitability architecture to the chapters that follow.

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A The Generalized Perpetual Inventory Method: Recursive Construction and Calibration

This appendix documents the recursive construction of U.S. nonfinancial corporate capital stocks via the Generalized Perpetual Inventory Method (GPIM) of Shaikh (2016, App. 6.5). The construction proceeds asset-by-asset for machinery and equipment (ME) and nonresidential structures (NRC), recovering four distinct quantities from BEA Fixed Assets data: the asset-specific implicit price index $P_{j,t}$, the gross real stock $K^{G,R}_{j,t}$, the net current-cost stock $K^{N,V}_{j,t}$, and the net-value schedule $V_j(v)$. All series are expressed in constant 2017 dollars with base-year normalization $P_{j,2017} = 100$.

A.1 The Survival Function

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage of age v still in physical operation is

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right], \quad v = 0, 1, \ldots, L_j, \tag{100}$$

where $\alpha_j > 1$ is the Weibull shape parameter and the scale parameter satisfies $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, with L_j denoting the mean service life. The shape parameter $\alpha_j > 1$ implies a progressively increasing hazard rate: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. This pattern is consistent with the empirical evidence of Nomura (2005), who finds that nearly half of 66 Japanese asset classes exhibit increasing hazard rates.

The calibrated parameters follow the BEA Fixed Assets methodology (Fraumeni et al., 1997):

$$L_{\text{ME}} = 14, \quad \alpha_{\text{ME}} = 1.7, \qquad L_{\text{NRC}} = 30, \quad \alpha_{\text{NRC}} = 1.6. \tag{101}$$

The corresponding maximum ages are $L_{\text{ME}} = 14$ years and $L_{\text{NRC}} = 30$ years, beyond which $S_j(v) = 0$ and the vintage is fully retired.

A.2 The Age-Price Schedule and Net-Value Function

The net-value schedule combines physical survival with financial age-price decay. For a vintage of age v, the net-value function is

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v, \tag{102}$$

where d_j is the geometric declining-balance depreciation rate. The GPIM separates the two dimensions that the BEA conflates in its geometric depreciation convention: physical survival follows the Weibull distribution (100), while financial age-price decay follows the geometric profile $(1 - d_j)^v$. This separation is necessary to construct distinct capacity and profitability measures (Shaikh, 2016, App. 6.3).

The declining-balance rates are calibrated as

$$d_{\text{ME}} = 0.110, \qquad d_{\text{NRC}} = 0.024, \tag{103}$$

consistent with the geometric depreciation rates documented by Fraumeni et al. (1997) for the corresponding asset classes.

A.3 The SFC Implicit Price Recursion

The implicit price index $P_{j,t}$ is the unique price level that closes the stock-flow identity for the net current-cost stock. In each period t, the BEA-reported current-cost net stock must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}, \tag{104}$$

where $I_{j,t}^{\text{NOM}}$ is current-year nominal gross investment and $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}}/(P_{j,t-v}/100)$ is real investment of vintage $t-v$ deflated by the previously recovered implicit price. The first term on the right-hand side is the current-year investment vintage (age zero, valued at the current price). The second term is the net-value-weighted sum of surviving real vintages revalued at the current price level.

Solving (104) for $P_{j,t}$ yields the forward recursion

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\displaystyle\sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}}. \tag{105}$$

Given the initial price $P_{j,0}$ and the sequence of nominal investment flows $\{I_{j,t}^{\text{NOM}}\}_{t=0}^{T}$, the recursion (105) determines $P_{j,t}$ uniquely for each $t = 1, \ldots, T$. The price index is "implicit" because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA's reported current-cost net stock.

A.4 Initial Conditions and Base-Year Normalization

The recursion (105) requires an initial price $P_{j,0}$ and initial real investment vintages. The base year is 2017, at which $P_{j,2017} = 100$ by normalization. The nominal investment series begins in 1901, sourced from BEA Fixed Assets Table FAAt201 (nonresidential equipment and structures, gross investment by asset class). The current-cost net stock anchors $K_{j,t}^{\text{NET,CC}}$ are sourced from BEA Fixed Assets Table FAAt401 (net stock by asset class, current-cost basis).[18]

The initialization proceeds backward from 2017. Given $P_{j,2017} = 100$ and the observed nominal investment series, the recursion (105) is iterated backward to recover $P_{j,t}$ for $t < 2017$. Alternatively, the recursion may be initialized at the earliest available year and iterated forward; the two approaches yield identical results given the BEA anchors.

A.5 The Gross Real Stock

The capital stock used to estimate productive capacity is the gross real stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v}^{\text{REAL}}. \tag{106}$$

[18] Data retrieved via the BEA API (`DatasetName`: FixedAssets). Table FAAt201 provides nominal gross investment by asset; Table FAAt401 provides current-cost net stock by asset; Table FAAt402 provides current-cost depreciation by asset.

This measure relies exclusively on the physical survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$. It therefore measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter. The OECD manual confirms that the gross stock, valued at "as new" prices, "is widely used as a broad indicator of the productive capacity of a country" (OECD, 2001, 31).

A.6 The Warm-Up Period and Estimation Sample

The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for nonresidential structures ($L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. The estimation sample is restricted to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations.

A.7 Validation

The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors to machine precision. The BEA's Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing.

A.8 Data Sources

Table 10 summarizes the BEA Fixed Assets tables used in the GPIM construction.

Table 10: BEA Fixed Assets tables used in GPIM construction

Table	**Content**	**Use**
FAAt201	Nominal gross investment by asset	$I_{j,t}^{\text{NOM}}$
FAAt401	Current-cost net stock by asset	$K_{j,t}^{\text{NET,CC}}$
FAAt402	Current-cost depreciation by asset	Validation

Notes: All tables are from the BEA Fixed Assets Accounts, retrieved via the BEA API (`DatasetName`: FixedAssets). Asset classes: ME = nonresidential equipment; NRC = nonresidential structures. Sectoral scope: nonfinancial corporate business.

B Appendix: United States Specification B — Full Estimation Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the aggregate capital model (Specification A). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 11: **Specification A, Set 1: Standard Wage Interaction (GVA Basis)**

Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.61^{***}	0.58^{***}	0.59^{***}	0.57	0.55	0.58^{***}	0.61^{***}	0.57	0.54
	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.07)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.03	-0.04	0.06	-0.08	-0.19	-0.23	0.01	-0.07	-0.20	-0.29
	(0.13)	(0.10)	(0.12)	(0.10)	(0.14)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.05	0.05	0.04	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	2.70	11.41	5.82	9.68	2.56	12.13	5.30	10.48	3.74	12.92
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.33	-2.73	-4.15^{**}	-3.56^{*}	-1.80	-2.15	-3.33	-2.66	-1.75	-2.12
Bandwidth ratio b	0.275	0.187	0.286	0.193	0.285	0.200	0.315	0.254	0.304	0.201	0.321	0.256
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 12: **Specification A, Set 1: Standard Wage Interaction (NVA Basis)**

Model	M1.7a	M1.7b	M1.8a	M1.8b	M1.9a	M1.9b	M1.10a	M1.10b	M1.11a	M1.11b	M1.12a	M1.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.57	0.56	0.59^{***}	0.71^{***}	0.57	0.63
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.02	-0.08	-0.02	-0.08	-0.19	-0.26	-0.09	-0.32^{*}	-0.20	-0.29
	(0.13)	(0.10)	(0.13)	(0.10)	(0.13)	(0.10)	(0.11)	(0.09)	(0.09)	(0.08)	(0.10)	(0.10)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.05	0.06^{*}	0.03	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	3.30	11.34	3.30	11.34	3.37	12.47	5.19	10.19	4.81	12.40
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-1.97	-2.40	-2.42	-3.61^{*}	-1.91	-2.31
Bandwidth ratio b	0.275	0.187	0.275	0.187	0.275	0.187	0.277	0.235	0.242	0.156	0.280	0.239
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 13: **Specification A, Set 2: Effective Wage Interaction (GVA Basis)**

Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.57^{***}	0.58^{***}	0.54^{***}	0.57	0.55	0.58^{***}	0.56^{***}	0.56	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	-0.00	-0.15	0.04	-0.20	-0.22	-0.37	0.01	-0.18	-0.20	-0.39
	(0.10)	(0.07)	(0.10)	(0.07)	(0.11)	(0.09)	(0.10)	(0.09)	(0.10)	(0.08)	(0.09)	(0.17)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.04	0.05	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	6.17	9.61	4.39	7.88	5.72	12.93	12.82	8.55	6.98	10.32
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.26	-2.96	-4.02^{**}	-2.93	-1.79	-2.31	-3.37	-2.87	-1.68	-2.20
Bandwidth ratio b	0.298	0.169	0.307	0.173	0.291	0.176	0.298	0.241	0.310	0.178	0.315	0.245
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 14: **Specification A, Set 2: Effective Wage Interaction (NVA Basis)**

Model	M2.7a	M2.7b	M2.8a	M2.8b	M2.9a	M2.9b	M2.10a	M2.10b	M2.11a	M2.11b	M2.12a	M2.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.57	0.55	0.58^{***}	0.65^{***}	0.57	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	0.00	-0.17	0.00	-0.17	-0.21	-0.35	-0.04	-0.30^{**}	-0.19	-0.49
	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.08)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.04	0.07^{*}	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	10.76	9.07	10.76	9.07	6.13	13.18	11.05	9.06	7.50	10.89
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.00	-2.72	-2.22	-4.07^{**}	-1.84	-2.58
Bandwidth ratio b	0.298	0.169	0.298	0.169	0.298	0.169	0.259	0.237	0.280	0.165	0.272	0.237
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 15: Custom Monte Carlo Simulated Critical Bounds for CMPR IM-OLS Inference (Specification A)

Statistic / Parameter	**Asymptotic 95%**	**Sim 90%**	**Sim 95%**	**Sim 99%**
Panel A: Baseline Sample ($T = 94$, 1931–2024) — No Structural Breaks ($m = 0$)				
t_{level} (k_t^{cap})	1.960	3.877	5.119	8.502
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.123	4.006	6.238
CT_{IM} (Type 2 Cointegration)	0.070	0.0580	0.0705	0.1005
Panel B: Baseline Sample ($T = 94$, 1931–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
t_{level} (k_t^{cap})	1.960	2.607	3.267	4.989
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.455	4.325	6.444
CT_{IM} (Type 2 Cointegration)	0.070	0.0482	0.0598	0.0895
Panel C: Supervisory Sample ($T = 76$, 1948–2024) — No Structural Breaks ($m = 0$)				
t_{level} (k_t^{cap})	1.960	3.900	5.155	8.387
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.139	4.018	6.206
CT_{IM} (Type 2 Cointegration)	0.070	0.0635	0.0780	0.1151
Panel D: Supervisory Sample ($T = 76$, 1948–2024) — Bai-Perron Level-Shift Breaks ($m \geq 1$)				
t_{level} (k_t^{cap})	1.960	2.687	3.371	5.188
t_{inter} $(k_t^{\text{cap}} \cdot C_t)$	N/A	3.427	4.277	6.213
CT_{IM} (Type 2 Cointegration)	0.070	0.0545	0.0696	0.1065

Notes: Critical bounds derived from a custom $N = 100,000$ iteration Monte Carlo simulation suite calibrated specifically to the $I(1) \times I(1)$ product integration geometry of the regressors and dynamic Bai-Perron break degrees of freedom. Panels A and B reflect the baseline 1931–2024 sample length ($T = 94$), while Panels C and D reflect the supervisory-adjusted Tier 2 wage share sample ($T = 76$). Bandwidth $b = M/T$ is fixed per Vogelsang & Wagner (2014) IM-OLS specification. Data Generating Process (H_0) constructs independent zero-drift random walks for base level variables $(k_t^{\text{cap}}, \omega_t)$, with interaction product processes algebraically assembled prior to estimation. Parameter t-statistics are centered around their true theoretical values ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to ensure pure size calibration rather than power distortion.

C Appendix: United States Specification B - Heterogeneous Capital Stock Empirical Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the heterogeneous capital model (Specification B). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1 - \iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate Baseline, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 16: Specification B, Set 1: Standard Wage Interaction (GVA Basis)

Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
$\hat{\alpha}$ (Intercept)	12.90	12.82	12.89	12.82	12.90	12.82	13.85	13.52	12.89	12.81	13.85	13.52
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.60	0.51	0.61	0.47	0.61	0.48	0.51	0.50	0.59	0.52	0.51	0.49
	(0.03)	(0.08)	(0.03)	(0.07)	(0.03)	(0.07)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$	0.38	0.67	0.40	0.44	0.39	0.53	0.38	1.09	0.36	0.60	0.34	1.11
	(0.14)	(0.26)	(0.14)	(0.22)	(0.14)	(0.23)	(0.12)	(0.20)	(0.14)	(0.33)	(0.11)	(0.20)
$\hat{\theta}_{\omega\kappa}$	-2.75	-4.89	7.95	0.05	4.01	-4.20	-16.40	-10.51	-9.89	-5.93	-18.27	-9.04
	(4.07)	(3.10)	(5.98)	(2.97)	(7.41)	(4.18)	(6.63)	(5.86)	(4.85)	(3.96)	(5.56)	(5.55)
$\hat{\theta}_{\kappa p}$	-0.68	1.38	-2.07	0.16	-1.28	0.64	-1.43	-0.35	0.01	1.63	-0.78	-0.35
	(0.58)	(0.82)	(0.87)	(0.64)	(0.61)	(0.65)	(0.56)	(0.47)	(0.57)	(0.99)	(0.54)	(0.47)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.05	0.03	0.04	0.03	0.05	0.03	0.03	0.02	0.05	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	16.11	9.55	16.40	12.60	16.25	11.41	9.98	2.78	16.37	10.87	8.42	3.26
t_{ADF} [H_0: no coint.]	-4.16*	-4.46**	-4.25*	-4.67**	-4.26*	-4.69**	-2.56	-3.22	-4.26*	-4.50**	-2.58	-3.17
Bandwidth ratio b	0.209	0.176	0.215	0.173	0.211	0.174	0.211	0.203	0.213	0.191	0.210	0.203
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 17: Specification B, Set 1: Standard Wage Interaction (NVA Basis)

Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
$\hat{\alpha}$ (Intercept)	12.90	12.82	12.90	12.82	12.90	12.82	13.87	13.57	12.90	12.82	13.88	13.57
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.60	0.51	0.60	0.51	0.60	0.51	0.51	0.52	0.59	0.50	0.51	0.50
	(0.03)	(0.08)	(0.03)	(0.08)	(0.03)	(0.08)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$	0.38	0.67	0.38	0.67	0.38	0.67	0.35	0.99	0.36	0.61	0.34	1.02
	(0.14)	(0.26)	(0.14)	(0.26)	(0.14)	(0.26)	(0.11)	(0.18)	(0.14)	(0.34)	(0.11)	(0.18)
$\hat{\theta}_{\omega\kappa}$	-2.75	-4.89	-2.79	-4.96	-2.75	-4.89	-13.06	-13.10	-7.06	-2.73	-12.13	-11.13
	(4.07)	(3.10)	(4.14)	(3.15)	(4.07)	(3.10)	(4.24)	(4.38)	(3.32)	(2.62)	(3.50)	(3.99)
$\hat{\theta}_{\kappa p}$	-0.68	1.38	-0.68	1.38	-0.68	1.38	-0.99	-0.37	0.33	1.50	-0.49	-0.30
	(0.58)	(0.82)	(0.58)	(0.82)	(0.58)	(0.82)	(0.52)	(0.42)	(0.65)	(0.98)	(0.53)	(0.43)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.05	0.03	0.05	0.03	0.05	0.03	0.02	0.02	0.05	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	16.11	9.55	16.11	9.55	16.11	9.55	8.41	6.15	16.35	11.90	7.50	4.26
t_{ADF} [H_0: no coint.]	-4.16*	-4.46**	-4.16*	-4.46**	-4.16*	-4.46**	-2.61	-3.55	-4.20*	-4.49**	-2.50	-3.44
Bandwidth ratio b	0.209	0.176	0.209	0.176	0.209	0.176	0.207	0.205	0.213	0.192	0.211	0.205
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 18: Specification B, Set 2: Effective Wage Interaction (GVA Basis)

Model	M2.25a	M2.26b	M2.29a	M2.30b	M2.33a	M2.34b	M2.37a	M2.38b	M2.41a	M2.42b	M2.45a	M2.46b
$\hat{\alpha}$ (Intercept)	12.82	12.77	12.83	12.78	12.79	12.76	13.68	13.66	12.80	12.77	13.67	13.61
	(0.07)	(0.06)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)	(0.05)	(0.07)	(0.06)	(0.05)	(0.06)
$\hat{\theta}_{\text{NRC}}$	0.61	0.58	0.61	0.55	0.61	0.62	0.55	0.51	0.61	0.60	0.55	0.70
	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.09)	(0.01)	(0.03)
$\hat{\theta}_{\kappa}$	0.32	0.39	0.32	0.44	0.31	0.14	0.36	0.02	0.31	0.36	0.34	0.72
	(0.13)	(0.29)	(0.14)	(0.30)	(0.14)	(0.38)	(0.08)	(0.11)	(0.14)	(0.29)	(0.08)	(0.10)
$\hat{\theta}_{\omega\kappa}$	-12.77	-10.13	-13.73	-12.11	-15.99	-12.29	-20.77	-19.87	-14.54	-11.15	-18.85	-15.55
	(3.81)	(3.43)	(4.45)	(3.82)	(4.94)	(4.22)	(4.11)	(3.99)	(4.60)	(3.63)	(3.32)	(3.54)
$\hat{\theta}_{\kappa p}$	0.18	0.92	0.35	1.44	-0.39	0.58	-1.37	-0.81	0.08	0.88	-0.96	0.10
	(0.51)	(0.69)	(0.58)	(0.79)	(0.45)	(0.60)	(0.42)	(0.39)	(0.52)	(0.66)	(0.40)	(0.41)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.04	0.03	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	15.18	6.02	14.58	3.71	13.20	11.19	14.75	6.00	12.33	9.93
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.36*	-4.52**	-4.40*	-4.77**	-4.12	-4.02	-4.41*	-4.69**	-4.13*	-3.97
Bandwidth ratio b	0.185	0.173	0.194	0.175	0.192	0.173	0.206	0.206	0.195	0.179	0.207	0.207
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 19: Specification B, Set 2: Effective Wage Interaction (NVA Basis)

Model	M2.27a	M2.28b	M2.31a	M2.32b	M2.35a	M2.36b	M2.39a	M2.40b	M2.43a	M2.44b	M2.47a	M2.48b
$\hat{\alpha}$ (Intercept)	12.82	12.77	12.82	12.77	12.82	12.77	13.72	13.69	12.83	12.78	13.73	13.73
	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.04)	(0.04)	(0.07)	(0.07)	(0.04)	(0.04)
$\hat{\theta}_{\text{NRC}}$	0.61	0.58	0.61	0.58	0.61	0.58	0.54	0.51	0.60	0.60	0.54	0.54
	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.08)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$	0.32	0.39	0.32	0.39	0.32	0.39	0.33	0.06	0.32	0.39	0.31	0.31
	(0.13)	(0.29)	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.09)	(0.13)	(0.29)	(0.06)	(0.06)
$\hat{\theta}_{\omega\kappa}$	-12.77	-10.13	-12.97	-10.28	-12.77	-10.13	-17.05	-16.76	-11.08	-7.93	-14.97	-14.97
	(3.81)	(3.43)	(3.86)	(3.48)	(3.81)	(3.43)	(2.34)	(2.55)	(3.48)	(2.96)	(2.01)	(2.01)
$\hat{\theta}_{\kappa p}$	0.18	0.92	0.18	0.92	0.18	0.92	-0.79	-0.71	0.69	1.41	-0.31	-0.31
	(0.51)	(0.69)	(0.51)	(0.69)	(0.51)	(0.69)	(0.32)	(0.31)	(0.63)	(0.75)	(0.33)	(0.33)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.03	0.02	0.03	0.02	0.03	0.02	0.02	0.02	0.03	0.03	0.02	0.02
$S_{\text{Type 2}}$ [H_0: coint.]	14.97	5.93	14.97	5.93	14.97	5.93	10.28	8.70	15.37	6.37	8.96	8.96
t_{ADF} [H_0: no coint.]	-4.33*	-4.58**	-4.33*	-4.58**	-4.33*	-4.58**	-4.71**	-4.57**	-4.34*	-4.60**	-4.24*	-4.24*
Bandwidth ratio b	0.185	0.173	0.185	0.173	0.185	0.173	0.208	0.208	0.193	0.181	0.208	0.208
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log real capacity). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2024). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 20: Custom Monte Carlo Simulated Critical Bounds Across Structural Break Regimes (Specification B)

Statistic / Parameter	**Asymptotic 95%**	**Sim 90%**	**Sim 95%**	**Sim 99%**	**Empirical (M2.44b)**
Panel A: No Structural Breaks ($m = 0$, Sample $T = 94$)					
t_{NRC} (k_t^{NRC})	1.960	1.842	2.195	2.910	7.50**
t_κ ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$)	1.960	1.895	2.241	2.980	1.34
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.115	2.512	3.340	-2.68**
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.080	2.475	3.295	1.88*
CT_{IM} (Type 2 Global Test)	18.50	11.20	13.45	18.90	6.37**
Panel B: One Endogenous Structural Break ($m = 1$, e.g., M2.40b)					
t_{NRC} (k_t^{NRC})	1.960	1.920	2.280	3.010	–
t_κ (κ_t)	1.960	1.970	2.330	3.080	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.180	2.590	3.430	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.150	2.550	3.390	–
CT_{IM} (Type 2 Global Test)	19.80	12.10	14.50	20.20	–
Panel C: Two Endogenous Structural Breaks ($m = 2$, e.g., M2.46b)					
t_{NRC} (k_t^{NRC})	1.960	1.995	2.370	3.120	–
t_κ (κ_t)	1.960	2.050	2.420	3.190	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.250	2.670	3.520	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.210	2.630	3.470	–
CT_{IM} (Type 2 Global Test)	21.20	13.00	15.50	21.60	–
Panel D: Three Endogenous Structural Breaks ($m = 3$, Preferred Model M2.44b)					
t_{NRC} (k_t^{NRC})	1.960	2.075	2.455	3.210	7.50**
t_κ (κ_t)	1.960	2.130	2.515	3.305	1.34
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.315	2.745	3.615	-2.68**
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.275	2.705	3.560	1.88*
CT_{IM} (Type 2 Global Test)	22.65	13.90	16.55	22.95	6.37**
Panel E: Four Endogenous Structural Breaks ($m = 4$, e.g., M2.18b, M2.34b)					
t_{NRC} (k_t^{NRC})	1.960	2.150	2.540	3.310	–
t_κ (κ_t)	1.960	2.210	2.610	3.420	–
$t_{\omega\kappa}$ ($\kappa_t \cdot C_t$)	N/A	2.380	2.820	3.710	–
$t_{\kappa p}$ ($\kappa_t \cdot p_t^\kappa$)	N/A	2.340	2.780	3.650	–
CT_{IM} (Type 2 Global Test)	24.10	14.80	17.60	24.30	–

Notes: Critical bounds derived from a 100,000-pass specification-dependent Monte Carlo suite tailored to Specification B's 4-variable stochastic design ($k^{\text{NRC}}, \kappa, \kappa \cdot C, \kappa \cdot p^\kappa$) across all structural break counts ($m = 0 \ldots 4$). The DGP under H_0 simulates independent zero-drift random walks for base variables and algebraically constructs product terms, with dynamic Bai-Perron break selection applied recursively. Simulated parameter t-statistics are centered around true DGP parameters ($t^* = |(\hat{\beta}^* - \beta_{\text{true}})/SE|$) to prevent artificial bound inflation. Empirical values reported for preferred Model M2.44b ($m = 3$). Bold empirical values denote significance based on the simulated 95% bound (* $p < 0.05$, ** $p < 0.01$).

D Appendix: Chile - External Constraint and Regime Shifting Results

Table 21: Comparative Cointegration Estimates across Net and Gross Wage Shares (Chile, 1943–2010)

	Net Wage Share (ω^{net})				**Gross Wage Share** (ω^{gross})			
	Baseline	Bai-Perron	Comp. Threshold	Parsimonious	Baseline	Bai-Perron	Comp. Threshold	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.96***	0.73***	0.67***	0.79***	0.99***	0.98***	0.70***	0.82***
	(0.01)	(0.12)	(0.09)	(0.07)	(0.00)	(0.05)	(0.08)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	0.03	0.56	0.53	0.12	-0.41	0.22	-0.14	-1.48*
	(0.25)	(0.50)	(0.40)	(0.39)	(0.29)	(0.27)	(0.44)	(0.54)
$\hat{\theta}_{\omega\kappa}$ (Wage Interaction)	-0.28	-0.16	-0.12	-0.01	0.01	-0.35	0.96	3.29*
	(0.22)	(0.21)	(0.17)	(0.22)	(0.38)	(0.26)	(0.73)	(1.07)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	—	-0.18	0.36*	0.34	—	5.13***	0.66	1.52**
	(0.00)	(0.28)	(0.13)	(0.16)	(0.00)	(1.00)	(0.31)	(0.45)
$\hat{\theta}_{\omega\kappa D}$ (Triple Interaction)	—	-0.09	-0.15	-0.17	—	-11.86***	-0.95	-3.15*
		(0.33)	(0.18)	(0.23)		(2.35)	(0.72)	(1.04)
$\hat{\theta}_{p}$ (Price Control $\tilde{p}$)	-0.05	-0.19**	-0.24***	—	-0.04	0.02	-0.20***	—
	(0.11)	(0.09)	(0.07)		(0.11)	(0.07)	(0.07)	
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	None	BP (1999)	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
CT_{IM} [H_0: Cointegration]	—	12.88	11.72	15.20	—	14.55	8.50	8.39
t_{ADF} [H_0: Unit Root]	-1.84	-2.72	-3.82*	-3.81*	-1.92	-2.98	-4.11**	-3.81*
Bandwidth ratio b	0.626	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68

Table 22: Comparative Cointegration Estimates across Net and Gross Rates of Exploitation (Chile, 1943–2010)

	Net Rate of Exploitation (e^{net})				**Gross Rate of Exploitation** (e^{gross})			
	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious	Baseline	Bai-Perron	Comp. Thresh-old	Parsimonious
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_{k^{\text{NR}}}$ (Structures k^{NR})	0.96***	0.73***	0.68***	0.80***	0.99***	0.98***	0.71***	0.82***
	(0.01)	(0.11)	(0.09)	(0.07)	(0.01)	(0.06)	(0.08)	(0.08)
$\hat{\theta}_{\kappa}$ (Mechanization Gap κ)	-0.21	0.45	0.43	0.15	-0.42***	-0.05	0.51	0.85
	(0.11)	(0.42)	(0.28)	(0.25)	(0.08)	(0.27)	(0.27)	(0.38)
$\hat{\theta}_{e\kappa}$ (Exploitation Interaction)	0.14	0.08	0.06	0.02	0.00	0.09	-0.19	-0.69*
	(0.11)	(0.10)	(0.07)	(0.09)	(0.10)	(0.07)	(0.17)	(0.23)
$\hat{\theta}_{\kappa D}$ (Regime Shift κD)	—	-0.27	0.21*	0.15	—	-2.39***	-0.02	-0.73
	(0.00)	(0.10)	(0.07)	(0.08)	(0.00)	(0.50)	(0.23)	(0.29)
$\hat{\theta}_{e\kappa D}$ (Triple Interaction)	—	0.05	0.09	0.12	—	1.83***	0.20	0.67*
		(0.15)	(0.07)	(0.10)		(0.36)	(0.16)	(0.22)
$\hat{\theta}_{p}$ (Price Control $\tilde{p}$)	-0.04	-0.18**	-0.22***	—	-0.04	0.02	-0.20***	—
	(0.11)	(0.09)	(0.07)		(0.11)	(0.07)	(0.07)	
Regime Threshold (D_t)	None	BP (1987)	$Z_t \leq 0.425$	$Z_t \leq 0.425$	None	BP (1999)	$Z_t \leq 0.425$	$Z_t \leq 0.425$
Price Control Included	Yes	Yes	Yes	No	Yes	Yes	Yes	No
CT_{IM} [H_0: Cointegration]	—	12.52	10.83	14.99	—	14.40	7.96	8.34
t_{ADF} [H_0: Unit Root]	-1.89	-2.73	-3.86*	-3.84*	-1.96	-3.05	-4.15**	-3.89*
Bandwidth ratio b	0.623	0.650	0.654	0.654	0.635	0.652	0.655	0.654
Sample Size T	68	68	68	68	68	68	68	68

E Appendix: Critical Values and Inferencefor the Cointegration Diagnostics

Three families of test statistic carry the cointegration verdict in this chapter, and they do not draw their critical values from the same source. The residual-based diagnostics use published constants. The fixed-b Wald statistics do not, because their limiting distribution is jointly indexed by the bandwidth ratio b and the number of integrated regressors k, so no single tabulated value covers the specifications estimated here.

Table 23: Critical Values for the Residual-Based Cointegration Diagnostics

Test	Source	10%	5%	1%
Residual ADF (t_{ADF})	MacKinnon (2010), Case 2, constant only, $m = 4$	-3.81	-4.10	-4.64
Shin LM (C_{Shin}), $k = 2$	Shin (1994), Table 1, demeaned C_μ	0.163	0.221	0.380
Shin LM (C_{Shin}), $k = 3$	Shin (1994), Table 1, demeaned C_μ	0.121	0.159	0.271

Notes: The residual ADF statistic tests the null of no cointegration, so rejection supports a cointegrating relation. The Shin LM statistic reverses the null: it tests the null of cointegration, so a value below the critical value fails to reject and is the favorable outcome. m denotes the number of regressors entering the MacKinnon response surface; k denotes the number of integrated regressors entering the Shin statistic.

For the fixed-b Wald statistics reported as CT_{IM} and for the coefficient significance stars in every results table, critical values are generated by simulation rather than read from a table. The protocol holds the estimated specification fixed and varies only the innovation draws. For each specification I generate integrated regressors as independent random walks of the same length as the estimation sample, impose the null under test, apply the same Frisch-Waugh-Lovell orthogonalization and the same deterministic components used in estimation, and compute the IM-OLS statistic on each replication. The empirical quantiles of the resulting distribution supply the thresholds. Replication counts are 100,000 for the United States specifications and 10,000 for the Chilean threshold models, where each pass carries the additional cost of a profile search over the threshold parameter.

Two features of this protocol matter for reading the tables. First, because the simulated null reproduces the bandwidth ratio actually used in estimation, the reported stars already incorporate the fixed-b correction and are not comparable to standard normal or chi-squared thresholds. Second, because the Chilean critical bounds are simulated under profile search, they are conservative relative to a fixed-threshold benchmark: the search itself consumes degrees of freedom, and the simulation charges for it.

F Appendix: Order of Integration and Stochastic Pre-Testing Battery for the United States and Chile

A In Cointegrating Multivariate Polynomial Regressions (CMPR) estimated via Integrated Modified OLS (IM-OLS) (Vogelsang and Wagner, 2014, 2024), valid asymptotic inference and pivotal fixed-b Wald distributions require all stochastic regressors to be integrated of order one ($I(1)$). When non-linear interaction terms ($k_t \times \omega_{t-1}$, $k_t \times \omega_{t-1} \times (1 - \iota_{t-1})$) or regime-dependent threshold regressors ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq \hat{\gamma}\}$) enter the long-run relation, establishing their stochastic integration order prior to estimation is an essential prerequisite. This appendix reports the complete unit-root and stationarity test batteries for both economies across all baseline series, non-linear interactions, and demand controls.

B Table 24 outlines the sample boundaries, degree-of-freedom spans, and zero-NA estimation windows across the empirical dataset. For the United States, the stochastic pre-testing battery spans 1929–2024 ($N_{\text{lev}} = 96, N_{\text{diff}} = 95$). For Chile, the estimation window spans 1943–2010 ($N_{\text{lev}} = 68, N_{\text{diff}} = 67$).

Table 24: Data Series Coverage, Time Horizons, and Estimation Windows

Economy / Data Series Scope	**Year Range**	N_{lev}	N_{diff}	**Empirical Function**
U.S. NIPA Compensation Shares	1929–2024	96	95	Distribution Series
U.S. Private Core Gross Capital (K^{cap})	1925–2024	100	99	GPIM Capital Series
U.S. Nonfinancial Corporate Output (Y^{NFC})	1929–2024	96	95	Chained Real Series
U.S. Unit Root Pre-Testing Sample	**1929–2024**	**96**	**95**	**Stochastic Audit**
U.S. Long-Run Estimation Window	**1931–2024**	**94**	**93**	**IM-OLS Regressions**
Chile Historical Macroeconomic Series	1900–2024	125	124	Historical Panel
Chile Descriptive Stylized Facts	1940–2010	71	70	Summary Statistics
Chile Long-Run Estimation Window	**1943–2010**	**68**	**67**	**Threshold CMPR**

Note: U.S. data are from BEA NIPA Tables and Fixed Assets Accounts. Chilean data are from Central Bank historical accounts and ClioLab databases. For Chile, the 1943 start corresponds to the enactment of Law No. 7,200 (1942), which abandoned the gold-standard convertibility rule and established state-led industrialization credit allocation alongside CORFO. Historical external and terms-of-trade series end in 2010.

F.1 United States: Unit Root Test Batteries and Interaction Properties

A We evaluate the stochastic properties of U.S. level variables and their interactions using five standard tests: Augmented Dickey-Fuller (ADF), DF-GLS (ERS), Phillips-Perron (PP), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and Zivot-Andrews (ZA) tests with endogenous structural breaks. For level specifications, test regressions include a deterministic trend. For first-differenced specifications,

regressions specify an intercept. In the Zivot-Andrews procedure, structural breaks are estimated endogenously in both intercept and trend (`model = "both"`).

B Table 25 reports the test battery for the corporate Intellectual Property Products (IPP) gross investment share ($\iota_t = I_t^{\text{IPP}}/I_t^{\text{TOTAL}}$), the capital composition ratio ($\kappa_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), private core gross capital ($k^{\text{cap}} = \ln[K^{\text{ME}} + K^{\text{NRC}}]$), and their double and triple interaction terms with eight distributive wage share variants (ω_i).

Table 25: Unit Root Test Battery for U.S. IPP Share and Capital Composition Interactions (1929–2024)

Variable / Measure	N_{lev}	N_{diff}	**ADF Level**	**ADF Diff**	**ERS Level**	**ERS Diff**	**KPSS Level**	**KPSS Diff**	**ZA Break (Year)**	**Verdict**
Corp IPP Share (ι_t)	96	95	-2.432	-7.129***	-2.104	-4.883***	0.444***	0.108	-4.529 (1976)	**I(1)**
Capital Composition (κ_t)	96	95	-0.196	-4.592***	-1.024	-2.737***	0.593***	0.761***	-3.632 (1974)	**I(1) with Break**
Private Core Capital (k^{cap})	96	95	-1.811	-5.744***	-2.041	-0.837	0.318***	0.237	-4.792 (1972)	**I(1)**
Panel B: IPP and Capital Composition Interactions										
$\iota_t \times \omega_{\text{CORP,GVA}}$	96	95	-3.296*	-7.220***	-3.008	-4.622***	0.331***	0.047	-4.573 (1984)	**I(1)**
$\iota_t \times \omega_{\text{CORP,NVA}}$	96	95	-3.128	-7.278***	-2.825	-4.682***	0.372***	0.053	-4.714 (1984)	**I(1)**
$\iota_t \times \omega_{\text{NFC,GVA}}$	96	95	-3.184*	-7.164***	-2.952	-4.881***	0.336***	0.051	-4.541 (1984)	**I(1)**
$\iota_t \times \omega_{\text{NFC,NVA}}$	96	95	-3.013	-7.254***	-2.764	-4.944***	0.378***	0.059	-4.666 (1984)	**I(1)**
$\iota_t \times \omega_{\text{ST,Tier1}}$	96	95	-3.013	-7.254***	-2.764	-4.944***	0.378***	0.059	-4.666 (1984)	**I(1)**
$\iota_t \times \omega_{\text{ST,Tier2}}$	96	95	-2.869	-7.480***	-2.622	-4.484***	0.428***	0.063	-4.917* (1984)	**I(1)**
$\iota_t \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-3.184*	-7.164***	-2.952	-4.881***	0.336***	0.051	-4.541 (1984)	**I(1)**
$\iota_t \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-3.027	-7.417***	-2.730	-4.479***	0.396***	0.057	-4.800 (1984)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{CORP,GVA}}$	96	95	-1.469	-6.952***	-1.525	-3.290***	0.532***	0.205	-4.610 (1985)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{CORP,NVA}}$	96	95	-1.408	-6.975***	-1.513	-3.198***	0.537***	0.217	-4.615 (1985)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{NFC,GVA}}$	96	95	-1.386	-6.952***	-1.480	-3.493***	0.533***	0.219	-4.634 (1985)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{NFC,NVA}}$	96	95	-1.316	-6.990***	-1.467	-3.399***	0.538***	0.232	-4.663 (1986)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier1}}$	96	95	-1.316	-6.990***	-1.467	-3.399***	0.538***	0.232	-4.663 (1986)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier2}}$	96	95	-1.465	-7.334***	-1.564	-3.194***	0.544***	0.201	-4.635 (1983)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-1.386	-6.952***	-1.480	-3.493***	0.533***	0.219	-4.634 (1985)	**I(1)**
$\kappa_t \times \iota_t \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-1.529	-7.275***	-1.569	-3.315***	0.537***	0.192	-4.578 (1983)	**I(1)**

Note: Primary estimation sample: 1929–2024 ($N_{\text{lev}} = 96$, $N_{\text{diff}} = 95$). Significance stars: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. **Star Interpretation Warning:** For ADF, ERS, PP, and ZA, H_0 is a *unit root*; stars indicate rejection of the unit root (supporting stationarity). For KPSS, H_0 is *stationarity*; stars indicate rejection of stationarity (confirming non-stationarity in levels; absence of stars in differences confirms differenced stationarity). Critical values: ADF Level $(-3.99, -3.43, -3.13)$, ADF Diff $(-3.46, -2.88, -2.57)$; ERS Level (-3.58), ERS Diff (-2.59); KPSS Level $(0.216, 0.146, 0.119)$, KPSS Diff $(0.739, 0.463, 0.347)$; ZA Level $(-5.57, -5.08, -4.82)$.

C Table 26 documents the integration properties of private core gross capital (k^{cap}) interacted with distributive wage shares and corporate IPP investment shares. Both Machinery and Equipment (k_t^{ME}) and Nonresidential Structures (k_t^{NRC}) are established $I(1)$ series. The apparent persistence in first-differenced capital composition ($\Delta\kappa_t$) under linear KPSS tests reflects the sharp structural deceleration of mechanization after 1973 (captured by the 1974 Zivot-Andrews break) rather than an $I(2)$ process (Perron, 1989). When capital is interacted with bounded wage shares and IPP shares ($k^{\text{cap}} \times \omega_i$ and $k^{\text{cap}} \times \omega_i \times \iota_t$), all double and triple interaction terms are confirmed to be strictly $I(1)$ ($ADF_{\text{diff}} < -6.89^{***}, KPSS_{\text{diff}} < 0.170$), ensuring valid CMPR estimation.

Table 26: Unit Root Test Battery for U.S. Core Capital and Distributive Interactions (1929–2024)

Variable / Measure	N_{lev}	N_{diff}	ADF Level	ADF Diff	ERS Level	ERS Diff	KPSS Level	KPSS Diff	ZA Break (Year)	Verdict
Panel C: Private Core Capital Interactions (K^{cap})										
$k^{\text{cap}} \times \omega_{\text{CORP,GVA}}$	96	95	-1.910	-7.797***	-1.149	-3.808***	0.423***	0.170	-4.847* (2001)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{CORP,NVA}}$	96	95	-2.547	-8.050***	-1.567	-3.005***	0.324***	0.108	-4.202 (1997)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{NFC,GVA}}$	96	95	-2.018	-7.307***	-1.386	-5.587***	0.413***	0.133	-4.791 (2003)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{NFC,NVA}}$	96	95	-2.645	-7.741***	-1.730	-5.153***	0.311***	0.086	-4.266 (2003)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier1}}$	96	95	-2.645	-7.741***	-1.730	-5.153***	0.311***	0.086	-4.266 (2003)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier2}}$	96	95	-3.037	-7.956***	-1.975	-4.444***	0.284***	0.065	-3.739 (1971)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier1,GVA}}$	96	95	-2.018	-7.307***	-1.386	-5.587***	0.413***	0.133	-4.791 (2003)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier2,GVA}}$	96	95	-2.775	-7.227***	-2.032	-5.237***	0.226***	0.061	-3.256 (1975)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{CORP,GVA}} \times \iota_t$	96	95	-2.682	-6.943***	-2.245	-5.225***	0.421***	0.098	-4.578 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{CORP,NVA}} \times \iota_t$	96	95	-2.561	-7.005***	-2.063	-5.341***	0.451***	0.113	-4.699 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{NFC,GVA}} \times \iota_t$	96	95	-2.559	-6.891***	-2.182	-5.357***	0.424***	0.109	-4.503 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{NFC,NVA}} \times \iota_t$	96	95	-2.434	-6.993***	-2.001	-5.469***	0.454***	0.125	-4.601 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier1}} \times \iota_t$	96	95	-2.434	-6.993***	-2.001	-5.469***	0.454***	0.125	-4.601 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier2}} \times \iota_t$	96	95	-2.304	-7.159***	-1.991	-5.156***	0.482***	0.122	-4.787 (1981)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier1,GVA}} \times \iota_t$	96	95	-2.559	-6.891***	-2.182	-5.357***	0.424***	0.109	-4.503 (1984)	**I(1)**
$k^{\text{cap}} \times \omega_{\text{ST,Tier2,GVA}} \times \iota_t$	96	95	-2.421	-7.087***	-2.137	-5.094***	0.459***	0.107	-4.644 (1984)	**I(1)**

Note: Primary estimation sample: 1929–2024 ($N_{\text{lev}} = 96$, $N_{\text{diff}} = 95$). Significance stars: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tests evaluated under identical specifications to Table 25.

F.2 Chile: Unit Root Test Batteries and Interaction Properties

A Table 27 reports the unit root battery for Chilean macroeconomic, distributive, and external constraint series over the 1943–2010 estimation window ($T = 68$). Real GDP (y_t), Nonresidential Structures (k_t^{NR}), Machinery and Equipment (k_t^{ME}), aggregate core capital (k_t^{cap}), and distributive wage shares ($\omega_t^{\text{net}}, \omega_t^{\text{gross}}$) are established as $I(1)$ non-stationary processes. The composite Balance-of-Payments index ($Z_t = 0.2\tilde{M}_t + 0.8\tilde{S}_t$) rejects non-stationarity in levels ($ADF = -6.661^{***}$), functioning as a stationary threshold state variable that partitions accumulation regimes.

Table 27: Unit Root Test Battery for Chilean Macroeconomic and Distributive Series (1943–2010)

Variable / Measure	N_{lev}	N_{diff}	**ADF Lev**	**ADF Diff**	**KPSS Lev**	**KPSS Diff**	**ZA Break**	**Verdict**
Log Real GDP (y_t)	68	67	-2.543	-6.619***	0.262***	0.129	-4.68 (1975)	**I(1)**
Structures Capital (k_t^{NR})	68	67	-1.485	-3.024**	0.485***	0.921***	-3.85 (1973)	**I(1)**
Machinery Capital (k_t^{ME})	68	67	-2.232	-2.945**	0.472***	0.924***	-4.12 (1982)	**I(1)**
Core Capital (k_t^{cap})	68	67	-1.521	-3.012**	0.480***	0.912***	-3.94 (1973)	**I(1)**
Capital Composition (κ_t)	68	67	-3.114	-2.889**	0.426***	0.780***	-4.31 (1982)	**I(1) with Break**
Net Wage Share (ω_t^{net})	68	67	-5.120***	-11.671***	0.253***	0.057	-4.89* (1973)	**I(1)**
Gross Wage Share (ω_t^{gross})	68	67	-5.120***	-11.671***	0.253***	0.057	-4.89* (1973)	**I(1)**
Import Coverage Ratio ($\tilde{M}_t$)	68	67	-4.986***	-10.901***	0.234***	0.060	-4.52 (1982)	**I(1)**
Terms of Trade ($\tilde{S}_t$)	68	67	-2.442	-10.171***	0.401***	0.348*	-4.11 (1975)	**I(1)**
Composite BoP Index (Z_t)	68	67	-6.661***	-11.292***	0.075	0.022	-5.42** (1982)	**I(0) Threshold**

Note: Sample spans 1943–2010 ($T = 68$). Significance stars: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. **Star Interpretation Warning:** For ADF, PP, and ZA, stars indicate rejection of the unit root null. For KPSS, stars indicate rejection of stationarity. Critical values: ADF Level $(-3.99, -3.45, -3.15)$, ADF Diff $(-3.46, -2.89, -2.57)$; KPSS Level $(0.216, 0.146, 0.119)$, KPSS Diff $(0.739, 0.463, 0.347)$; ZA Level $(-5.57, -5.08, -4.82)$.

B Table 28 documents the integration orders of Chilean interaction terms. Raw structures (k_t^{NR}) exhibit moderate differenced persistence; however, FWL-orthogonalization (ξ_t^{ω}) and threshold conditioning ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$) ensure that all entered regressors are strictly $I(1)$ in differences, eliminating higher-order polynomial contamination.

Table 28: Unit Root Test Battery for Chilean Interaction and Threshold Regressors (1943–2010)

Interaction / Regressor	$N_{\mathbf{lev}}$	$N_{\mathbf{diff}}$	**ADF Lev**	**ADF Diff**	**KPSS Lev**	**KPSS Diff**	**Verdict**
Raw Interaction ($k_t \times \omega_t^{\text{net}}$)	68	67	-2.442	-10.171***	0.401***	0.348*	**I(1)**
FWL Interaction ($\xi_t^{\omega} = (k_t \times \omega_t^{\text{net}})^{\perp}$)	68	67	-4.936***	-10.758***	0.229***	0.059	**I(1)**
Distribution-FX ($\omega_t^{\text{net}} \times Z_t$)	68	67	-4.986***	-10.901***	0.234***	0.060	**I(1)**
Threshold Mechanization ($k_t \times \omega_{t-1} \times \mathbf{1}\{Z_t \leq 0.425\}$)	68	67	-3.393*	-10.105***	0.189**	0.048	**I(1)**

Note: Sample spans 1943–2010 ($T = 68$). Significance stars: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Evaluated with trend models in levels and drift models in differences.